MATHEMATICS

AREA UNDER CURVE

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AREA UNDER CURVE

JEE EXERCISES

Exercise J-I

AREA BOUNDED BETWEEN TWO CURVES

1. The area between the curve $xy = a^2$, x-axis, $x = a$ and $x = 2a$ is :

(A) $a \log 2$ (B) $a^2 \log 2$ (C) $2a \log 2$ (D) None of these

वक्र $xy = a^2$, x-अक्ष, $x = a$ तथा $x = 2a$ के मध्य क्षेत्रफल है :

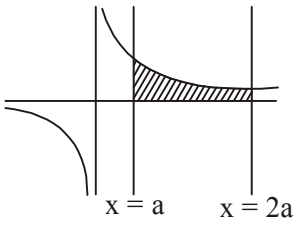
(A) $a \log 2$ (B) $a^2 \log 2$ (C) $2a \log 2$ (D) कोई नहीं

Ans. B

Sol. $xy = a^2$ x-axis $x = a$ and $x = 2a$

$$A = \int_a^{2a} y\,dx$$

$$= \int_a^{2a} \frac{a^2}{x}dx = a^2 \log x\Big|_a^{2a}$$



$$= a^2 \log \frac{2a}{a}$$

$$= a^2 \log 2$$

2. Area under the curve $y = \sin 2x + \cos 2x$ between $x = 0$ and $x = \frac{\pi}{4}$, is :

(A) 2 sq. units (B) 1 sq. units (C) 3 sq. units (D) 4 sq. units

वक्र $y = \sin 2x + \cos 2x$ तथा $x = 0$ तथा $x = \frac{\pi}{4}$, के मध्य क्षेत्रफल है :

(A) 2 वर्ग इकाई (B) 1वर्ग इकाई (C) 3 वर्ग इकाई (D) 4 वर्ग इकाई

Ans. B

Sol. $y = \sin 2x + \cos 2x$ $x = 0$ & $\frac{\pi}{4}$

$$A = \int_0^{\frac{\pi}{4}} (\sin 2x + \cos 2x)dx$$

$$= -\frac{\cos 2x}{2} + \sin \frac{2x}{2}\Big|_0^{\frac{\pi}{4}}$$

$$= \frac{1}{2} + \frac{1}{2} = 1$$

3. The area bounded between the curve $|y| = 1 - x^2$ is :

(A) 2/3 (B) 4/3 (C) 8/3 (D) None of these

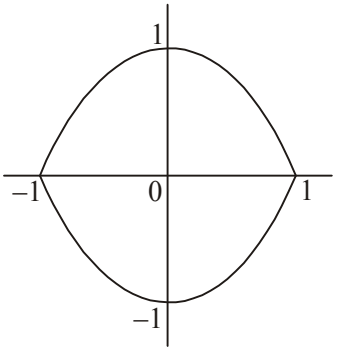
वक्र $|y| = 1 - x^2$ के द्वारा परिबद्ध क्षेत्रफल है :

(A) 2/3 (B) 4/3 (C) 8/3 (D) इनमें से कोई नहीं

Ans. C

Sol. $|y| = 1 - x^2$

$y = 1 - x^2 \Rightarrow x^2 = -(y - 1)$



$|y| = 1 - x^2$

$$A = 2\int_{-1}^{1} y\,dx$$

$$= 2\int_{-1}^{1} \left(1 - x^2\right) dx$$

$$= 2\left(x - \frac{x^3}{3}\right)_{-1}^{1} = \left(2 - \frac{1}{3} - \left(-1 + \frac{1}{3}\right)\right) = \frac{8}{3}$$

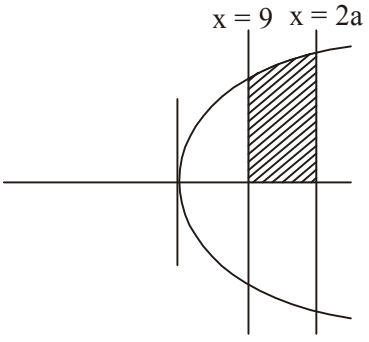
4. The area bounded by the parabola $y^2 = 4ax$, x-axis and $x = 2a$ and latus rectum is :

परवलय $y^2 = 4ax$, x-अक्ष तथा $x = 2a$ तथा नाभि लम्ब के मध्य क्षेत्रफल है :

(A) $2a^2\left(\sqrt{2} - 1\right)$ (B) $4a^2\left(2\sqrt{2} - 1\right)$ (C) $\frac{4}{3}a^2\left(2\sqrt{2} - 1\right)$ (D) $\frac{8a^2}{3}\left(2\sqrt{2} - 1\right)$

Ans. C

Sol. $y^2 = 4ax$ x-axis $x = 2a$ & latus rectum



$$y = \int_{a}^{2a} \sqrt{4ax}\,dx$$

$$= 2\sqrt{a}\left[\frac{2}{3}x^{\frac{3}{2}}\right]_{a}^{2a} 4$$

$$= 2\sqrt{a}\left[\frac{2}{3}\cdot 2\sqrt{2}a^{\frac{3}{2}} - \frac{2}{3}a^{\frac{3}{2}}\right]$$

$$= \frac{4}{3}\left[2\sqrt{2}a^2 - a^2\right]$$

$$= \frac{4}{3}a^2\left[2\sqrt{2}-1\right]$$

5. The area bounded between parabola $x^2 = 4y$ and $y = |x|$ is :

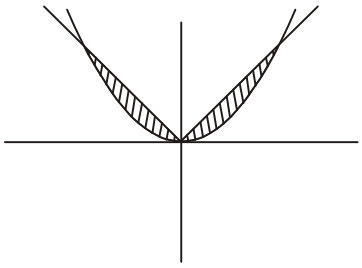
परवलय $x^2 = 4y$ तथा $y = |x|$ के मध्य क्षेत्रफल है :

(A) 2/3 (B) 4/3 (C) 8/3 (D) 16/3

Ans. D

Sol. $x^2 = 4y$

$$\text{Area} = 2\int_0^4 (y_1 - y_2)dx$$



$$= 2\int_0^4 \left(x - \frac{x^2}{4}\right)dx$$

$$= 2\left(\frac{x^2}{2} - \frac{x^3}{12}\right)\Bigg|_0^4$$

$$= 2\left[\left(8 - \frac{16}{3}\right) - 0\right]$$

$$= 2 \times \frac{8}{3}$$

$$= \frac{16}{3}$$

6. The area of the figure bounded by the parabola $y = x^2 + 1$ and the straight line $x + y = 3$ is :

$y = x^2 + 1$ तथा सरल रेखा $x + y = 3$ के मध्य क्षेत्रफल है-

(A) 3/2 (B) 5/2 (C) 7/2 (D) 9/2

Ans. D

Sol. $y = x^2 + 1$ $\qquad$ $x + y = 3$

Point of Intersection $\qquad$ $3 - x = x^2 + 1$

$x^2 + x - 2 = 0$

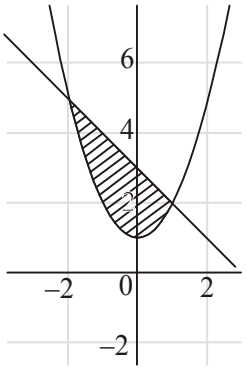
$(x + 2)(x - 1) = 0$

$x = -2, 1$

$$A = \int_{-2}^{1} (y_1 - y_2)\,dx$$

$$= \int_{-2}^{1} \left((3-x)-\left(x^2+1\right)\right)dx$$

$$= \int_{-2}^{1} \left(2-x-x^2\right)dx$$



$$= 2x - \frac{x^2}{2} - \frac{x^3}{3}\Big|_{-2}^{1}$$

$$A = \left(2-\frac{1}{2}-\frac{1}{3}\right)-\left(-4-\frac{4}{2}+\frac{8}{3}\right)$$

$$= \frac{12-3-2}{6} - \frac{-24-12+16}{6}$$

$$= \frac{7}{6}+\frac{20}{6}$$

$$= \frac{27}{6}$$

$$= \frac{9}{2}$$

7. Common area between the parabolas $y = 2x^2$ and $y = x^2 + 4$ is :

(A) 16/3 (B) 8/3 (C) 32/3 (D) None of these

परवलयों $y = 2x^2$ तथा $y = x^2 + 4$ का उभयनिष्ठ क्षेत्रफल है :

(A) 16/3 (B) 8/3 (C) 32/3 (D) इनमें से कोई नहीं

Ans. C

Sol. $y = 2x^2$ $y = x^2 + 4$

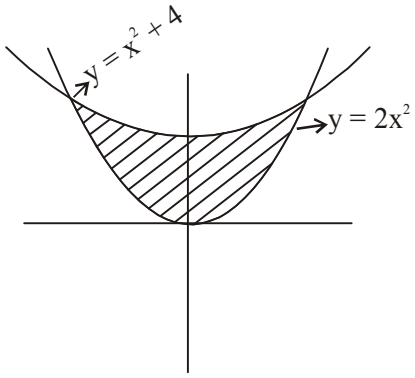
Point of intersection

$2x^2 = x^2 + 4$

$x^2 = 4$

$x = \pm 2$

$$\text{Area} = 2\int_0^2 (y_2 - y_1)\,dx$$



$$A = 2\int_0^2 \left(\left(x^2 + 4\right) - \left(2x^2\right)\right)dx$$

$$= 2\int_0^2 \left(4 - x^2\right)dx$$

$$= 2\left[4x - \frac{x^3}{3}\right]_0^2$$

$$= 2\left(8 - \frac{8}{3}\right)$$

$$= \frac{32}{3}$$

8. Area bounded by the curves $y = 2^x$, $y = 2x - x^2$, $x = 0$ and $x = 2$ is :

वक्रों $y = 2^x$, $y = 2x - x^2$, $x = 0$ तथा $x = 2$ के मध्य क्षेत्रफल है :

(A) $\frac{3}{\log 2} - \frac{4}{3}$ (B) $\frac{3}{\log 2} + \frac{4}{3}$ (C) $3\log 2 + \frac{4}{3}$ (D) $3\log 2 - \frac{4}{3}$

Ans. A

Sol. $y = 2x$ $y = 2x - x^2$ $x = 0$ $x = 2$

$y = 2^x$

$y = 2x - x^2$

$-y = x^2 - 2x$

$-(y-1) = (x-1)^2$

$$\text{Area} = \int_0^2 (y_2 - y_1)\,dx$$

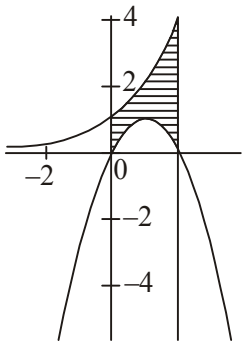
$$= \int_0^2 \left(2^x - \left(2x - x^2\right)\right) dx$$

$$= \int_0^2 \left(2^x - 2x + x^2\right) dx$$

$$= \frac{2^x}{\ln 2} - x^2 + \frac{x^3}{3}\Big|_0^2$$

$$= \frac{4}{\ln 2} - 4 + \frac{8}{3} - \frac{1}{\ln 2}$$

$$= \frac{3}{\ln 2} - \frac{4}{3}$$



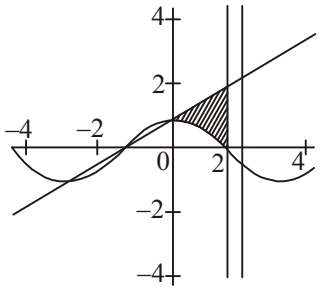
9. The area of the closed figure bounded by $y = \cos x$, $y = 1 + (2/\pi)\, x$ and $x = \pi/2$ is :

$y = \cos x$, $y = 1 + (2/\pi)\, x$ तथा $x = \pi/2$ के मध्य क्षेत्रफल है :

(A) $\frac{3\pi}{4}$ (B) $\frac{\pi}{2}$ (C) $\frac{3\pi - 4}{4}$ (D) $\frac{\pi - 4}{4}$

Ans. C

Sol. $y = \cos x$ $y = 1 + \frac{2}{\pi} x$ $x = \frac{\pi}{2}$



$$A = \int_0^{\frac{\pi}{2}} \left(y_2 - y_1\right) dx$$

$$= \int_0^{\frac{\pi}{2}} \left(\left(1 + \frac{2}{\pi} x\right) - \cos x\right) dx$$

$$= x + \frac{x^2}{\pi} - \sin x\Big|_0^{\frac{\pi}{2}}$$

$$= \frac{\pi}{2} + \frac{\pi}{4} - 1$$

$= \frac{3\pi}{4} - 1$

10. The area enclosed between the curves $y = \log_e (x + e)$, $x = \log_e \left(\frac{1}{y}\right)$ and the x-axis is :

वक्रों $y = \log_e (x + e)$, $x = \log_e \left(\frac{1}{y}\right)$ तथा x-अक्ष द्वारा बने क्षेत्र का क्षेत्रफल होगा :

(A) 1 (B) 2 (C) 3 (D) 4

Ans. B

Sol. $y = \log_e (x + e)$ $x = \log_e \left(\frac{1}{y}\right)$ x-axis

$x = \log_e \left(\frac{1}{y}\right)$

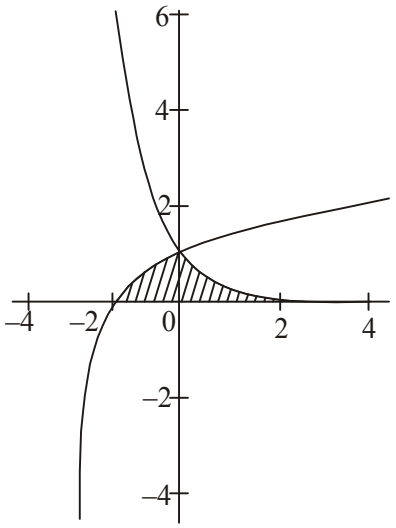
$A = \int_{1-e}^{0} \log_e (x + e)\,dx - \int_{0}^{\infty} e^{-x}\,dx$

$= (x + e)\log_e \left(\frac{x+e}{e}\right)\Big|_{1-e}^{0} - -e^{-x}\Big|_{0}^{\infty}$

$= \left(-\log_e \frac{1}{e}\right) - (-1)$

$= 1 + 1$

$= 2$



11. Area of the circle $(x - 2)^2 + (y - 3)^2 = 32$ which lies below the line $y = x + 1$ is :

(A) $\int_{-2}^{6} \left[(x+1) + \sqrt{32 - (x-2)^2} + 3\right] dx$ (B) $\int_{-2}^{6} \left[\left\{\sqrt{32 - (x-2)^2} + 3\right\} - (x+1)\right] dx$

(C) 16π (D) None of these

वृत्त $(x - 2)^2 + (y - 3)^2 = 32$ जो कि रेखा $y = x + 1$ के नीचे स्थित है, का क्षेत्रफल है :

(A) $\int_{-2}^{6} \left[(x+1) + \sqrt{32 - (x-2)^2} + 3\right] dx$ (B) $\int_{-2}^{6} \left[\left\{\sqrt{32 - (x-2)^2} + 3\right\} - (x+1)\right] dx$

(C) 16π (D) इनमें से कोई नहीं

Ans. C

Sol. $(x-2)^2+(y-3)^2=32$ $\quad y=x+1$

Point of intersection

$(x-2)^2+(x-2)^2=32$

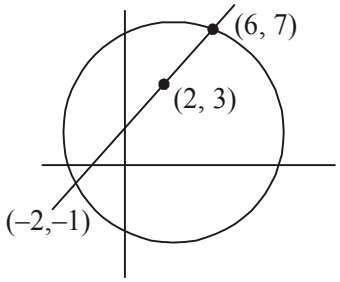
$(x-2)^2=16$

$x-2=\pm4$

$x=6,-2$

$x=6 \quad y=7 \quad (6,7)$

$x=-2 \quad y=-1 \quad (-2,-1)$



center is on the line, so line is diameter

$$A=\frac{\pi r^2}{2}=\frac{\pi\times32}{2}=16\pi$$

12. If $y=2\sin x+\sin 2x$ for $0\le x\le 2\pi$, then the area enclosed by the curve and the x-axis is :
(A) 6 (B) 8 (C) 12 (D) none

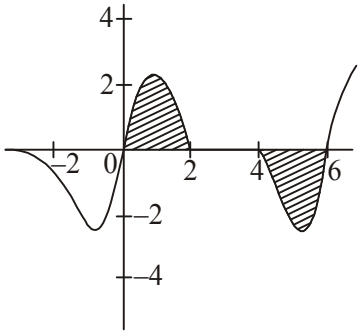
यदि $y=2\sin x+\sin 2x$, $0\le x\le 2\pi$ के लिये, तो वक्र तथा x-अक्ष द्वारा परिबद्ध क्षेत्रफल होगा :
(A) 6 (B) 8 (C) 12 (D) कोई नहीं

Ans. B

Sol. $y=2\sin x+\sin 2x$ $\quad$ x-axis $\quad x\in[0,2\pi]$

$y=2\sin x(1+\cos x)$

$$=2\sin x\cdot 2\cos^2\frac{x}{2}$$



$$A=\int_0^{\pi}(2\sin x+\sin 2x)\,dx+(-)\int_{\pi}^{2\pi}2\sin x+\sin 2x$$

$$=-2\cos x-\frac{\cos 2x}{2}\Big|_0^{\pi}-\left[-2\cos x-\frac{\cos 2x}{2}\right)_{\pi}^{2\pi}$$

$$=+2-\frac{1}{2}+2+\frac{1}{2}-\left[-2-\frac{1}{2}-\left(2-\frac{1}{2}\right)\right)$$

$=8$

13. The area of the closed figure bounded by $x = -1$, $x = 2$ and $y = \begin{cases} -x^2 + 2 \ , & x \le 1 \\ 2x - 1 \ , & x > 1 \end{cases}$ and the abscissa axis, is :

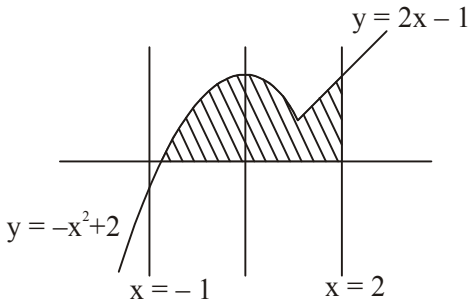
वक्र $x = -1$, $x = 2$, $y = \begin{cases} -x^2 + 2 \ , & x \le 1 \\ 2x - 1 \ , & x > 1 \end{cases}$ तथा x-अक्ष द्वारा परिबद्ध क्षेत्रफल होगा :

(A) $\frac{7}{3}$ (B) $\frac{10}{3}$ (C) $\frac{13}{3}$ (D) $\frac{16}{3}$

Ans. D

Sol. $y = \begin{cases} -x^2 + 2 & x \le 1 \\ 2x - 1 & x > 1 \end{cases}$, $x = -1, x = 2$

$$A = \int_{-1}^{1} \left(-x^2 + 2\right) dx + \int_{1}^{2} (2x - 1)\, dx$$



$$= 2\int_{0}^{-1} \left(-x^2 + 2\right) dx + x^2 - x\Big|_{1}^{2}$$

$$= 2\left(\frac{-x^3}{3} + 2x \right)_{0}^{1} + \left((4 - 2) - (1 - 1)\right)$$

$$= 2\left(-\frac{1}{3} + 2 \right) + 2$$

$$= \frac{10}{3} + 2$$

$$= \frac{16}{3}$$

14. Area between curve $y = \frac{x}{1 + x^2}$ and x axis from $x = -1$ to $x = 1$ is :

वक्र $y = \frac{x}{1 + x^2}$ तथा x- अक्ष, $x = -1$ से $x = 1$ तक परिबद्ध क्षेत्रफल होगा :

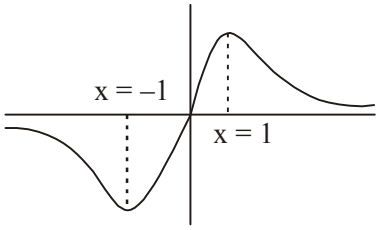
(A) $\ln 2$ (B) $\ln 4$ (C) $\pi/2$ (D) π

Ans. A

Sol. $y = \frac{x}{1+x^2}$ x-axis $x = -1, 1$

$$\frac{dy}{dx} = \frac{1+x^2-2x^2}{\left(1+x^2\right)^2}$$

$$= \frac{1-x^2}{\left(1+x^2\right)^2}$$



$$A = 2\int_0^1 y\,dx$$

$$= 2\int_0^1 \frac{x}{1+x^2}\,dx$$

$$= \log\left(1+x^2\right)\Big|_0^1$$

$$= \log 2$$

15. The area of the figure bounded by the curves $y = e^x$, $y = e^{-x}$ and the straight line $x = 1$ is :

(A) $e + \frac{1}{e}$ (B) $e - \frac{1}{e}$ (C) $e + \frac{1}{e} - 2$ (D) none

वक्र $y = e^x$, $y = e^{-x}$ तथा सरल रेखा $x = 1$ द्वारा परिबद्ध क्षेत्रफल होगा :

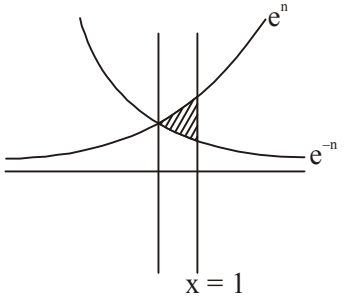
(A) $e + \frac{1}{e}$ (B) $e - \frac{1}{e}$ (C) $e + \frac{1}{e} - 2$ (D) कोई नहीं

Ans. C

Sol. $y = e^x$ $y = e^{-x}$ $x = 1$

$$A = \int_0^1 e^x dx - \int_0^1 e^{-x} dx$$

$$= \left(e^x\right)_0^1 - \left(-e^{-x}\right)_0^1$$



$$= (e-1) - \left(-\frac{1}{e}+1\right)$$

$$= e + \frac{1}{e} - 2$$

16. The area bounded by $y = x^2$ and $y = 2 - x^2$ is :

वक्र $y = x^2$ तथा $y = 2 - x^2$ द्वारा परिबद्ध क्षेत्रफल होगा :

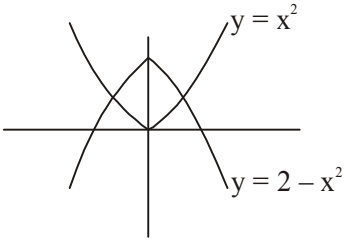
(A) $\frac{8}{3}$ (B) $\frac{16}{3}$ (C) $\frac{32}{3}$ (D) $\frac{64}{3}$

Ans. A

Sol. $y = x^2$ $y = 2 - x^2$

Point of intersection $x = \pm 1$

$$A = 2\int_0^1 (y_2 - y_1)\,dx$$



$$= 2\int_0^1 \left(2 - x^2 - x^2\right) dx$$

$$= 2\left(2x - \frac{2x^3}{3}\right)_0^1$$

$$= 2\left(2 - \frac{2}{3}\right)$$

$$= \frac{8}{3}$$

17. The area bounded by the curves $y^2 = 3x$ & $y^2 = 36 - 9x$ is :

(A) 16 (B) 18 (C) 21.5 (D) none

वक्र $y^2 = 3x$ तथा $y^2 = 36 - 9x$ द्वारा परिबद्ध क्षेत्रफल होगा :

(A) 16 (B) 18 (C) 21.5 (D) कोई नहीं

Ans. A

Sol. $y^2 = 3x$ $y^2 = 36 - 9x$

$= -9(x-4)$

Point of intersection

$3x = 36 - 9x$

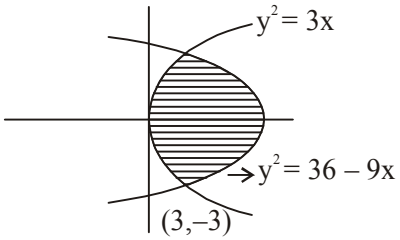
$12x = 36$

$x = 3$

$y^2 = 9$

$y = \pm 3$

$(3, 3) \qquad (3, -3)$



$$A = 2\int_0^3 (x_2 - x_1)\,dy$$

$$= 2\int_0^3 \left(\frac{36 - y^2}{9}\right) - \left(\frac{y^2}{3}\right) dy$$

$$= 2\left[\left.\frac{36y - \frac{y^3}{3}}{9}\right|_0^3 - \left.\frac{y^3}{9}\right|_0^3\right]$$

$$= 2\left[\frac{108 - 9}{9} - 3\right]$$

$$= 2\left[\frac{99}{9} - 3\right]$$

$= 2\,[11 - 3]$

$= 16$

18. The ratio in which the area enclosed by the curve $y = \cos x \left(0 \le x \le \frac{\pi}{2}\right)$ in the first quadrant is divided by the curve $y = \sin x$, is :

(A) $\left(\sqrt{2} - 1\right) : 1$ (B) $\left(\sqrt{2} + 1\right) : 1$ (C) $\sqrt{2} : 1$ (D) none

प्रथम चतुर्थांश में वक्र $y = \cos x\left(0 \le x \le \frac{\pi}{2}\right)$ द्वारा परिबद्ध क्षेत्रफल को वक्र $y = \sin x$ जिस अनुपात में विभाजित करता है, वह

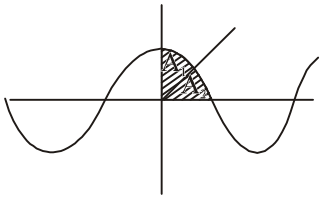
vuqikr gksxk:

(A) $\left(\sqrt{2}-1\right):1$ (B) $\left(\sqrt{2}+1\right):1$ (C) $\sqrt{2}:1$ (D) कोई नही

Ans. C

Sol. $y = \cos x$ $y = \sin x$

$$\frac{A_1}{A_2} = ?$$



$$A_1 = \int_0^{\frac{\pi}{4}} (\cos x - \sin x)$$

$$= \sin x + \cos x\Big|_0^{\frac{\pi}{4}}$$

$$= \sqrt{2} - 1$$

$$A_2 = \int_0^{\frac{\pi}{4}} \sin x dx + \int_0^{\frac{\pi}{2}} \cos x dx$$

$$= -\cos x\Big|_0^{\frac{\pi}{4}} + \sin x\Big|_{\frac{\pi}{4}}^{\frac{\pi}{2}}$$

$$= -\frac{1}{\sqrt{2}} + 1 + 1 - \frac{1}{\sqrt{2}}$$

$$= 2 - \sqrt{2}$$

$$= \sqrt{2}\left(\sqrt{2}-1\right)$$

$$\frac{A_2}{A_1} = \frac{\sqrt{2}\left(\sqrt{2}-1\right)}{\sqrt{2}-1} = \frac{\sqrt{2}}{1}$$

19. The area bounded by the curve $y^2 + x^4 = x^2$ is :

(A) $\frac{2}{3}$ (B) $\frac{4}{3}$ (C) $\frac{8}{3}$ (D) none

वक्र $y^2 + x^4 = x^2$ से घिरा हुआ क्षेत्रफल होगा :

(A) $\frac{2}{3}$ (B) $\frac{4}{3}$ (C) $\frac{8}{3}$ (D) कोई नहीं

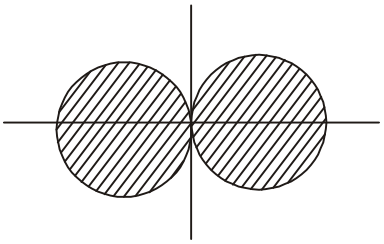
Ans. B

Sol. $y^2 + x^4 = x^2$

$y^2 = x^2 - x^4$

$y^2 = x^2 (1 - x^2)$

$y = \pm x\sqrt{1 - x^2}$



Symmetric about x-axis and y-axis both

$$A = 4\int_0^1 y\,dx$$

$$= 4\int_0^1 x\sqrt{1 - x^2}\,dx$$

$$= 2\left(-\frac{2}{3}\left(1 - x^2\right)^{\frac{3}{2}} \right)_0^1$$

$$= 2\left(\frac{2}{3}\right) = \frac{4}{3}$$

20. Area bounded by the curves $y = |x| - 2$ and $y = 1 - |x-1|$ is equal to :

(A) 4 sq. units (B) 6 sq. units (C) 2 sq. units (D) 8 sq. units

वक्रों $y = |x| - 2$ तथा $y = 1 - |x-1|$ के मध्य घिरा हुआ क्षेत्रफल होगा :

(A) 4 वर्ग ईकाई (B) 6 वर्ग ईकाई (C) 2 वर्ग ईकाई (D) 8 वर्ग ईकाई

Ans. A

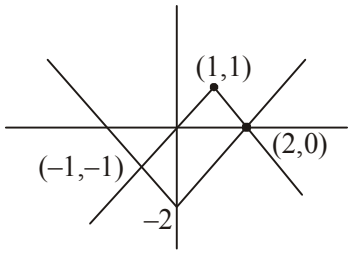
Sol. $y = |x| - 2$ $y = 1 - |x - 1|$

I $y = |x| - 2$

$$y = \begin{cases} x-2 & x \geq 0 \\ -x-2 & x < 0 \end{cases}$$

II $y = 1 - |x-1|$

$$y = \begin{cases} 2-x & x \geq 1 \\ x & x < 1 \end{cases}$$



$A = l \times b$

$= 2\sqrt{2} \times \sqrt{2} = 4$

21. Area bounded by the parabola $y = x^2 - 2x + 3$ and tangents drawn to it from the point P(1, 0) is equal to :

(A) $4\sqrt{2}$ sq. units (B) $\frac{4\sqrt{2}}{3}$ sq. units (C) $\frac{8\sqrt{2}}{3}$ sq. units (D) $\frac{16}{3}\sqrt{2}$ sq. units

परवलय $y = x^2 - 2x + 3$ तथा बिन्दु P(1, 0) से इस पर खींची गई स्पर्श रेखा के मध्य का क्षेत्रफल होगा :

(A) $4\sqrt{2}$ वर्ग ईकाई (B) $\frac{4\sqrt{2}}{3}$ वर्ग ईकाई (C) $\frac{8\sqrt{2}}{3}$ वर्ग ईकाई (D) $\frac{16}{3}\sqrt{2}$ वर्ग ईकाई

Ans. B

Sol. $y = x^2 - 2x + 3$ tangent from P (1, 0)

$y - 2 = (x-1)^2$

Tangent

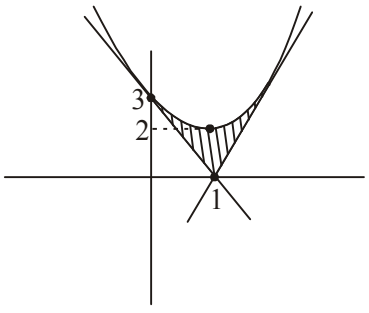
$$x = \frac{y-2}{m} + \frac{m}{4}$$

(1, 0)

$$= \frac{-2}{m} + \frac{m}{4}$$

$0 = -8 + m^2$

$m = \pm 2\sqrt{2}$



Tangents

$y = 2\sqrt{2}(x-1)$ & $y = -2\sqrt{2}(x-1)$

Point of intersection of $y - 2 = (x-1)^2$ & $y = \pm 2\sqrt{2}(x-1)$

$$y-2=\frac{y^2}{8}\Rightarrow y^2-8m+16=0$$

$$y=4$$

$$A=2\int_1^{1+\sqrt{2}}(y_1-y_2)dx$$

$$=2\int_1^{1+\sqrt{2}}\left[2+(x-1)^2-2\sqrt{2(x-1)}\right]dx$$

$$x-1=t$$

$$=2\int_0^{\sqrt{2}}\left(2+t^2-2\sqrt{2}t\right)dt$$

$$=2\left(2t+\frac{t^3}{3}-\sqrt{2}t^2\right)_0^{\sqrt{2}}$$

$$=2\left(2\sqrt{2}+\frac{2\sqrt{2}}{3}-2\sqrt{2}\right)$$

$$=\frac{4\sqrt{2}}{3}$$

22. Area bounded by the curves $y=\sin x$, tangent drawn to it at $x=0$ and the line $x=\frac{\pi}{2}$, is equal to :

(A) $\frac{\pi^2-4}{2}$ sq. units (B) $\frac{\pi^2-8}{8}$ sq. units (C) $\frac{\pi^2-2}{4}$ sq. units (D) $\frac{\pi^2-2}{2}$ sq. units

वक्रों $y=\sin x$, तथा इस पर $x=0$ पर स्पर्श रेखा तथा रेखा $x=\frac{\pi}{2}$ के मध्य का क्षेत्रफल होगा :

(A) $\frac{\pi^2-4}{2}$ वर्ग ईकाई (B) $\frac{\pi^2-8}{8}$ वर्ग ईकाई (C) $\frac{\pi^2-2}{4}$ वर्ग ईकाई (D) $\frac{\pi^2-2}{2}$ वर्ग ईकाई

Ans. B

Sol. $y=\sin x$ Tangent at $x=0$ & line $x=\frac{\pi}{2}$

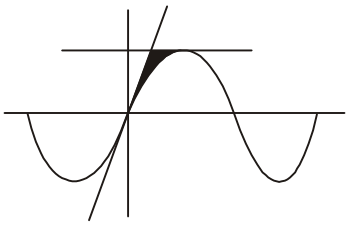
$$\frac{dy}{dx} = \cos x$$

$$m_T\left(x = 0\right) = 1$$

at $x = 0$ $\quad y = 0$

Tangent $y = mx$

$y = x$



$$A = \int_0^{\frac{\pi}{2}} \left(-\sin x + x\right) dx$$

$$= \left(+\cos x + \frac{x^2}{2} \right)\Bigg|_0^{\frac{\pi}{2}}$$

$$= \left(+\cos\frac{\pi}{2} + \frac{\pi^2}{8} \right) - (1)$$

$$= +\frac{\pi^2}{8} - 1 = \frac{\pi^2 - 8}{8}$$

23. Value of the parameter a such that the area bounded by $y = a^2x^2 + ax + 1$, coordinate axes and the line $x = 1$, attains it's least value, is equal to :

प्राचल 'a' के किस मान के लिए वक्र $y = a^2x^2 + ax + 1$, निर्देशांक अक्षों को $x = 1$ से घिरे हुये क्षेत्रफल का न्यूनतम मान होगा :

(A) $-\frac{1}{4}$ (B) $-\frac{1}{2}$ (C) $-\frac{3}{4}$ (D) -1

Ans. C

Sol. $y = a^2x^2 + ax + 1$ x-axis & y-axis line $x = 1$

$D = a^2 - 4(a^2)$

$= -3a^2 < 0$

$$A = \int_0^1 y\,dx$$

$$A = \int_0^1 \left(a^2x^2 + ax + 1\right)dx$$

$$A = \frac{a^2x^3}{3} + \frac{ax^2}{2} + x\Big|_0^1$$

$$A = \frac{a^2}{3} + \frac{a}{2} + 1$$

$$\frac{dA}{da} = \frac{2a}{3} + \frac{1}{2} = 0$$

$$a = -\frac{3}{4}$$

$$A = \frac{9}{16\times 3} - \frac{3}{8} + 1$$

$$= \frac{3}{16} - \frac{6}{16} + \frac{16}{16}$$

$$= \frac{13}{16}$$

24. Area enclosed by the curve $y = f(x)$ defined parametrically as $x = \frac{1-t^2}{1+t^2}$, $y = \frac{2t}{1+t^2}$ is equal to :

(A) π sq. units (B) $\frac{\pi}{2}$ sq. units (C) $\frac{3\pi}{4}$ sq. units (D) $\frac{3\pi}{2}$ sq. units

यदि वक्र $y = f(x)$ प्राचलिक रूप में $x = \frac{1-t^2}{1+t^2}$, $y = \frac{2t}{1+t^2}$ प्रकार से परिभाषित है, तो इस से घिरा हुआ क्षेत्रफल होगा :

(A) π वर्ग इकाई (B) $\frac{\pi}{2}$ वर्ग ईकाई (C) $\frac{3\pi}{4}$ वर्ग ईकाई (D) $\frac{3\pi}{2}$ वर्ग ईकाई

Ans. A

Sol. $y = f(x)$

$$x = \frac{1-t^2}{1+t^2} \quad \& \quad y = \frac{2t}{1+t^2}$$

Let $t = \tan\theta$

$$x = \frac{1-\tan^2\theta}{1+\tan^2\theta}$$

$$x = \cos 2\theta$$

and $$y = \frac{2\tan\theta}{1+\tan^2\theta}$$

$$y = \sin 2\theta$$

$x^2 + y^2 = 1$

Area = πr^2

$= \pi \times 1$

$= \pi$

25. The area of the region bounded by the curves $y = |x - 1|$ and $y = 3 - |x|$ is:

(A) 6 sq. units (B) 2 sq. units (C) 3 sq. units (D) 4 sq. units

वक्र $y = |x - 1|$ तथा $y = 3 - |x|$ के द्वारा परिबद्ध क्षेत्रफल है :

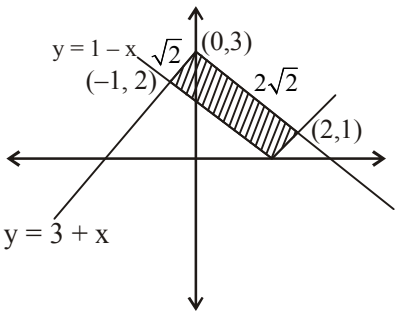
(A) 6 वर्ग इकाई (B) 2 वर्ग इकाई (C) 3 वर्ग इकाई (D) 4 वर्ग इकाई

Ans. D

Sol. $y = |x - 1|$ $y = 3 - |x|$

$$y = \begin{cases} x-1 & x \geq 1 \\ 1-x & x < 1 \end{cases}$$

$$y = \begin{cases} 3-x & x > 0 \\ 3+x & x \leq 0 \end{cases}$$



Area = $\sqrt{2} \times 2\sqrt{2}$

$= 4$

26. The area enclosed between the curve $y = \log_e(x + e)$ and the coordinate axes is :

उस क्षेत्रफल, जो कि वक्र $y = \log_e(x + e)$ और अक्षों के बीच में बना है, उसका मान होगा :

(A) 1 (B) 2 (C) 3 (D) 4

Ans. A

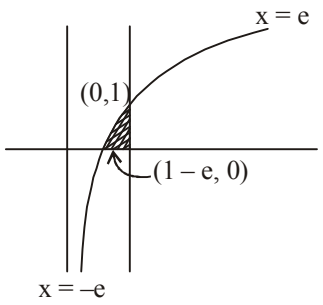
Sol. $y = \log_e (x + e)$

$$A = \int_{1-e}^{0} \log_e (x+e)\,dx$$

$$= (x+e)\log_e \left(\frac{x+e}{e}\right)_{1-e}^{0}$$

$= (1 - e + e)\,(1)$

$= 1$



27. The parabolas $y^2 = 4x$ and $x^2 = 4y$ divide the square region bounded by the lines $x = 4$, $y = 4$ and the coordinate axes. If S_1, S_2, S_3 are respectively the areas of these parts numbered from top to bottom; then $S_1 : S_2 : S_3$ is :

परवलय $y^2 = 4x$ तथा $x^2 = 4y$, रेखाओं $x = 4$, $y = 4$ तथा निर्देशांक अक्षों से घिरे वर्गाकार क्षेत्र को बांटते हैं। यदि ऊपर से नीचे तक r hu [k Mad sØe' k/kksa esa Øe'k% Qy S_1, S_2 तथा S_3 हैं, तो $S_1 : S_2 : S_3$ का मान है :

(A) 1 : 2 : 1 (B) 1 : 2 : 3 (C) 2 : 1 : 2 (D) 1 : 1 : 1

Ans. D

Sol. $y^2 = 4x$ $x^2 = 4y$

$$S_1 = \int_0^4 x\,dy$$

$$= \int_0^4 \frac{y^2}{4}\,dy$$

$$= \frac{y^3}{3x^4}\Big|_0^4$$

$$= \frac{64}{3\times 4} = \frac{16}{3}$$

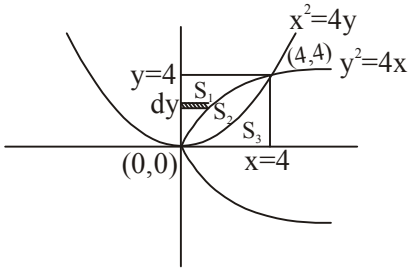
from symmetry

$$S_2 = \frac{16}{3}$$

$S_2 = 16 - S_1 - S_3$

$$= 16 - \frac{16}{3} - \frac{16}{3}$$

$$= 16 - \frac{32}{3}$$



$= \frac{16}{3}$

$S_1 : S_2 : S_3 = 1 : 1 : 1$

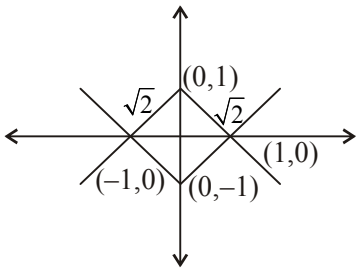
28. The area bounded by the curves $y = |x| - 1$ and $y = -|x| + 1$ is :

वक्रों $y = |x| - 1$ और $y = -|x| + 1$ से परिबद्ध क्षेत्रफल है :

(A) 1 (B) 2 (C) $2\sqrt{2}$ (D) 4

Ans. B

Sol. $y = |x| - 1$ $y = -|x| + 1$



$$y = \begin{cases} x - 1 & x > 0 \\ x - 1 & x \le 0 \end{cases}$$

$A = \sqrt{2} \times \sqrt{2} = 2$

29. If the area bounded by $y = ax^2$ and $x = ay^2$, $a > 0$, is 1, then a =

यदि $y = ax^2$ और $x = ay^2$, $a > 0$ से परिबद्ध क्षेत्रफल 1 है, तब a =

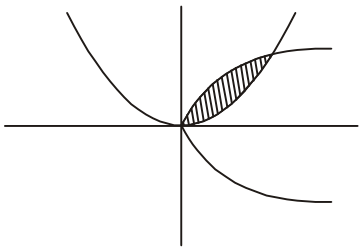
(A) 1 (B) $\frac{1}{\sqrt{3}}$ (C) $\frac{1}{3}$ (D) $-\frac{1}{\sqrt{3}}$

Ans. B

Sol. $y = ax^2$ $x = ay^2$

$x^2 = \frac{y}{a}$ $y^2 = \frac{x}{a}$

Enclosed area $= \frac{16}{3}\left(\frac{1}{4a} \times \frac{1}{4a}\right)$



$A = \frac{1}{3a^2}$

$1 = \frac{1}{3a^2}$

$a = \pm\frac{1}{\sqrt{3}}$

$\because a > 0$ $a = \frac{1}{\sqrt{3}}$

30. The area bounded by the parabolas $y = (x + 1)^2$ and $y = (x - 1)^2$ and the line $y = 1/4$ is :

(A) 4 sq. units (B) $\frac{1}{6}$ sq. units (C) $\frac{4}{3}$ sq. units (D) $\frac{1}{3}$ sq. units

परवलयों $y = (x + 1)^2$ और $y = (x - 1)^2$ और रेखा $y = 1/4$ से परिबद्ध क्षेत्रफल है :

(A) 4 वर्ग इकाई (B) $\frac{1}{6}$ वर्ग इकाई (C) $\frac{4}{3}$ वर्ग इकाई (D) $\frac{1}{3}$ वर्ग इकाई

Ans. D

Sol. $y = (x + 1)^2$

$y = (x - 1)^2$

$$y = \frac{1}{4}$$

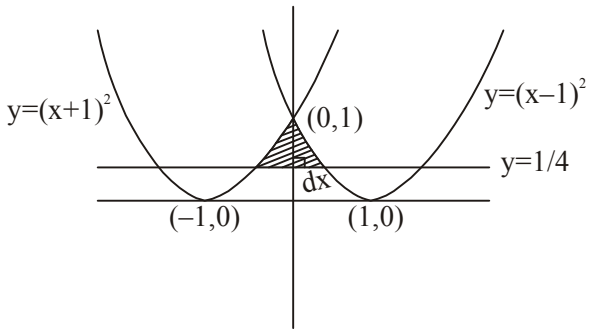
$$A = \int_{\frac{1}{4}}^{1} \left(-\sqrt{y} + 1\right) dy$$

$$\because y = (x-1)^2$$

$$x = \begin{cases} -\sqrt{y} + 1 & \text{for} \quad x < 1 \\ \sqrt{y} + 1 & \text{for} \quad x \geq 1 \end{cases}$$

$$= 2\int_{\frac{1}{4}}^{1} \left(-\sqrt{y} + 1\right) dy$$

$$= 2\left(-\frac{2}{3} y^{\frac{3}{2}+y} \right)_{\frac{1}{4}}^{1}$$



$$= 2\left[\left(\frac{-2}{3} + 1 \right) - \left(-\frac{2}{3} \times \frac{1}{8} + \frac{1}{4} \right) \right]$$

$$= 2\left[\frac{1}{3} - \left(-\frac{1}{12} + \frac{1}{4} \right) \right]$$

$$= 2\left[\frac{1}{3} - \left(\frac{1}{6} \right) \right]$$

$$= 2\left(\frac{1}{6} \right) = \frac{1}{3}$$

31. The area of the region between the curves $y = \sqrt{\frac{1+\sin x}{\cos x}}$ and $y = \sqrt{\frac{1-\sin x}{\cos x}}$ bounded by the lines $x = 0$ and $x = \frac{\pi}{4}$ is :

वक्रों $y = \sqrt{\frac{1+\sin x}{\cos x}}$ और $y = \sqrt{\frac{1-\sin x}{\cos x}}$ तथा रेखाओं $x = 0$ और $x = \frac{\pi}{4}$ से परिबद्ध क्षेत्र का क्षेत्रफल है :

(A) $\int_0^{\sqrt{2}-1} \frac{t}{(1+t^2)\sqrt{1-t^2}}dt$ (B) $\int_0^{\sqrt{2}-1} \frac{4t}{(1+t^2)\sqrt{1-t^2}}dt$

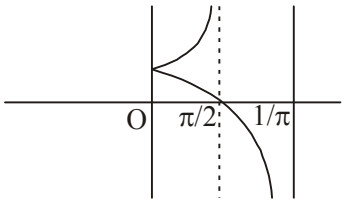
(C) $\int_0^{\sqrt{2}+1} \frac{4t}{(1+t^2)\sqrt{1-t^2}}dt$ (D) $\int_0^{\sqrt{2}+1} \frac{t}{(1+t^2)\sqrt{1-t^2}}dt$

Ans. B

Sol. Area $= \int_0^{\pi/4} \left(\sqrt{\frac{1+\sin x}{\cos x}} - \sqrt{\frac{1-\sin x}{\cos x}} \right) dx$

$$= \int_0^{\pi/4} \frac{\left(\cos\frac{x}{2}+\sin\frac{x}{2}\right)-\left(\cos\frac{x}{2}-\sin\frac{x}{2}\right)}{\sqrt{\cos^2\frac{x}{2}-\sin^2\frac{x}{2}}}dx$$

$$= \int_0^{\pi/4} \frac{2\sin\frac{x}{2}}{\sqrt{\cos^2\frac{x}{2}-\sin^2\frac{x}{2}}}dx = \int_0^{\pi/4} \frac{2\tan\frac{x}{2}}{\sqrt{1-\tan^2\frac{x}{2}}}dx$$



Let $\tan\frac{x}{2} = t$

$$\sec^2\frac{x}{2}dx = 2dt \Rightarrow dx = \frac{2dt}{(1+t^2)}$$

$\therefore$ Area $= \int_0^{\sqrt{2}-1} \frac{4t}{(1+t^2)\sqrt{1-t^2}}dt$

32. The area enclosed between the curves $y^2 = x$ and $y = |x|$ is :

वक्र $y^2 = x$ तथा $y = |x|$ द्वारा परिबद्ध क्षेत्रफल है :

(A) $\frac{2}{3}$ (B) 1 (C) $\frac{1}{6}$ (D) $\frac{1}{3}$

Ans. C

Sol. $y^2 = x$ $y = |x|$

$$A = \int_0^1 \left(y - y^2\right) dy$$

$$= \frac{y^2}{2} - \frac{y^3}{3}\Big|_0^1$$

$$= \frac{1}{2} - \frac{1}{3}$$

$$= \frac{1}{6}$$

33. The area of the plane region bounded by the curves $x + 2y^2 = 0$ and $x + 3y^2 = 1$ is equal to :

वक्रों $x + 2y^2 = 0$ तथा $x + 3y^2 = 1$ के बीच घिरे तलीय क्षेत्र का क्षेत्रफल है :

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $\frac{4}{3}$ (D) $\frac{5}{3}$

Ans. C

Sol. $y^2 = -\frac{x}{2}$ $y^2 = -\frac{1}{3}(x-1)$

$x = -2y^2$ $x = 1 - 3y^2$

POI

$$y^2 = -\frac{x}{2} = -\frac{1}{3}(x-1)$$

$3x = 2x - 2$

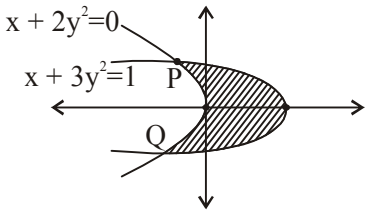
$x = -2$

$y^2 = 1 \Rightarrow y = \pm 1$

P (–2,1)

Q (–2,–1)

$$I = \int_{y_1}^{y_2} (x_2 - x_1)\,dy$$



$$= 2\int_0^1 \left((1-3y^2)-(-2y^2)\right)dy$$

$$= 2\int_0^1 (1-y^2)\,dy$$

$$= 2\left(y - \frac{y^3}{3}\right)_0^1$$

$$= \frac{4}{3}$$

34. The area of the region bounded by the parabola $(y-2)^2 = x-1$, the tangent to the parabola at the point (2, 3) and the x – axis is : **[AIEEE2009]**

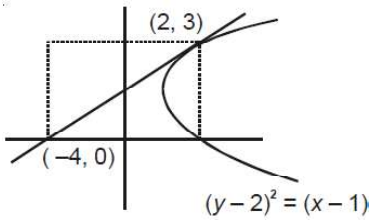
परवलय $(y-2)^2 = x-1$, परवलय के बिन्दु (2, 3) पर खींची गई स्पर्श रेखा तथा x – अक्ष द्वारा परिबद्ध क्षेत्र का क्षेत्रफल है : **[AIEEE2009]**

(A) 3 (B) 6 (C) 9 (D) 12

Ans. C

Sol. The equation of tangent at (2, 3) to the given parabola is $x = 2y - 4$

Required area $= \int_0^3 \left\{(y-2)^2 + 1 - 2y + 4\right\}dy$



$$= \left[\frac{(y-2)^3}{3} - y^2 + 5y\right]_0^3$$

$$= \frac{1}{3} - 9 + 15 + \frac{8}{3}$$

= 9 sq. units

35. The area (in square units) bounded by the curves $y = \sqrt{x}$, $2y - x + 3 = 0$ x-axis and lying in the first quadrant is: **[JEE(Main)2013]**

वक्रों $y = \sqrt{x}$, $2y - x + 3 = 0$ तथा x- अक्ष से घिरे उस क्षेत्र, जो प्रथम चतुर्थांश में स्थित है, का (वर्ग इकाई में) क्षेत्रफल है : **[JEE(Main)2013]**

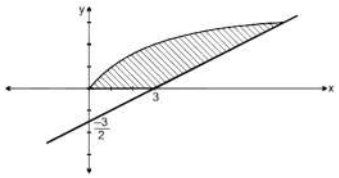
(A) 9 (B) 36 (C) 18 (D) 27/4

Ans. A

Sol. $y = \sqrt{x}$ ...(1)

and $2y - x + 3 = 0$ ...(2)

On solving both $y = -1, 3$



$$\text{Required area} = \int_0^3 \left\{(2y+3) - y^2\right\} dy$$

$$= y^2 + 3y - \frac{y^3}{3}\Big|_0^3$$

$= 9 + 9 - 9$

$= 9$

36. Let $g(x) = \cos x^2$, $f(x) = \sqrt{x}$, and α, β $(\alpha < \beta)$ be the roots of the quadratic equation $18x^2 - 9\pi x + \pi^2 = 0$. Then the area (in sq. unit) bounded by the curve $y = (gof)(x)$ and the lines $x = \alpha$, $x = \beta$ and $y = 0$ is :

[JEE(Main)2018]

माना $g(x) = \cos x^2$, $f(x) = \sqrt{x}$, तथा α, β $(\alpha < \beta)$ द्विघाती समीकरण $18x^2 - 9\pi x + \pi^2 = 0$ के मूल हैं। तो वक्र $y=(gof)(x)$ तथा j§ kv lsx $x = \alpha$, $x = \beta$ तथा $y = 0$ द्वारा घिरे क्षेत्र का क्षेत्रफल (वर्ग इकाइयों में) है : **[JEE(Main)2018]**

(A) $\frac{1}{2}\left(\sqrt{3} - \sqrt{2}\right)$ (B) $\frac{1}{2}\left(\sqrt{2} - 1\right)$ (C) $\frac{1}{2}\left(\sqrt{3} - 1\right)$ (D) $\frac{1}{2}\left(\sqrt{3} + 1\right)$

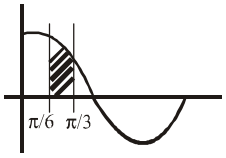
Ans. C

Sol. $gof(x) = \cos x$

$18x^2 - \pi x + \pi^2 = 0$

$(6x - \pi)(3x - \pi) = 0$

$x = \frac{\pi}{6}, x = \frac{\pi}{3}$



$$A = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \cos x \, dx = \frac{\sqrt{3} - 1}{2}$$

37. Let the straight line $x = b$ divide the area enclosed by $y = (1 - x^2)$, $y = 0$ and $x = 0$ into two parts $R_1 (0 \le x \le b)$ and $R_2 (b \le x \le 1)$ such that $R_1 - R_2 = \frac{1}{4}$. Then b equals : **[JEE-2011]**

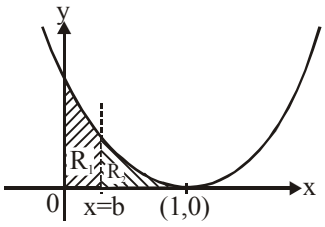
यदि रेखा x = b वक्रों $y = (1 - x^2)$, y = 0 तथा x = 0 को दो भागों में $R_1(0 \le x \le b)$ तथा $R_2(b \le x \le 1)$ में इस प्रकार विभाजित

d j r kgSd $R_1 - R_2 = \frac{1}{4}$ तो b का मान होगा : **[JEE-2011]**

(A) 3/4 (B) 1/2 (C) 1/3 (D) 1/4

Ans. B

Sol. $\because \quad R_1 - R_2 = \frac{1}{4}$



$$\Rightarrow \int_0^b (1-x)^2 dx - \int_b^1 (1-x)^2 dx = \frac{1}{4}$$

$$\Rightarrow -\left(\frac{(1-x)^3}{3}\right)_0^b + \left(\frac{(1-x)^3}{3}\right)_b^1 = \frac{1}{4}$$

$$\Rightarrow -\left\{\frac{(1-b)^3}{3} - \frac{1}{3}\right\} - \frac{(1-b)^3}{3} = \frac{1}{4}$$

$$\Rightarrow \frac{1}{3} - \frac{2}{3}(1-b)^3 = \frac{1}{4} \Rightarrow \frac{2}{3}(1-b)^3 = \frac{1}{12}$$

$$\Rightarrow (1-b)^3 = \frac{1}{8} \Rightarrow 1 - b = \frac{1}{2}$$

$$\Rightarrow b = \frac{1}{2}$$

SHIFTING OF ORIGIN BASED QUESTIONS

38. The line y = mx bisects the area enclosed by the curve $y = 1 + 4x - x^2$ and the lines x = 0, x = 2 and y = 0. Then the value of m is :

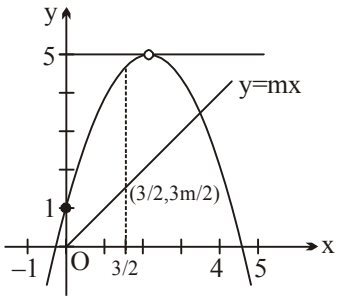
(A) $\frac{13}{6}$ (B) $\frac{6}{13}$ (C) $\frac{11}{6}$ (D) none

यदि रेखा y = mx, वक्र $y = 1 + 4x - x^2$ तथा रेखाऐं x = 0, x = 2 तथा y = 0 द्वारा परिबद्ध क्षेत्रफल को समद्विभाजित करती है, तो m का मान होगा :

(A) $\frac{13}{6}$ (B) $\frac{6}{13}$ (C) $\frac{11}{6}$ (D) कोई नही

Ans. A

Sol. $2 \cdot \frac{1}{2}\left(\frac{3}{2}\cdot\frac{3m}{2}\right) = \int_0^{3/2}(1+4x-x^2)\,dx$



$$\frac{9m}{4} = x+2x^2-\frac{x^3}{3}\Big]_0^{3/2} = \frac{3}{2}+\frac{9}{2}-\frac{27}{8}\cdot\frac{1}{3} = 6-\frac{9}{8} = \frac{39}{8}$$

$$m = \frac{13}{6}$$

39. Area enclosed by the curve $|x-2|+|y+1|=1$ is equal to :

(A) 4 sq. units (B) 6 sq. units (C) 2 sq. units (D) 8 sq. units

वक्र $|x-2|+|y+1|=1$ से घिरा हुआ क्षेत्रफल होगा :

(A) 4 वर्ग ईकाई (B) 6 वर्ग ईकाई (C) 2 वर्ग ईकाई (D) 8 वर्ग ईकाई

Ans. C

Sol. $|x-2|+|y-1|=1$

$(0,0) \to (2,1)$

$|x|+|y|=1$

Area = $\sqrt{2}\times\sqrt{2}$

$= 2$

AREA REPRESENTED BY INEQUALITIES

40. The area of the region of the plane bounded by $\max(|x|,|y|) \le 1$ and $xy \le \frac{1}{2}$ is :

$\max(|x|,|y|) \le 1$ तथा $xy \le \frac{1}{2}$ द्वारा निरूपित क्षेत्र का क्षेत्रफल होगा :

(A) $\frac{1}{2}+\ln 2$ (B) $\frac{31}{4}$ (C) $1+2\ln 2$ (D) $3+\ln 2$

Ans. D

Sol. $\max(|x|,|y|) \le 1$ $\qquad xy \le \frac{1}{2}$

$$\text{Area} = 2\times 2 - 2\int_{\frac{1}{2}}^{1}\left(1-\frac{1}{2}x\right)dx$$

$$= 4 - 2\left(x-\frac{1}{2}\ell nx\right)_{\frac{1}{2}}^{1}$$

$$= 4 - 2 + 2\left(\frac{1}{2}-\frac{1}{2}\ell n\frac{1}{2}\right)$$

$$= 2 + 1 - \ell n\frac{1}{2}$$

$$= 3 + \ell n\, 2$$

41. Area of the region which consists of all the points satisfying the conditions $|x-y|+|x+y|\le 8$ and $xy \ge 2$, is equal to:

(A) $4(7 - l n8)$ sq. units (B) $4\,(9 - l n8)$ sq. units

(C) $2(7 - l n8)$ sq. units (D) $2\,(9 - l n\,8)$ sq. units

प्रतिबंधों $|x - y| + |x + y| \le 8$ तथा $xy \ge 2$ को संतुष्ट करने वाले सभी बिन्दु से बना हुआ कुल क्षेत्रफल होगा :

(A) 4;7 $l n8)$ वर्ग इकाई (B) 4 ;9 $l n8)$ वर्ग इकाई

(C) 2;7 $l n8)$ वर्ग इकाई (D) 2 ;9 $l n\,8)$ वर्ग इकाई

Ans. A

Sol. $|x-y|+|x+y| \le 8$ $xy \ge 2$

$$\text{Required Area} = 2\times\int_{\frac{1}{2}}^{4}\left(4-\frac{2}{x}\right)dx$$

$$= 2\left(4x - 2\ell nx\right)_{\frac{1}{2}}^{4}$$

$$= 4\left(8 - \ell n4 - 1 + \ell n\left(\frac{1}{2}\right)\right)$$

$$= 4(7 - \ell n8)$$

42. A point P moves in xy plane in such a way that $[|x|] + [|y|] = 1$, where [.] denotes the greatest integer function. Area of the region representing all possible positions of the point P is equal to :

(A) 4 sq. units (B) 16 sq. units (C) $2\sqrt{2}$ sq. units (D) 8 sq. units

एक बिन्दु P xy तल में इस प्रकार गति करता है कि [|x|] + [|y|] = 1, जहाँ [.] महत्म पूर्णांक फलन है, तो P के सभी सम्भव मानों से घिरा gq k{kQy gkx k:

(A) 4 वर्ग ईकाई (B) 16 वर्ग ईकाई (C) $2\sqrt{2}$ वर्ग ईकाई (D) 8 वर्ग ईकाई

Ans. D

Sol. $\left[|x|\right]+\left[|y|\right]=1$

1st Quadrant

[x] + [y] = 1

2nd Quadrant

[–x] + [y] = 1

3rd Quadrant

[–x] + [–y] = 1

4th Quadrant

[x] + [–y] = 1

Area = 2 × 2 = 4

43. The area defined by $|y| \le e^{-|x|} - \frac{1}{2}$ in cartesian co-ordinate system, is :

कार्तिय–निर्देशांक निकाय में $|y| \le e^{-|x|} - \frac{1}{2}$ से परिभाषित क्षेत्रफल होगा :

(A) (4 – 2*l*n 2) (B) (4 – *l*n 2) (C) (2 – *l*n 2) (D) (2 – 2 *l*n 2)

Ans. D

Sol. For $y \ge 0$,

$$\therefore\ y = e^{-|x|} - \frac{1}{2} = \begin{cases} e^{-x} - \frac{1}{2} \text{ if } x \ge 0 \\ e^{x} - \frac{1}{2} \text{ if } x < 0 \end{cases} \quad \text{.....(1)}$$

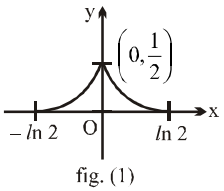


fig. (1)

and for $y < 0$,

$$-y = \frac{1}{2} - e^{-|x|} = \begin{cases} \frac{1}{2} - e^{-x} & \text{if } x \geq 0 \\ \frac{1}{2} - e^{x} & \text{if } x < 0 \end{cases} \quad \text{.....(2)}$$

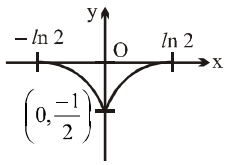
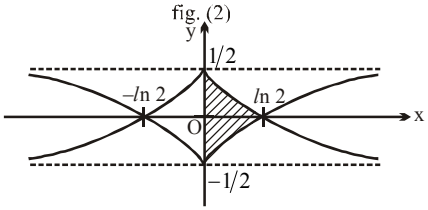


fig. (2)



Combining (1) and (2) we have the required area given as

$$A = 4\int_0^{\ln 2}\left(e^{-x} - \frac{1}{2}\right)dx = 4\left(\frac{1}{2} - \frac{1}{2}\ln 2\right) = (2 - 2\ln 2)$$

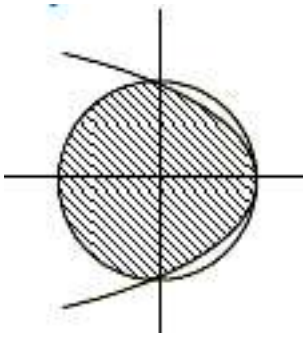
44. The area of the region described by $A = \{(x, y) : x^2 + y^2 \leq 1 \text{ and } y^2 \leq 1 - x\}$ is : **[JEE(Main)2014]**

$A = \{(x, y) : x^2 + y^2 \leq 1$ तथा $y^2 \leq 1 - x\}$ के द्वारा प्रदत्त क्षेत्र का क्षेत्रफल है : **[JEE(Main)2014]**

(A) $\frac{\pi}{2} - \frac{2}{3}$ (B) $\frac{\pi}{2} + \frac{2}{3}$ (C) $\frac{\pi}{2} + \frac{4}{3}$ (D) $\frac{\pi}{2} - \frac{4}{3}$

Ans. C

Sol.



Required area $= \pi - 2\int_0^1\left(\sqrt{1-x^2} - \sqrt{1-x}\right)dx$

$$= \pi - 2\left[\frac{x\sqrt{1-x^2}}{2} + \frac{1}{2}\sin^{-1}x + \frac{2}{3}(1-x)^{3/2}\right]_0^1 = \pi - 2\left[\frac{\pi}{4} - \frac{2}{3}\right] = \frac{\pi}{2} + \frac{4}{3}$$

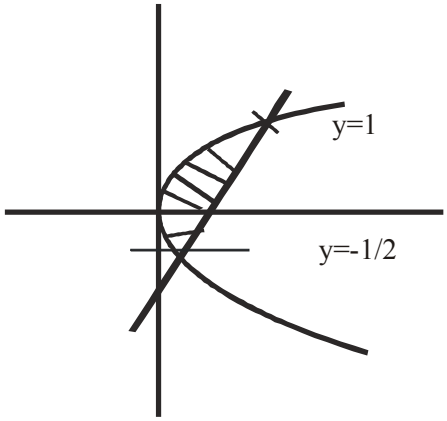
45. The area (in sq. units) of the region described by $[(x, y) : y^2 \leq 2x \text{ and } y \geq 4x - 1]$ is : **[JEE(Main)2015]**

$[(x, y) : y^2 \leq 2x$ तथा $y \geq 4x–1]$ द्वारा परिभाषित क्षेत्र का क्षेत्रफल (वर्ग इकाईयों) में हैं : **[JEE(Main)2015]**

(A) $\frac{1}{32}$ (B) $\frac{5}{64}$ (C) $\frac{15}{64}$ (D) $\frac{9}{32}$

Ans. D

Sol. $y^2 \leq 2x$ $y \geq 4x - 1$



$$\text{Req Area} = \int_{-\frac{1}{2}}^{1}\left(\frac{y+1}{4}-\frac{y^2}{2}\right)dy$$

$$=\frac{9}{32}$$

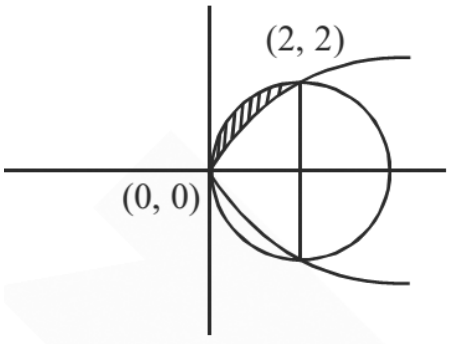
46. The area (in sq. units) of the region $\{(x, y\} : y^2 \geq 2x$ and $x^2 + y^2 \leq 4x, x \geq 0, y \geq 0\}$ is : **[JEE(Main)2016]**

क्षेत्र $\{(x, y\} : y^2 \geq 2x$ तथा $x^2 + y^2 \leq 4x, x \geq 0, y \geq 0\}$ का क्षेत्रफल (वर्ग इकाईयों में) है : **[JEE(Main)2016]**

(A) $\pi - \frac{4\sqrt{2}}{3}$ (B) $\frac{\pi}{2} - \frac{2\sqrt{2}}{3}$ (C) $\pi - \frac{4}{3}$ (D) $\pi - \frac{8}{3}$

Ans. D

Sol.



$$\text{Area} = \frac{1}{4}(\text{Area of circle}) - \int_0^2 \sqrt{2x}\,dx$$

$$= \pi - \frac{8}{3}$$

47. The area (in sq. units) of the region $\{(x, y) : x \geq 0, x + y \leq 3, x^2 \leq 4y$ and $y \leq 1 + \sqrt{x}\}$ is :

[JEE(Main)2017]

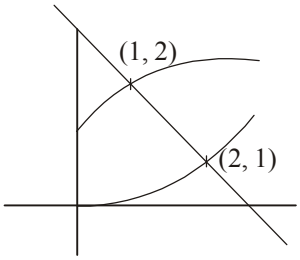
क्षेत्र $\{(x, y) : x \geq 0, x + y \leq 3, x^2 \leq 4y$ तथा $y \leq 1 + \sqrt{x}\}$ का क्षेत्रफल होगा : **[JEE(Main)2017]**

(A) $\frac{7}{3}$ (B) $\frac{5}{2}$ (C) $\frac{59}{12}$ (D) $\frac{3}{2}$

Ans. B

Sol. $\text{Area} = \int_0^1 \left\{\left(1+\sqrt{x}\right) - \frac{x^2}{4}\right\} dx + \int_1^2 \left\{(3-x) - \frac{x^2}{4}\right\} dx$

$= \frac{5}{2}$



DETERMINATION OF UNKNOWN PARAMETERS

48. If A is the area between the curve y = sin x and x-axis in the interval [0, $\pi/2$], then the area between y = sin 2x and x- axis in this interval will be :

(A) A (B) 2A (C) A/2 (D) None of these

यदि वक्र y = sin x तथा x-अक्ष के मध्य अन्तराल [0, $\pi/2$] में क्षेत्रफल A हो, तो y = sin 2x तथा x-अक्ष के मध्य इसी अन्तराल में {lsk{Qy gkskk:

(A) A (B) 2A (C) A/2 (D) इनमें से कोई नहीं

Ans. A

Sol. $y = \sin x$

$$A = \int_0^{\frac{\pi}{2}} \sin x dx$$

$$= -\cos x\Big|_0^{\frac{\pi}{2}}$$

$= 1$

$$A' = \int_0^{\frac{\pi}{2}} \sin 2x dx$$

$$= -\frac{\cos 2x}{2}\Big|_0^{\frac{\pi}{2}}$$

$$= -\frac{1}{2}[-1-1] = 1 = A$$

49. If $(a, 0); a > 0$ is the point where the curve $y = \sin 2x - \sqrt{3}\sin x$ cuts the x-axis first, A is the area bounded by this part of the curve , the origin and the positive x-axis, then :

यदि $(a, 0); a > 0$ वह बिन्दु है जहाँ वक्र $y = \sin 2x - \sqrt{3}\sin x$, x-अक्ष को प्रथम बार काटता है तथा वक्र के इस भाग, मूलबिन्दु तथा /kukRed x-अक्ष द्वारा घिरा क्षेत्रफल A है, तब :

(A) $4A + 8\cos a = 7$ (B) $4A + 8\sin a = 7$ (C) $4A - 8\sin a = 7$ (D) $4A - 8\cos a = 7$

Ans. A

Sol. $(a, 0)$ lies on the given curve

$\therefore \quad 0 = \sin 2a - \sqrt{3}\sin a \quad \Rightarrow \quad \sin a = 0$ or $\cos a = \sqrt{3}/2$

$\Rightarrow \quad a = \frac{\pi}{6}$ (as $a > 0$ and the first point of intersection with positive X-axis)

and $A = \int_0^{\pi/6} (\sin 2x - \sqrt{3}\sin x)\,dx = \left(-\frac{\cos 2x}{2} + \sqrt{3}\cos x\right)_0^{\pi/6}$

$$= \left(-\frac{1}{4} + \frac{3}{2}\right) - \left(-\frac{1}{2} + \sqrt{3}\right) = \frac{7}{4} - \sqrt{3} = \frac{7}{4} - 2\cos a \quad \Rightarrow \quad 4A + 8\cos a = 7$$

AREA BOUNDED BY INVERSE OF A FUNCTION

50. Area of the region enclosed between the curves $x = y^2 - 1$ and $x = |y|\sqrt{1-y^2}$ is :

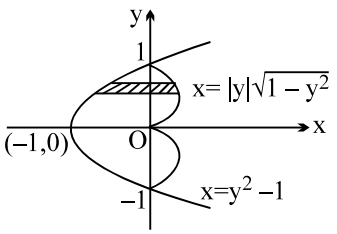
वक्रों $x = y^2 - 1$ तथा $x = |y|\sqrt{1-y^2}$ द्वारा परिबद्ध क्षेत्रफल होगा :

(A) 1 (B) 4/3 (C) 2/3 (D) 2

Ans. D

Sol. $A = 2\int_0^1 \left[y\sqrt{1-y^2} - (y^2 - 1)\right] dy$

$= 2$



DETERMINATION OF FUNCTION

51. The area bounded by the curve $y = f(x)$, the x-axis & the ordinates $x = 1$ & $x = b$ is $(b-1)\sin(3b+4)$. Then $f(x)$ is :

(A) $(x-1)\cos(3x+4)$ (B) $\sin(3x+4)$

(C) $\sin(3x+4)+3(x-1).\cos(3x+4)$ (D) none

यदि वक्र $y = f(x)$, x-अक्ष तथा कोटि $x = 1$ तथा $x = b$ द्वारा परिबद्ध क्षेत्रफल $(b-1)\sin(3b+4)$ है, तो $f(x)$ होगा :

(A) $(x-1)\cos(3x+4)$ (B) $\sin(3x+4)$

(C) $\sin(3x+4)+3(x-1).\cos(3x+4)$ (D) कोई नहीं

Ans. C

Sol. $\int_1^b f(x)\,dx = (b-1)\sin(3b+4)$

Apply Leibnitz

$f(b) = (b-1)\cos(3b+4)3 + \sin(3b+4)$

52. If the area bounded by the x-axis, curve $y = f(x)$ and the lines $x = 1$, $x = b$ is equal to $\sqrt{b^2+1} - \sqrt{2}$ for all $b > 1$, then $f(x)$ is :

यदि x-अक्ष, वक्र $y = f(x)$ तथा रेखाओं $x = 1$, $x = b$ के द्वारा परिबद्ध क्षेत्रफल $\sqrt{b^2+1} - \sqrt{2}$; सभी $b > 1$ के लिऐ हो, तो $f(x)$ होगा :

(A) $\sqrt{(x-1)}$ (B) $\sqrt{(x+1)}$ (C) $\sqrt{(x^2+1)}$ (D) $\dfrac{x}{\sqrt{1+x^2}}$

Ans. D

Sol. $\int_1^b f(x)\,dx = \sqrt{b^2+1} - \sqrt{2}$

$$f(b) = \frac{2b}{2\sqrt{b^2+1}} = \frac{b}{\sqrt{b^2+1}}$$

53. Let $f(x)$ be a non-negative continuous function such that the area bounded by the curve $y = f(x)$, x-axis and the ordinates $x = \dfrac{\pi}{4}$ and $x = \beta > \dfrac{\pi}{4}$ is $\left(\beta\sin\beta + \dfrac{\pi}{4}\cos\beta + \sqrt{2}\beta\right)$. Then $f\left(\dfrac{\pi}{2}\right)$ is :

माना f (x) एक इस प्रकार का अऋणात्मक सतत् फलन है, कि वक्र y = f (x), x-अक्ष तथा रेखाओं $x = \frac{\pi}{4}$ तथा $x = \beta > \frac{\pi}{4}$ से आबद्ध क्षेत्र का क्षेत्रफल $\left(\beta \sin\beta + \frac{\pi}{4}\cos\beta + \sqrt{2}\beta\right)$ है, तो $f\left(\frac{\pi}{2}\right)$ का मान है :

(A) $\left(\frac{\pi}{4} + \sqrt{2} - 1\right)$ (B) $\left(\frac{\pi}{4} - \sqrt{2} + 1\right)$ (C) $\left(1 - \frac{\pi}{4} - \sqrt{2}\right)$ (D) $\left(1 - \frac{\pi}{4} + \sqrt{2}\right)$

Ans. D

Sol. $\int_{\frac{\pi}{4}}^{\beta} f(x)\,dx = \beta\sin\beta + \frac{\pi}{4}\cos\beta + \sqrt{2}\beta$

$f(\beta) = \beta\cos\beta + \sin\beta - \frac{\pi}{4}\sin\beta + \sqrt{2}$

$f\left(\frac{\pi}{2}\right) = 0 + 1 - \frac{\pi}{4} + \sqrt{2}$

Exercise J-II

SUBJECTIVE / INTEGER TYPE QUESTIONS

AREA BOUNDED BETWEEN TWO CURVES

1. Find the area bounded on the right by the line $x+y=2$, on the left by the parabola $y=x^2$ and below by the x-axis in first quadrant.

   क्षेत्रफल ज्ञात कीजिए, जो दांयी ओर से सरल रेखा $x+y=2$ से, बांयी ओर से परवलय $y=x^2$ से, नीचे की तरफ से x-अक्ष से परिबद्ध हो (प्रथम चतुर्थांश में)।

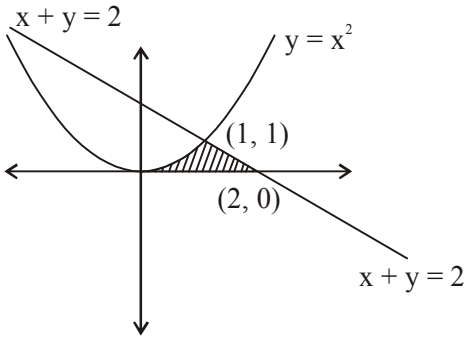
Ans. 5/6 sq. units

Sol. Line $x + y = 2$ and $y = x^2$

POI $x + y = 2$

$y = x^2$

(0, 0) and (1, 1)



$$\text{Area} = \int_0^1 y_1 dx + \int_1^0 y_2 dx$$

$$A = \int_0^1 x^2 dx + \int_1^2 (2-x)\,dx$$

$$= \frac{x^3}{3}\Big|_0^1 + 2x - \frac{x^2}{2}\Big|_1^2$$

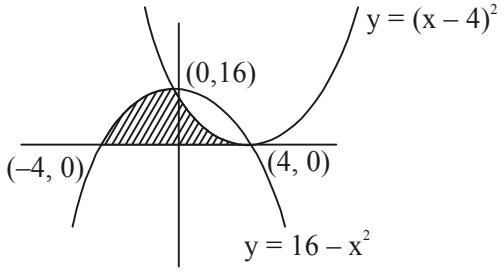
$$= \frac{1}{3} + 4 - 2 - 2 + \frac{1}{2} = \frac{5}{6}$$

2. Find the area of the region bounded by curves $f(x) = (x-4)^2$, $g(x) = 16 - x^2$ and the x - axis.

   वक्रों $f(x) = (x-4)^2$, $g(x) = 16 - x^2$ और x-अक्ष से परिबद्ध क्षेत्र का क्षेत्रफल ज्ञात कीजिए।

Ans. 64

Sol. $f(x) = (x-4)^2$ $g(x) = 16 - x^2$ x–axis



$$A = \int_{-4}^{0} \left(16 - x^2\right) dx + \int_{0}^{4} \left(x - 4\right)^2 dx$$

$$= 16x - \frac{x^3}{3}\Bigg|_{-4}^{0} + \frac{\left(x-4\right)^3}{3}\Bigg|_{0}^{4}$$

$$= 64 + \frac{16}{3} - \frac{16}{3} = 64$$

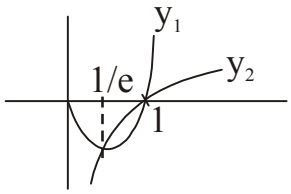
3. Compute the area of the region bounded by the curves y = e. x. $ln\,x$ & y = $ln\,x$/(e.x) where ln e = 1.

वक्रों y = e. x. $ln\,x$ और y = $ln\,x$/(e.x) जहाँ ln e = 1 से परिबद्ध क्षेत्र का क्षेत्रफल ज्ञात कीजिए।

Ans. $(e^2 - 5)/4$ e sq. units

Sol. $y_1 = e \cdot x \cdot \ell nx$

$$y_2 = \frac{\ell nx}{ex}$$



$$A = \int_{\frac{1}{e}}^{1} \left(\frac{\ell nx}{ex} - ex\ell nx \right) dx$$

$$= \frac{1}{e} \frac{\left(\ell nx\right)^2}{2}\Bigg|_{\frac{1}{e}}^{1} - e\left[\frac{x^2}{2}\ell nx - \frac{x^2}{4} \right]_{\frac{1}{e}}^{1}$$

$$= \frac{1}{e}\left(-\frac{1}{2}\right) - e\left[-\frac{1}{4} + \frac{1}{2e^2} + \frac{1}{4e^2}\right]$$

$$= -\frac{1}{2e} + \frac{e}{4} - \frac{3}{4e}$$

$$= \frac{e}{4} - \frac{5}{4e}$$

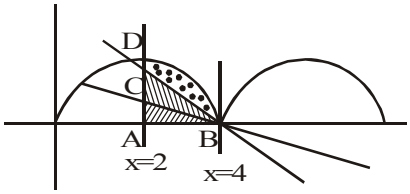
$$= \frac{e^2 - 5}{4e}$$

4. A figure is bounded by the curves $y = \left|\sqrt{2}\sin\frac{\pi x}{4}\right|$, $y = 0$, $x = 2$ & $x = 4$. At what angles to the positive x–axis straight lines must be drawn through $(4, 0)$ so that these lines partition the figure into three parts of the same size.

एक क्षेत्र वक्रों $y = \left|\sqrt{2}\sin\frac{\pi x}{4}\right|$, $y = 0$, $x = 2$ और $x = 4$ से परिबद्ध है। तब x–अक्ष की धनात्मक दिशा किस कोण से सरल रेखा जो कि $(4, 0)$ से जाती है खिचीं जाए की वह क्षेत्र को समान आकार के तीन भागों में विभाजित कर दे।

Ans. $\pi - \tan^{-1}\frac{2\sqrt{2}}{3\pi}$; $\pi - \tan^{-1}\frac{4\sqrt{2}}{3\pi}$

Sol. Let equation of line is $y = mx - 4m$



$$A = \int_2^4 \sqrt{2}\sin\frac{\pi}{4}x\,dx = \left[-\sqrt{2}\frac{4}{\pi}\cos\frac{\pi x}{4}\right]_2^4 = \frac{4\sqrt{2}}{\pi} \quad \text{... (i)}$$

$$\text{Also area of } \Delta ABC = \frac{1}{2}.2.(-2m_1) = -2m_1 \quad \text{... (ii)}$$

from (i) and (ii)

$$-2m_1 = \frac{4\sqrt{2}}{3\pi} \Rightarrow m_1 = \frac{-2\sqrt{2}}{3\pi}$$

$$\Rightarrow \tan(\pi - \theta_1) = \frac{-2\sqrt{2}}{3\pi} \Rightarrow \pi - \theta_1 = \tan^{-1}\frac{2\sqrt{2}}{3\pi}$$

$\Rightarrow q_1 = \pi - \tan^{-1}\frac{2\sqrt{2}}{3\pi}$ or $\frac{1}{2}.(2)(-2m_2) = \frac{8\sqrt{2}}{3\pi}$

$\Rightarrow m_2 = \frac{-4\sqrt{2}}{3\pi} \Rightarrow \tan(\pi - \theta_2) = \frac{-4\sqrt{2}}{3\pi}$

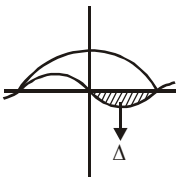
$\Rightarrow q_2 = \pi - \tan^{-1}\frac{4\sqrt{2}}{3\pi}$

5. Find the area bounded by the curves $y = \sqrt{1-x^2}$ and $y = x^3 - x$. Also find the ratio in which the y-axis divided this area.

वक्रों $y = \sqrt{1-x^2}$ और $y = x^3 - x$ से परिबद्ध क्षेत्रफल ज्ञात कीजिए। साथ ही वह अनुपात भी बताईये जबकि y-अक्ष क्षेत्रफल को विभाजित करता है।

Ans. $\frac{\pi}{2};\ \frac{\pi-1}{\pi+1}$

Sol. $\Delta = \left|\int_0^1 (x^3 - x)dx\right| = \frac{1}{4}$



$\text{Area} = \frac{\pi}{4} + \frac{1}{4} + \frac{\pi}{4} - \frac{1}{4} = \frac{\pi}{2}$

$\text{required ratio} = \frac{\frac{\pi}{4}+\frac{1}{4}}{\frac{\pi}{4}-\frac{1}{4}}$

6. The line $3x + 2y = 13$ divides the area enclosed by the curve, $9x^2 + 4y^2 - 18x - 16y - 11 = 0$ into two parts. Find the ratio of the larger area to the smaller area.

यदि सरल रेखा $3x + 2y = 13$, वक्र $9x^2 + 4y^2 - 18x - 16y - 11 = 0$ के बन्द क्षेत्र को दो भागों में विभाजित करती है तब बड़े आकार के क्षेत्रफल तथा छोटे क्षेत्र के क्षेत्रफलों का अनुपात ज्ञात कीजिए।

Ans. $\frac{3\pi + 2}{\pi - 2}$

Sol. $3x + 2y = 13$

$9x^2 + 4y^2 - 18x - 16y - 11 = 0$

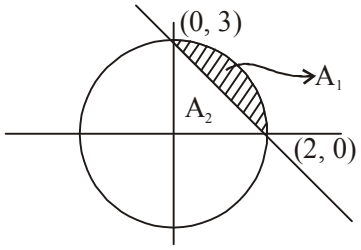
$9(x-1)^2 + 4(y-2)^2 - 36 = 0$

$$\frac{(x-1)^2}{4}+\frac{(y-2)^2}{9}=1$$

$x-1=x \qquad y-2=y$

$\frac{x^2}{4}+\frac{y^2}{9}=1$ and $3(x+1)+2(y+2)=13$

$$3x+2y=6$$



$$A_2=\frac{1}{2}\times 2\times 3=3$$

Area of ellipse = $\pi ab = \pi \times 2 \times 3$

$$= 6\pi$$

$$A_1=\frac{6\pi}{4}-3$$

$$=\frac{3\pi-6}{2}$$

Non-shaded ellipse area

$$=\frac{3}{4}\times 6\pi+3=\frac{9\pi+6}{2}$$

$$\text{Ratio}=\frac{\frac{9\pi+6}{2}}{\frac{3\pi-6}{2}}=\frac{3\pi+2}{\pi-2}$$ Ans.

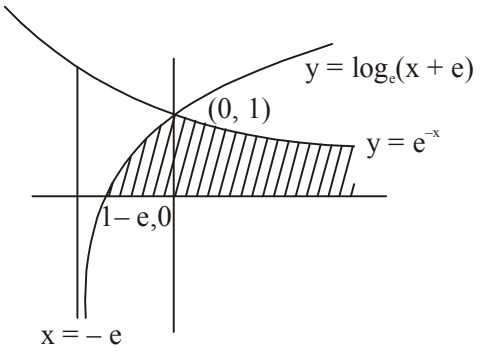
7. Find the area enclosed between the curves : $y=\log_e(x+e)$, $x=\log_e(1/y)$ & the x–axis.

वक्रों $y=\log_e(x+e)$, $x=\log_e(1/y)$ और x–अक्ष से परिबद्ध क्षेत्र का क्षेत्रफल ज्ञात कीजिए।

Ans. 2 sq. units

Sol. $y = \log_e (x + e)$ and x-axis

$x = \log_e (y_y) = y = e^{-x}$



$$\int_{1-e}^{0} \log_e (x+e)\,dx + \int_{0}^{\infty} e^{-x}dx$$

$$= (x+e)\log_e\left(\frac{x+e}{e}\right)\Bigg|_{1-e}^{0} + -e^{-x}\Big|_{0}^{\infty}$$

A = O – 1(–1) + 1 = 2 Ans.

8. Compute the area of the curvilinear triangle bounded by the y-axis & the curve, y = tan *x* & y=(2/3)cos x.

वक्रों y = tan *x* और y=(2/3)cos x और y-अक्ष से परिबद्ध वक्रीय त्रिभुज का क्षेत्रफल ज्ञात कीजिए।

Ans. $\frac{1}{3} + \ell n\left(\frac{\sqrt{3}}{2}\right)$ sq. units

Sol. $y = \tan x \qquad y = \frac{2}{3}\cos x \qquad$ y-axis

POI

$$\tan x = \frac{2}{3}\cos x$$

$$3\tan x = 2\cos x$$

$$3\sin x = 2\cos^2 x$$

$3\sin x = 2 - 2\sin^2 x$

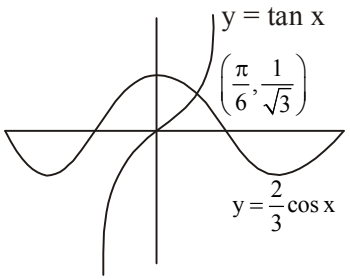
$2\sin^2 x + 3\sin x - 2 = 0$

$2\sin^2 x + 4\sin x - \sin x - 2 = 0$

$(2\sin x - 1)(\sin x + 2) = 0$

$\sin x = -1$

$x = \frac{\pi}{6}$



$$A = \int_0^{\frac{\pi}{6}} \left(\frac{2}{3}\cos x - \tan x\right) dx$$

$$= \frac{2}{3}\sin x - |n|\sec x\Big|_0^{\frac{\pi}{6}}$$

$$= \frac{2}{3}\times\frac{1}{2} - |n|\frac{2}{\sqrt{3}}\Big| = \frac{1}{3} + |n|\frac{\sqrt{3}}{2}\Big|$$

9. Find the area bounded by the curve $y = x e^{-x}$; $xy = 0$ and $x = c$ where c is the x-coordinate of the curve's inflection point.

वक्रों $y = x e^{-x}$; $xy = 0$ और $x = c$ से परिबद्ध क्षेत्रफल ज्ञात कीजिए। जहाँ c वक्र के नति परिवर्तन बिन्दु का x-निर्देशांक है।

Ans. $1 - 3e^{-2}$

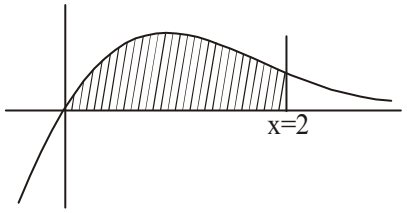
Sol. $y = xe^{-x}$

$y' = e^{-x} - x e^{-x} = (1 - x)e^{-x} \uparrow$ for $x < 1$

$y'' = -e^{-x} - [e^{-x} - xe^{-x}] = e^{-x}[-1 - 1 + x]$

$= (x - 2)e^{-x}$

for point of inflection $y'' = 0 \quad \Rightarrow \quad x = 2$



$$A = \int_0^2 xe^{-x}\,dx = -xe^{-x}\Big|_0^2 + \int_0^2 e^{-x}$$

$= (-2e^{-2}) - \left(e^{-x}\right)_0^2$

$= -2e^{-2} - (e^{-2} - 1) = 1 - e^{-2} - 2e^{-2} = 1 - 3e^{-2}$

10. Find the area bounded by the curve $y = x e^{-x^2}$, the x-axis, and the line x = c where y(c) is maximum.

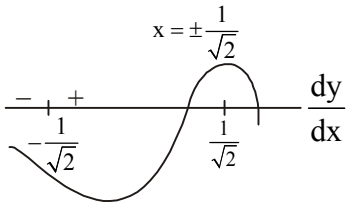
वक्र $y = x e^{-x^2}$, x-अक्ष तथा रेखा x = c परिबद्ध क्षेत्रफल ज्ञात कीजिए। (जहाँ y(c) अधिकतम है।)

Ans. $\frac{1}{2}(1 - e^{-1/2})$

Sol. $y = xe^{-x^2}$, x-axis $\quad x = c$

$$\frac{dy}{dx} = -2x^2e^{-x^2} + e^{-x^2} = 0$$

$$x = \pm\frac{1}{\sqrt{2}}$$



$$\left.\frac{dy}{dx}\right|_{max} \text{ at } x = \frac{1}{\sqrt{2}} \qquad c = \frac{1}{\sqrt{2}}$$

$$A = \int_0^{\frac{1}{\sqrt{2}}} xe^{-x^2}dx$$

$x^2 = t \quad \Rightarrow 2xdx = dt$

$$= \int_0^{\frac{1}{2}} e^{-t}\frac{dt}{2}$$

$$= \frac{1}{2}\left[-e^{-t}\right]_0^{\frac{1}{2}}$$

$$= \frac{1}{2}\left(1 - e^{-\frac{1}{2}}\right) \text{ Ans.}$$

11. Consider a circle $x^2 + (y-1)^2 = 1$ and the parabola $y = -\frac{x^2}{4}$. The common tangents to the two curves constitute a triangle ABC, the point A and B being on the x-axis and C on the y-axis. CA produced touches the parabola at P and CB produced touches the parabola at Q.

(a) Find the equation of the common tangent BC.

(b) Find the area of the portion between the upper arc of the circle and the common tangents QC and PC.

(c) Find the area enclosed by the parabola $y = -\frac{x^2}{4}$, the x-axis and the lines AP and BQ.

माना वृत्त $x^2 + (y-1)^2 = 1$ और परवलय $y = -\frac{x^2}{4}$ दिया गया है। यदि वक्रों की उभयनिष्ठ स्पर्शरेखाऐं एक त्रिभुज ABC का निमार्ण करता है, जहाँ बिन्दु A और B, x-अक्ष पर है और C, y-अक्ष पर, यदि CA का बढ़ाया भाग परवलय का P पर स्पर्श करता है और CB का बढ़ाया भाग परवलय को Q पर स्पर्श करता है।

(a) उभयनिष्ठ स्पर्शरेखा BC ज्ञात कीजिए।

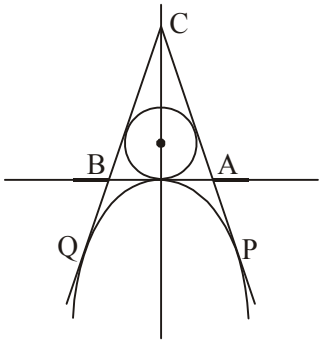
(b) वृत्त के ऊपरी चाप और उभयनिष्ठ स्पर्श रेखाओं QC और PC से परिबद्ध क्षेत्र का क्षेत्रफल ज्ञात कीजिए।

(c) परवलय $y = -\frac{x^2}{4}$, x-अक्ष और रेखाओं AP और BQ से परिबद्ध क्षेत्रफल ज्ञात कीजिए।

Ans. (a) $\sqrt{3}x - y + 3 = 0$ (b) $\left(\sqrt{3} - \frac{\pi}{3}\right)$ (c) $\sqrt{3}$

Sol. $x^2 + (y-1)^2 = 1$ and $x^2 = -4y$

Common tangent at parabola



$x = \frac{y}{m} - m$

$y = mx + m^2$

COT from (0, 1)

$\left|\frac{1-m^2}{\sqrt{1+m^2}} = 1\right|$

$m^4 - 2m^2 + 1 = 1 + m^2$

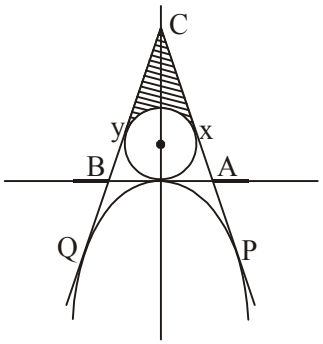
$m^4 = m^2$

$m = O, m = \sqrt{3}, m = -\sqrt{3}$

AB : $y = 0$ BC : $y = \sqrt{3}x + 3$ AC : $y = -\sqrt{3}x + 3$

$A\left(-\sqrt{3},0\right)$ $B\left(\sqrt{3},0\right)$ C(0, 3)

Point of contact on circle



12. If f (x) is monotonic in (a, b) then prove that the area bounded by the ordinates at x = a ; x = b ; y = f (x) and y = f(c), c ∈ (a, b) is minimum when $c = \frac{a+b}{2}$. Hence if the area bounded by the graph of $f(x)=\frac{x^3}{3}-x^2+a$, the straight lines x = 0, x = 2 and the x-axis is minimum then find the value of 'a'.

यदि f (x); (a, b) में एकदिष्ट फलन है, तो सिद्ध कीजिये कि x = a ; x = b ; y = f (x) और y = f (c), c ∈ (a, b) से परिबद्ध क्षेत्रफल न्यूनतम होगा यदि $c = \frac{a+b}{2}$ । अब यदि वक्र $f(x)=\frac{x^3}{3}-x^2+a$, सरल रेखा x = 0, x = 2 और x-अक्ष से परिबद्ध क्षेत्रफल न्यूनतम है, तो 'a' का मान ज्ञात कीजिये।

Ans. $a = \frac{2}{3}$

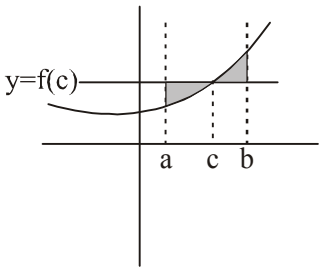
Sol. f(x)↑ (a, b)

Shaded area

$$A=\int_a^c\left(f(c)-(x)\right)dx+\int_c^b\left(f(x)-f(c)\right)dx$$

$$A=f(c)(c-a)-f(b-c)+\int_c^a f(n)+\int_c f(x)$$

$$A=f(c)(2c-a-b)+\underbrace{\int_c^a f(n)+\int_c^b f(x)}_{+ive}$$



f(c) (2c – a – b) = 0

$$c = \frac{a+b}{2}$$

$$f(x) = \frac{x^3}{3} - x^2 + a$$

$$f'(x) = x^2 - 2x$$

$-x = 0,\ x = 2$

Area will be minimum when graph will be symmetric about root α.

$$\alpha = \frac{0+2}{2} = 1$$

$f(1) = 0$

$$-\frac{2}{3} + a = 0$$

$$a = \frac{2}{3}$$

13. Find the area of the region bounded by the curves $y = x^2$, $y = |2 - x^2|$ and $y = 2$, which lies to the right of the line $x = 1$.

वक्रों $y = x^2$, $y = |2 - x^2|$ और रेखा $y = 2$ से परिबद्ध उस क्षेत्र का क्षेत्रफल ज्ञात कीजिये जो कि रेखा $x = 1$ के दायीं ओर है।

Ans. $\left(\frac{20}{3} - 4\sqrt{2}\right)$ sq. units

Sol. The given curves are $y = x2$

which is an upward parabola with vertex at (0, 0)

$y = |2 - x2|$

$$\text{or } y = \begin{cases} 2 - x^2 & \text{if} \quad -\sqrt{2} < x < \sqrt{2} \\ x^2 - 2 & \text{if} \quad x < -\sqrt{2} \text{ or } x > \sqrt{2} \end{cases}$$

or $x2 = -(y - 2)$; $\quad -\sqrt{2} < x < \sqrt{2}$ ... (2)

a downward parabola with vertex at (0, 2)

$x2 = y + 2$; $\quad x < -\sqrt{2}$, $x > \sqrt{2}$ ... (3)

On upward parabola with vertex at (0, –2)

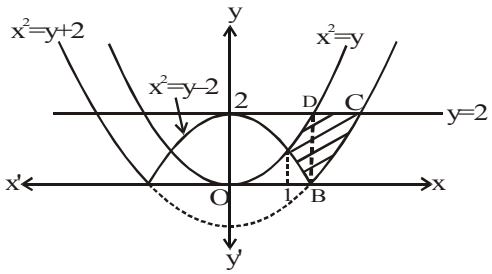
$y = 2$ ... (4)

Straight line parallel to x-axis

$x = 1$ ... (5)

Straight line parallel to y-axis

The graph of these curves is as follows.



$\therefore$ Required area = BCDEB

$$= \int_1^{\sqrt{2}} [x^2 - (2 - x^2)]dx + \int_{\sqrt{2}}^{2} [2 - (x^2 - 2)]dx$$

$$= \int_1^{\sqrt{2}} (2x^2 - 2)dx + \int_{\sqrt{2}}^{2} (4 - x^2)dx = \left(\frac{20}{3} - 4\sqrt{2}\right) \text{sq. units}$$

14. Find the area bounded by the curves $x^2 = y$, $x^2 = -y$ and $y^2 = 4x - 3$.

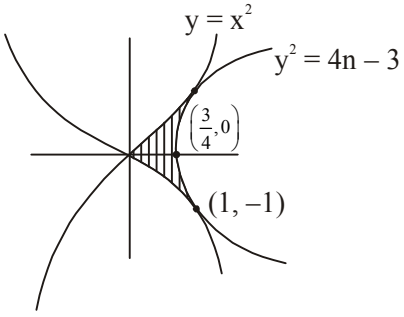
वक्रो $x^2 = y$, $x^2 = -y$ और $y^2 = 4x - 3$ से परिबद्ध क्षेत्र का क्षेत्रफल ज्ञात कीजिये।

Ans. $\frac{1}{3}$ sq. units

Sol. $x^2 = y$

$x^2 = -y$

$y^2 = x\left(x - \frac{3}{4}\right)$



POI

$y^2 = 4x - 3$ and $x^2 = y$

$x^4 = 4x - 3$

$x^4 - 4x + 3 = 0$

$(x - 1)(x^3 + x^2 + x - 3) = 0$

$$(x-1)(x-1)\left(\underbrace{x^2 + 2x + 3}_{D<0}\right) = 0$$

$n = 1$

$$A = 2\left[\int_0^1 x^2 dx - \int_{\frac{3}{4}}^1 \sqrt{4x-3}dx\right]$$

$$= 2\left[\frac{x^3}{3}\Big|_0^1 - \frac{2}{3}\frac{(4x-3)^{\frac{3}{2}}}{4}\Big|_{\frac{3}{4}}^1\right]$$

$$= 2\left(\frac{1}{3}-\frac{1}{6}\right)$$

$$= 2\times\frac{1}{6} = \frac{1}{3}$$

15. Let f(x) be a quadratic polynomial and a, b, c be distinct real numbers such that

$$\begin{bmatrix} 4a^2 & 4a & 1 \\ 4b^2 & 4b & 1 \\ 4c^2 & 4c & 1 \end{bmatrix}\begin{bmatrix} f(-1) \\ f(1) \\ f(2) \end{bmatrix} = \begin{bmatrix} 3a^2+3a \\ 3b^2+3b \\ 3c^2+3c \end{bmatrix}.$$

Let V be the point of maximum of the curve $y = f(x)$. If A and B are the points on this curve such that the curve meets the positive x-axis at A and the chord AB subtends a right angle at V, then find the area enclosed by the curve and the chord AB.

माना f(x) एक वर्ग बहुपद है और a, b, c तीन अलग–अलग वास्तविक संख्याएँ है, इस प्रकार है कि

$$\begin{bmatrix} 4a^2 & 4a & 1 \\ 4b^2 & 4b & 1 \\ 4c^2 & 4c & 1 \end{bmatrix}\begin{bmatrix} f(-1) \\ f(1) \\ f(2) \end{bmatrix} = \begin{bmatrix} 3a^2+3a \\ 3b^2+3b \\ 3c^2+3c \end{bmatrix}.$$

माना $y = f(x)$ का उच्चिष्ठ बिन्दु V है। यदि A और B वक्र पर दो बिन्दु इस प्रकार है कि वक्र x-अक्ष को बिन्दु A पर मिलता है तथा जीवा AB, V पर समकोण बनाती है, तब वक्र और जीवा AB से परिबद्ध क्षेत्र का क्षेत्रफल ज्ञात कीजिए।

Ans. $\frac{125}{3}$ sq. units

Sol. We have, $\begin{bmatrix} 4a^2 & 4a & 1 \\ 4b^2 & 4b & 1 \\ 4c^2 & 4c & 1 \end{bmatrix}\begin{bmatrix} f(-1) \\ f(1) \\ f(2) \end{bmatrix} = \begin{bmatrix} 3a^2+3a \\ 3b^2+3b \\ 3c^2+3c \end{bmatrix}$

$\Rightarrow 4a2f(-1) + 4af(1) + f(2) = 3a2 + 3a$

$4b2f(-1) + 4bf(1) + f(2) = 3b2 + 3b$

$4c2f(-1) + 4cf(1) + f(2) = 3c2 + 3c$

Consider the equation

$4x2f(-1) + 4xf(1) + f(2) = 3x2 + 3x$

or $[4f(-1) -3]x2 + [4f(1) - 3]\, x + f(2) = 0$

Then clearly this equation is satisfied by

$x = a, b, c$

A quadratic equation satisfied by more than two values of x means it is an identity and hence

$4f(-1) - 3 = 0 \quad \Rightarrow \quad f(-1) = 3/4$

$4f(1) - 3 = 0 \quad \Rightarrow \quad f(1) = 3/4$

$f(2) = 0 \quad \Rightarrow \quad f(2) = 0$

Let $f(x) = px2 + qx + r$ [$f(x)$ being a quad. equation]

$$f(-1) = \frac{3}{4} \quad \Rightarrow \quad p - q + r = \frac{3}{4}$$

$$f(1) = \frac{3}{4} \quad \Rightarrow \quad p + q + r = \frac{3}{4}$$

$$f(2) = 0 \quad \Rightarrow \quad 4p + 2q + r = 0$$

Solving the above we get $q = 0$, $p = \frac{-1}{4}$, $r = 1$

$$\therefore\ f(x) = -\frac{1}{4}\, x2 + 1$$

It's maximum value occur at $f'(x) = 0$

i.e., $x = 0$ then $f(x) = 1 \qquad \therefore\ V(0, 1)$

A $(-2, 0)$ is the pt. where curve meet x-axis

Let B be the pt. $\left(h, \frac{4 - h^2}{4}\right)$

As $\quad \angle AVB = 90°$

$mAV \times mBV = -1$

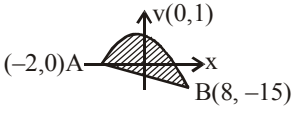
$$\Rightarrow \quad \frac{1}{2} \times \left(\frac{-h}{4}\right) = -1$$

$\Rightarrow \quad h = 8 \qquad \therefore\ B(8, -15)$

Equation of chord AB is

$$y + 15 = \frac{0 - (-15)}{-2 - 8}(x - 8)$$

$\Rightarrow 3x + 2y + 6 = 0$



Required area is

the area of shadded
region given by

$$=\int_{-2}^{8}\left[\left(-\frac{x^2}{4}+1\right)-\left(\frac{-6-3x}{2}\right)\right]dx$$

$$=\frac{125}{3}\text{ sq. units.}$$

DETERMINATION OF UNKNOWN PARAMETERS

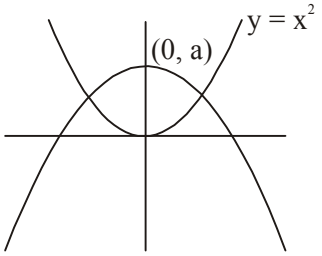
16. If the area enclosed by the parabolas $y = a - x^2$ and $y = x^2$ is $18\sqrt{2}$ sq. units. Find the value of 'a'.

यदि वक्रों $y = a - x^2$ और $y = x^2$ से परिबद्ध क्षेत्रफल $18\sqrt{2}$ वर्ग ईकाई है। तब a का मान ज्ञात कीजिए।

Ans. $a = 9$

Sol. $y = a - x^2$

$y = x^2$



$a \cdot x^2 = x^2$

$$x^2 = \frac{a}{2}$$

$$A = 2\int_{0}^{\sqrt{\frac{a}{2}}}\left(a\cdot x^2 - x^2\right)dx$$

$$18\sqrt{2} = 2\cdot\left[ax - \frac{2x^3}{3}\right]_0^{\sqrt{\frac{a}{2}}}$$

$$9\sqrt{2} = \frac{a^{\frac{3}{2}}}{\sqrt{2}} - \frac{2\cdot a^{\frac{3}{2}}}{3\cdot 2\sqrt{2}}$$

$$18 = \frac{3a^{\frac{3}{2}} - a^{\frac{3}{2}}}{3}$$

$$a^{\frac{3}{2}} = 27$$

$a = 9$

17. Find the values of m (m > 0) for which the area bounded by the line $y = mx + 2$ and $x = 2y - y^2$ is : (i) 9/2 square units and (ii) minimum. Also find the minimum area.

m (m > 0) के मानों को ज्ञात कीजिए जबकि सरल रेखा $y = mx + 2$ और वक्र $x = 2y - y^2$ से परिबद्ध क्षेत्रफल : ; पद्ध9ध्2 वर्ग इकाई है (ii) न्यूनतम है। साथ ही न्यूनतम क्षेत्रफल ज्ञात कीजिए।

Ans. (i) $m = 1$ (ii) $m = \infty$; $A_{min} = 4/3$

Sol. $y = mn + 2$; $m > 0$ and $x = 2y - y^2$

POI

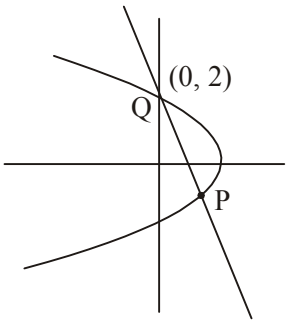
$y = mn + 2$

$$x = \frac{y-2}{m} = 2y - y^2$$

$$y - 2 = m(y-2)(-y)$$

$$y = -\frac{1}{m}, 2$$

$$x = -\frac{1-2m}{m^2}, 0$$



$$P\left(\frac{2m-1}{m^2}, -\frac{1}{3}\right) \text{ and } Q(0, 2)$$

$$\text{Area} = \int_{-\frac{1}{m}}^{2} \left[\left(2y - y^2\right) - \left(\frac{y-2}{m}\right)\right] dy$$

$$A = y^2 - \frac{y^2}{3} - \frac{y^2}{2m} + \frac{2y}{m}\Bigg|_{-\frac{1}{m}}^{2}$$

$$= \left(4 - \frac{8}{3} - \frac{2}{m} + \frac{4}{m}\right) - \left(\frac{1}{m^2} + \frac{1}{3m^3} - \frac{1}{2m^3} - \frac{2}{m^2}\right)$$

$$= \frac{4}{3} + \frac{2}{m} + \frac{1}{m^2} + \frac{1}{6m^3}$$

(i) $A = \frac{9}{2}$

$$\frac{9}{2} = \frac{4}{3} + \frac{2}{m} + \frac{1}{m^2} + \frac{1}{6m^3}$$

$27m^3 = 8m^3 + 12m^2 + 6m + 1$

$19m^3 - 12m^2 - 6m - 1 = 0$

$(m-1)(19m^2 + 7m + 1) = 0$

$D < 0$

$m = 1$

(ii) minimum area

$\frac{dA}{dm} = \frac{-2}{m^2} - \frac{2}{m^3} - \frac{1}{2m^4} = 0$

$\frac{4m^2 + 4m + 1 = 0}{2m^4}$

$(2m+1)^2 = 0$ or $m = \frac{1}{0}$

$m = -\frac{1}{2}$ or $m = \frac{1}{0}$

$m \neq -\frac{1}{2}$

So $m = \frac{1}{0}$

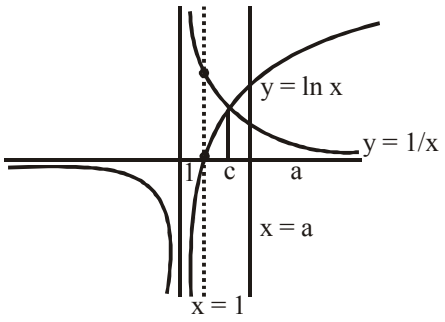
$Amin = \frac{4}{3}$

18. Consider two curves $C_1 : y = \frac{1}{x}$ and $C_2 : y = \ln x$ on the xy plane. Let D_1 denotes the region surrounded by C_1, C_2 and the line $x = 1$ and D_2 denotes the region surrounded by C_1, C_2 and the line $x = a$. If $D_1 = D_2$. Find the value of 'a'.

ब $_1 : y = \frac{1}{x}$ और $C_2 : y = \ln x$, xy समतल में दो वक्र है। माना D_1 वह क्षेत्र है जो कि C_1, C_2 और रेखा $x = 1$ से परिबद्ध है और D_2, C_1, C_2 और रेखा $x = a$ से परिबद्ध क्षेत्र है। यदि $D_1 = D_2$ हो तब a का मान ज्ञात कीजिए।

Ans. e

Sol.



$D_1 = D_2$

$$\int_1^C \left(\frac{1}{n} - \ln x\right) dx = \int_c^a \left(\ln x - \frac{1}{x}\right) dx = 0$$

$$\int_1^C \left(\frac{1}{x} - \ln x\right) dx + \int_c^a \left(\frac{1}{x} - \ln x\right) dx = 0$$

$$\int \left(\frac{1}{x} - \ln x\right) dx = 0$$

$$\ln x - x\left|x\left(\frac{x}{e}\right)\right|_1^a = 0$$

$$\ln a - a\ln\left(\frac{a}{e}\right) + \ln\left(\frac{1}{e}\right) = 0$$

$\ln a - a \ln a + a - 1 = 0$

$(\ln a - 1)(a - 1) = 0$

$\ln a = 1$ & $a = 1$

$a = e$

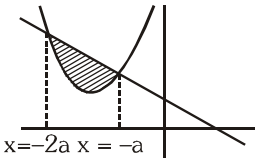
19. Find the value(s) of the parameter 'a' ($a > 0$) for each of which the area of the figure bounded by the straight line, $y = \frac{a^2 - ax}{1 + a^4}$ & the parabola $y = \frac{x^2 + 2ax + 3a^2}{1 + a^4}$ is the greatest.

; झ ०द्धके मानों का ज्ञात कीजिए जबकि सरल रेखा $y = \frac{a^2 - ax}{1 + a^4}$ और परवलय $y = \frac{x^2 + 2ax + 3a^2}{1 + a^4}$ से परिबद्ध क्षेत्रफल अधिकतम है।

Ans. $a = 3^{1/4}$

Sol. $A = \int_{-a}^{-2a} \frac{a^2 - ax - (x^2 + 2ax + 3a^2)}{1 + a^4} dx$

$= \frac{3}{2} \frac{a^3}{1 + a^4}$



Now $f(a) = \frac{3}{2} \frac{a^3}{1 + a^4}$

$\Rightarrow f'(a) = 0$

$\Rightarrow (1 + a4)\ 3a2 - a3\ 4a3 = 0$

$\Rightarrow$ amin = 0, amax = 31/4

20. For what value of 'a' is the area bounded by the curve $y = a^2x^2 + ax + 1$ and the straight line $y = 0$, $x = 0$ and $x = 1$ the least?

के किस मान के लिए वक्र $y = a^2x^2 + ax + 1$ और सरल रेखा $y = 0$, $x = 0$ और $x = 1$ से परिबद्ध क्षेत्रफल न्यूनतम होगा।

Ans. $a = -3/4$

Sol. Required area $= \int_0^1 ydx$

$$A = \int_0^1 \left(a^2x^2 + ax + 1\right) dx$$

$$A = \frac{a^2x^3}{3} + \frac{ax^2}{2} + x \Big|_0^1$$

$$A = \frac{a^2}{3} + \frac{a}{2} + 1$$

For min

$$\frac{dA}{da} = 0$$

$$\frac{2a}{3} + \frac{1}{2} = 0$$

$$\Rightarrow a = -\frac{3}{4}$$

21. Find the positive value of 'a' for which the parabola $y = x^2 + 1$ bisects the area of the rectangle with vertices (0, 0), (a, 0), (0, $a^2 + 1$) and (a, $a^2 + 1$).

का वह धनात्मक मान ज्ञात कीजिए जिसके लिए परवलय $y = x^2 + 1$ चतुर्भुज जिनके शीर्ष (0, 0), (a, 0), (0, $a^2 + 1$) और (a, $a^2 + 1$) से परिबद्ध क्षेत्रफल को समद्विभाजित करता है।

Ans. $\sqrt{3}$

Sol. Area of rectangle = $a(a^2 + 1)$

$= a^3 + a$

$$A = \int_0^a y\,dx$$

$$\frac{a^3 + a}{2} = \frac{a^3}{3}$$

$$a = 0, a = \pm\sqrt{3} \Rightarrow a = \sqrt{3}$$

22. Find the value of 'c' for which the area of the figure bounded by the curve, $y = 8x^2 - x^5$, the straight lines $x = 1$ & $x = c$ & the abscissa axis is equal to 16/3.

ब का मान ज्ञात कीजिए जबकि वक्र $y = 8x^2 - x^5$, सरल रेखाओं $x = 1$ और $x = c$ तथा भुज अक्ष से परिबद्ध क्षेत्रफल 16/3 वर्ग ईकाई है।

Ans. $C = -1$ or $\left(8 - \sqrt{17}\right)^{1/3}$

Sol. C-I $C < 1$

$$\int_c^1 \left(8x^2 + x^5\right) dx = \frac{16}{3}$$

$$\frac{8x^3}{3} - \frac{x^6}{6}\Bigg|_c^1 = \frac{16}{3}$$

$$\frac{8}{3} - \frac{1}{6} - \frac{8c^3}{3} + \frac{c^6}{6} = \frac{16}{3}$$

$$\frac{-8}{3}c^3 + \frac{c^6}{6} = \frac{17}{6} = \frac{17}{6}$$

$$c^6 - 16c^3 - 17 = 0$$

$c^3 = -1, 17$

$C = -1, 17^{1/3}$

$C < 1$

$C = -1$

C - II

$C \geq 1$

$$\int_1^C \left(8x^2 - x^5\right) dx = \frac{16}{3}$$

$$\frac{8x^3}{3} - \frac{x^6}{6}\Big|_1^C = \frac{16}{3}$$

$$\frac{8C^3}{3} - \frac{C^6}{6} = \frac{47}{6}$$

$C^6 - 16C^3 + 47 = 0$

$C^3 = t$

$t^2 - 16t + 47 = 0$

$$t = \frac{16 \pm \sqrt{256 - 188}}{2}$$

$t = 8 \pm \sqrt{64 - 47}$

$t = 8 \pm \sqrt{17}$

$C^3 = 8 \pm \sqrt{17}$

$C = \left(8 + \sqrt{17}\right)^{1/3}, \left(8 - \sqrt{17}\right)^{1/3}$

23. Consider one side AB of a square ABCD, (read in order) on the line $y = 2x - 17$, and the other two vertices C, D on the parabola $y = x^2$.

(a) Find the minimum intercept of the line CD on y-axis.

(b) Find the maximum possible area of the square ABCD.

(c) Find the area enclosed by the line CD with minimum y-intercept and the parabola $y = x^2$

माना वर्ग ABCD (क्रम में) की एक भुजा AB, सरल रेखा $y = 2x - 17$ पर है तथा अन्य दो शीर्ष C, D परवलय $y = x^2$ पर

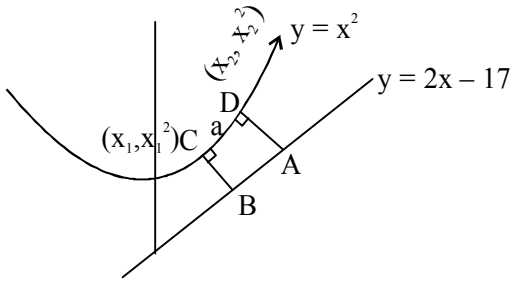
(a) सरल रेखा CD का न्यूनतम y-अन्तःखण्ड ज्ञात कीजिए।

(b) वर्ग ABCD का अधिकतम सम्भव क्षेत्रफल ज्ञात कीजिए।

(c) वह क्षेत्रफल ज्ञात कीजिए जो कि रेखा CD का न्यूनतम y-अन्तःखण्ड और परवलय $y = x^2$ से परिबद्ध है

Ans. (a) 3 (b) 1280 (c) $\frac{32}{3}$

Sol.



(a) As ABCD is a squre $\therefore$ AB || CD

Equation of line CD $\equiv y = 2x + c$

$$\text{distance between AB \& CD} = \frac{|c+17|}{\sqrt{2^2+1}} = \frac{|c+17|}{\sqrt{5}}$$

be 'a'

To find x coordinates of C and D solve

$y = x^2$ and $y = 2x + c$

$x^2 = 2x + c$

$x^2 - 2x - c = 0 \Leftarrow x_1, x_2$

$\Rightarrow x_1 + x_2 = 2$

Now length of CD

$$= \sqrt{(x_2 - x_1)^2 + (x_2^2 - x_1^2)^2}$$

$$= \sqrt{(x_2 - x_1)^2 + (x_2 - x_1)^2 (x_2 + x_1)^2}$$

$$= \sqrt{(x_2 - x_1)^2 \{1 + 4\}}$$

$$= \sqrt{5}\,|x_2 - x_1|$$

Now $\quad |x_2 - x_1| = \dfrac{\sqrt{D}}{|a'|}$

$$= \sqrt{4 + 4c}$$

$$= 2\sqrt{1 + c}$$

$$\therefore \qquad a = 2\sqrt{5}\sqrt{1 + c} = \frac{|c + 17|}{\sqrt{5}}$$

$4.5^2 (1 + c) = (c + 17)^2$

$100(1 + c) = c^2 + 34c + 289$

$100 + 100\ c = c^2 + 34c + 289$

$c^2 - 66C + 189 = 0$

$c = 63, 3$

$\therefore\ c_{min} = 3$

$c_{max} = 63$

(b) Maximum area $= a^2$

$$= \left(2\sqrt{5}\sqrt{1 + c}\right)^2$$

$$= 4.5(1 + c)$$

$$= 20(1 + 63)$$

$$= 20 \times 64$$

$$= 1280$$

(c) Equation of line CD with min y intercept i.e. c = 3

i.e. c = 3

$y = 2x + 3$ and $y = x^2$

Point of intersection

$x^2 = 2x + 3$

$\therefore\ x^2 - 2x - 3 = 0$

$x = 3, -1$

$$\therefore \quad \int_{-1}^{3} \left((2x+3) - x^2 \right) dx$$

$$-\frac{x^3}{3} + x^2 + 3x \Big|_{-1}^{3}$$

$$\Rightarrow \frac{32}{3}$$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 24 to 26

Consider the polynomial $f(x) = 1 + 2x + 3x^2 + 4x^3$. Let s be the sum of all distinct real roots of f(x) and let $t = |s|$.

माना $f(x) = 1 + 2x + 3x^2 + 4x^3$ एक बहुपदीय फलन है तथा S, f(x) के सभी वास्तविक भिन्न–भिन्न मूलों का योग है और माना $t = |s|$.

24. The real number s lies in the interval : **[JEE-2010]**

वास्तविक संख्या s अन्तराल में होगी : **[JEE-2010]**

(A) $\left(-\frac{1}{4}, 0\right)$ (B) $\left(-11, -\frac{3}{4}\right)$ (C) $\left(-\frac{3}{4}, -\frac{1}{2}\right)$ (D) $\left(0, \frac{1}{4}\right)$

Ans. C

Sol. $\because f'(x) = 2 + 6x + 12x2 > 0 \ \forall \ x \in R$

$\therefore$ f(x) is strictly increasing in R

$\because f(0) = 1, f(-1) = -2, f\left(-\frac{1}{2}\right) = \frac{1}{4}$ and $f\left(-\frac{3}{4}\right) = -\frac{1}{2}$

$\therefore$ f(x) = 0 has only one real root lying in $\left(-\frac{3}{4}, -\frac{1}{2}\right)$

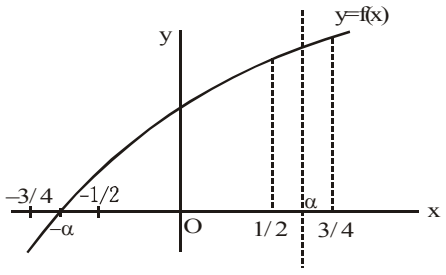
25. The area bounded by the curve y = f(x) and the lines x = 0, y = 0 and x = t, lies in the interval : **[JEE-2010]**

y = f(x) सरल रेखाओं x = 0, y = 0 और x = t से परिबद्ध क्षेत्रफल किस अन्तराल में होगा : **[JEE-2010]**

(A) $\left(\frac{3}{4}, 3\right)$ (B) $\left(\frac{21}{64}, \frac{11}{16}\right)$ (C) (9, 10) (D) $\left(0, \frac{21}{64}\right)$

Ans. A

Sol.



Let real root is $-\alpha$

$\Rightarrow \quad t = |s| = \alpha$

Required area

$$A = \int_0^{\alpha} f(x)dx \text{ and } \int_0^{1/2} f(x)dx < A < \int_0^{3/4} f(x)dx$$

$$\Rightarrow \left|x + x^2 + x^3 + x^4\right|_0^{1/2} < A < \left|x + x^2 + x^3 + x^4\right|_0^{3/4} < \left|4x\right|_0^{3/4}$$

$$\Rightarrow \frac{15}{16} < A < 3$$

26. The function $f'(x)$ is : **[JEE-2010]**

(A) increasing in $\left(-t, -\frac{1}{4}\right)$ and decreasing in $\left(-\frac{1}{4}, t\right)$

(B) decreasing in $\left(-t, -\frac{1}{4}\right)$ and increasing in $\left(-\frac{1}{4}, t\right)$

(C) increasing in $(-t, t)$

(D) decreasing in $(-t, t)$

फलन $f'(x)$ है : **[JEE-2010]**

(A) $\left(-t, -\frac{1}{4}\right)$ में वर्धमान और $\left(-\frac{1}{4}, t\right)$ में ह्रासमान

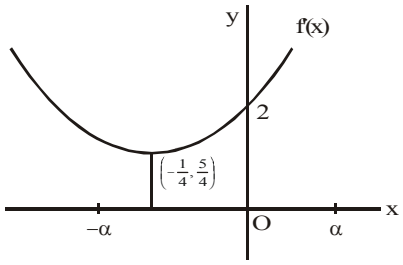
(B) $\left(-t, -\frac{1}{4}\right)$ में ह्रासमान और $\left(-\frac{1}{4}, t\right)$ में वर्धमान

(C) ; x के सभी में वर्धमान

(D) ; x के सभी में ह्रासमान

Ans. B

Sol.



$f'(x) = 2(6x2 + 3x + 1)$

$\Rightarrow$ $f'(x)$ is decreasing in $\left(-\alpha, -\frac{1}{4}\right)$ increasing in $\left(-\frac{1}{4}, \alpha\right)$

or $f'(x)$ is decreasing in $\left(-t, -\frac{1}{4}\right)$ and increasing in $\left(-\frac{1}{4}, t\right)$

Paragraph for Question nos. 27 to 29

Consider three real-valued functions f, g and h defined on R (the set of real numbers). Let $f(x) = 2x^3 + 3\left(1-\frac{3a}{2}\right)x^2 + 3(a^2-a)x + b$ where $a, b \in R$ and $g(x) = \frac{f'(x)}{6}$. Also h(x) is such that $h''(x) = 6x - 4$ and h(x) has a local minimum value 5 at $x = 1$.

वास्तविक मान वाले फलनों f, g तथा h जो कि R (वास्तविक संख्याओं के समुच्चय) पर परिभाषित है, माना $f(x) = 2x^3 + 3\left(1-\frac{3a}{2}\right)x^2 + 3(a^2-a)x + b$ जहाँ $a, b \in R$ तथा $g(x) = \frac{f'(x)}{6}$. तथा h(x) इस प्रकार है कि $h''(x) = 6x - 4$ तथा h(x) का, $x = 1$ पर स्थानीय निम्निष्ठ 5 है।

27. The true set of values of a for which f(x) has negative point of local minimum, is :

a का वास्तविक समुच्चय जिनके लिये f(x) के पास स्थानीय निम्निष्ठ का ऋणात्मक बिन्दु है :

(A) $(-\infty, 0)$ (B) $(1, \infty)$ (C) $(0, 1)$ (D) $(1, \infty) - \{2\}$

Ans. A

Sol. We have

$f(x) = 2x3 + 3\left(1-\frac{3a}{2}\right)x2 + 3(a2-a)x + b$

So, $f'(x) = 6x2 + 6\left(1-\frac{3a}{2}\right)x + 3(a2-a)$

Now, for f(x) to have negative point of local minimum, we must have both roots of the equation $f'(x) = 0$ real (unequal) and negative.

$\therefore$ Discriminant $> 0 \Rightarrow 36\left[\left(1-\frac{3a}{2}\right)^2 - 2(a^2-a)\right] > 0 \Rightarrow \left(\frac{a}{2}-1\right)^2 > 0 \Rightarrow a \in R - \{2\}$ .....(1)

Also, sum of roots $< 0 \Rightarrow \frac{-6\left(1-\frac{3a}{2}\right)}{6} < 0 \Rightarrow \frac{3a}{2} - 1 < 0 \Rightarrow a < \frac{2}{3}$ .....(2)

And product of roots $> 0 \Rightarrow \frac{3(a^2-a)}{6} > 0 \Rightarrow a(a-1) > 0 \Rightarrow a \in (-\infty, 0) \cup (1, \infty)$ .....(3)

$\therefore$ From (1) $\cap$ (2) $\cap$ (3), we get $a \in (-\infty, 0)$

28. The complete set of values of a for which vertex of parabola $y = g(x)$ has negative ordinate, is :

a के वास्तविक मानो का समुच्चय जिनके लिये, परवलय $y = g(x)$ का शीर्ष ऋणात्मक कोटि का है :

(A) $\left(\frac{1}{2},1\right)$ (B) $(0, \infty)$ (C) $R - \{2\}$ (D) $(1, \infty) - \{2\}$

Ans. C

Sol. We have

$$g(x) = \frac{f'(x)}{6} = x^2 + \left(1 - \frac{3a}{2}\right)x + \left(\frac{a^2 - a}{2}\right)$$

As discriminant of $g(x) = \left(\frac{a}{2} - 1\right)^2 > 0 \; \forall \; a \in R - \{2\}$

$\Rightarrow$ Ordinate of the vertex of parabola $y = g(x)$ lies below x-axis.

So, $a \in R - \{2\}$

29. The area bounded by $y = h(x)$ between $x = 0$ and $x = 2$, is :

$y = h(x)$, से $x = 0$ एवं $x = 2$ के मध्य का क्षेत्रफल है :

(A) $\frac{23}{3}$ (B) $\frac{20}{3}$ (C) $\frac{40}{3}$ (D) $\frac{32}{3}$

Ans. D

Sol. We have

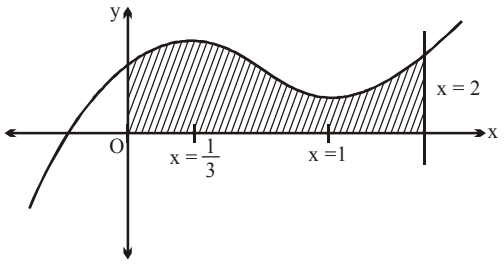
$h''(x) = 6x - 4 \Rightarrow h'(x) = 3x2 - 4x + c$

As $h'(1) = 0$, so $0 = 3 - 4 + c \Rightarrow c = 1$

$\therefore$ $h'(x) = 3x2 - 4x + 1 \Rightarrow h(x) = x3 - 2x2 + x + k$

As $h(1) = 5$, so $5 = 1 - 2 + 1 + k \Rightarrow k = 5$.

Hence, $h(x) = x3 - 2x2 + x + 5$



Clearly, required area

$$= \int_0^2 h(x)dx = \int_0^2 \left(x^3 - 2x^2 + x + 5\right)dx = \left(\frac{x^4}{4} - \frac{2x^3}{3} + \frac{x^2}{2} + 5x\right)_0^2 = \frac{32}{3} \text{ (square units).}$$

Paragraph for Question Nos. 30 to 32

Consider the functions defined implicitly by the equation $y^3 - 3y + x = 0$ on various intervals in the real line. If $x \in (-\infty, -2) \cup (2, \infty)$, the equation implicitly defines a unique real valued differentiable function $y = f(x)$. If $x \in (-2, 2)$, the equation implicitly defines a unique real valued differentiable function $y = g(x)$ satisfying $g(0) = 0$.

माना एक फलन अस्पष्ट रूप में समीकरण $y^3 - 3y + x = 0$ द्वारा वास्तविक संख्या रेखा के अन्तरालों के लिये परिभाषित है यदि $x \in (-\infty, -2) \cup (2, \infty)$ तब समीकरण एक अद्वितीय वास्तविक मान वाला अवकलनीय फलन $y = f(x)$ द्वारा परिभाषित है और यदि $x \in (-2, 2)$ तब समीकरण, अद्वितीय वास्तविक अवकलनीय फलन $y = g(x)$, $g(0) = 0$, द्वारा परिभाषित है।

30. If $f(-10\sqrt{2}) = 2\sqrt{2}$, then $f''(-10\sqrt{2}) =$

यदि $f(-10\sqrt{2}) = 2\sqrt{2}$, तो $f''(-10\sqrt{2}) =$

(A) $\frac{4\sqrt{2}}{7^3 3^2}$ (B) $-\frac{4\sqrt{2}}{7^3 3^2}$ (C) $\frac{4\sqrt{2}}{7^3 3}$ (D) $-\frac{4\sqrt{2}}{7^3 3}$

Ans. B

Sol. $y3 - 3y + x = 0$

$3y2y' - 3y' + 1 = 0 \qquad y' = \frac{-1}{3(y^2 - 1)}$

$f(-10\sqrt{2}) = 2\sqrt{2}$

$f'(-10\sqrt{2}) = -\frac{1}{3(7)} = -\frac{1}{21}$

$6y(y')2 + 3y2y'' - 3y'' = 0$

$y'' = -\frac{2y(y')^2}{y^2 - 1}$

$f''(-10\sqrt{2}) = \frac{-2(2\sqrt{2})}{441 \times 7} = \frac{-4\sqrt{2}}{7^3 3^2}$

31. The area of the region bounded by the curve $y = f(x)$, the x-axis, and the lines $x = a$ and $x = b$, where $-\infty < a < b < -2$, is :

वक्र $y = f(x)$, x-अक्ष और रेखाओं $x = a$ और $x = b$ जहाँ $-\infty < a < b < -2$, द्वारा परिबद्ध क्षेत्रफल है :

(A) $\int_a^b \frac{x}{3\left((f(x))^2 - 1\right)} dx + b f(b) - a f(a)$ (B) $-\int_a^b \frac{x}{3\left((f(x))^2 - 1\right)} dx + b f(b) - a f(a)$

(C) $\int_a^b \frac{x}{3\left((f(x))^2 - 1\right)} dx - b f(b) + a f(a)$ (D) $-\int_a^b \frac{x}{3\left((f(x))^2 - 1\right)} dx - b f(b) + a f(a)$

Ans. A

Sol. $\int_a^b f(x)dx = [xf(x)]ab - \int_a^b xf'(x)dx$

$= bf(b) - af(a) + \int_a^b \frac{x}{3\left[(f(x))^2 - 1\right]} dx$

$= \int_a^b \frac{x}{3\left[(f(x))^2 - 1\right]} dx + bf(b) - af(a)$

32. $\int_{-1}^{1} g'(x)\,dx =$

(A) $2g(-1)$ (B) 0 (C) $-2\,g(1)$ (D) $2\,g(1)$

Ans. D

Sol. $y^3 - 3y + x = 0$

when $x \in (-2, 2)$ $y = g(x)$ and $g(0) = 0$

$(g(x))^3 - 3g(x) + x = 0$

$(g(-x))^3 - 3g(-x) - x = 0$

$+ \quad + \quad +$

$(g(x))^3 + (g(-x))^3 - 3(g(x) + g(-x)) = 0$

$\{g(x) + g(-x)\}^2 \{(g(x)^2 + (g(-x))^2 - g(x).g(-x) - 3\} = 0$

$\Rightarrow g(x) + g(-x) = 0$

$\Rightarrow g(x) = -g(-x)$ _____(A)

$\therefore$ $g(x)$ is odd

Now $\int_{-1}^{1} g'(x)\,dx = g(x)\Big|_{-1}^{1}$

$= g(1) - g(-1)$ _____(1)

Put $x = 1$ in A

$g(1) = -g(-1)$

$g(-1) = -g(1)$

$\therefore$ Equation becomes $g(1) - (-g(1))$

$\Rightarrow$ $2g(1)$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

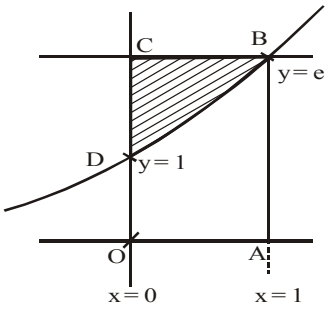
33. Area of the region bounded by the curve $y = e^x$ and lines $x = 0$ and $y = e$ is : **[JEE-2009]**

वक्र $y = e^x$ और रेखाओं $x = 0$ और $y = e$ द्वारा परिबद्ध क्षेत्रफल है : **[JEE-2009]**

(A) $e - 1$ (B) $\int_1^e \ln(e+1-y)dy$ (C) $e - \int_0^1 e^x dx$ (D) $\int_1^e \ln y \, dy$

Ans. BCD

Sol. $A = \int_1^e \ln y \, dy$



Apply

$= \int_1^e \ln(e+1-y)dy$

$A = ar(OABC) - ar(OABD)$

$= e - \int_1^e e^x dx$.

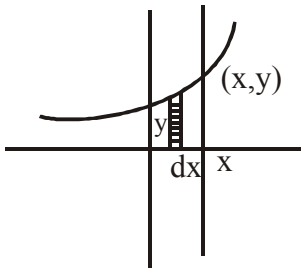
34. The area bounded by a curve, the axis of co-ordinates & the ordinate of some point of the curve is equal to the length of the corresponding arc of the curve . If the curve passes through the point $P(0, 1)$ then the equation of this curve can be :

वक्र, निर्देशी अक्षों तथा वक्र के किसी बिन्दु की कोटि द्वारा घिरा क्षेत्रफल, वक्र की संगत चाप की लम्बाई के बराबर है तथा वक्र बिन्दु $P(0, 1)$ से गुजरता है तब वक्र का समीकरण होगा :

(A) $y = \frac{1}{2}(e^x - e^{-x} + 2)$ (B) $y = \frac{1}{2}(e^x + e^{-x})$ (C) $y = 1$ (D) $y = \frac{2}{e^x + e^{-x}}$

Ans. BC

Sol.



$A = \int_0^x y dx = \int_0^x \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$

$$y^2 = 1 + \left(\frac{dy}{dx}\right)^2$$

Case-I $y \neq 1$ Case-II $y = 1$

$$\int \frac{dy}{\sqrt{y^2-1}} = \pm\int dx \qquad \because y \neq 1$$

$$\ln|y + \sqrt{y^2-1}| = x + C$$

$$(0,1) \Rightarrow C = 0$$

$$\ell n\left|y + \sqrt{y^2-1}\right| = x$$

$$y + \sqrt{y^2-1} > 0$$

$$y + \sqrt{y^2-1} = e^x$$

$$y - \sqrt{y^2-1} = e^{-x}$$

add $y = \frac{e^x + e^{-x}}{2}$

35. Let T be the triangle with vertices (0, 0), (0, c^2) and (c, c^2) and let R be the region between $y = cx$ and $y = x^2$ where $c > 0$ then :

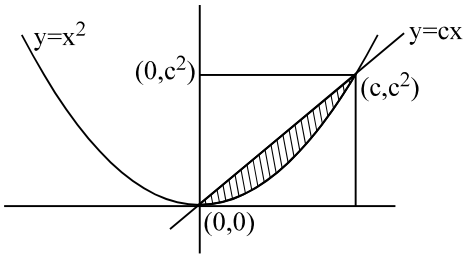
यदि T एक त्रिभुज है जिसके शीर्ष (0, 0), (0, c^2) तथा (c, c^2) है तथा यदि R, $y = cx$ एवं $y = x^2$ द्वारा घिरा क्षेत्र है तो : (जहाँ $c > 0$)

(A) Area (R) = $\frac{c^3}{6}$ (B) Area of R = $\frac{c^3}{3}$

(C) $\lim\limits_{c\to 0^+} \frac{\text{Area (T)}}{\text{Area (R)}} = 3$ (D) $\lim\limits_{c\to 0^+} \frac{\text{Area (T)}}{\text{Area (R)}} = \frac{3}{2}$

Ans. AC

Sol. Area (T) = $\frac{c \cdot c^2}{2} = \frac{c^3}{2}$



$$\text{Area (R)} = \frac{c^3}{2} - \int_0^c x^2\,dx$$

$$= \frac{c^3}{2} - \frac{c^3}{3} = \frac{c^3}{6}$$

$$\therefore \quad \lim_{c\to 0^+} \frac{\text{Area (T)}}{\text{Area (R)}} = \lim_{c\to 0^+} \frac{c^3}{2}\cdot\frac{6}{c^3} = 3$$

36. Suppose f is defined from $R \to [-1, 1]$ as $f(x) = \frac{x^2-1}{x^2+1}$ where R is the set of real number. Then the statement which does not hold is :

(A) f is many one onto

(B) f increases for $x > 0$ and decrease for $x < 0$

(C) minimum value is not attained even though f is bounded

(D) the area included by the curve $y = f(x)$ and the line $y = 1$ is π sq. units

माना f, $R \to [-1, 1]$ में परिभाषित है $f(x) = \frac{x^2-1}{x^2+1}$ जहाँ R वास्तविक सख्याएं है तब कौनसा कथन असत्य है :

(A) f बहुएकैकी आच्छादक है

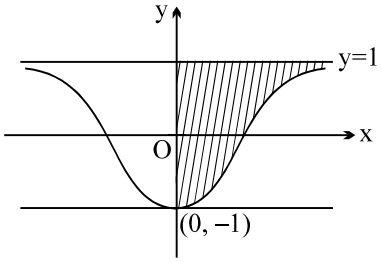
(B) f, $x > 0$ में वर्धमान तथा $x < 0$ में ह्रासमान

(C) यदि f परिबद्ध है तो भी न्यूनतम मान गृहण नहीं करता

(D) वक्र $y = f(x)$ तथा रेखा $y = 1$ द्वारा परिबद्ध क्षेत्रफल π वर्ग ईकाई है

Ans. ACD

Sol. $y = f(x) = \frac{x^2-1}{x^2+1} = 1 - \frac{2}{x^2+1}$



$$f'(x) = \frac{4x}{(x^2+1)^2}$$

$x > 0$, f is increasing and $x < 0$ f is decreasing $\Rightarrow$ (B) is true

range is $[-1, 1)$ $\Rightarrow$ into $\Rightarrow$ (A) is false;

minimum value occurs at $x = 0$ and $f(0) = -1$ $\Rightarrow$ (C) is false

$$A = 2\int_0^\infty \left(1 - \frac{x^2-1}{x^2+1}\right)dx = 4\int_0^\infty \frac{dx}{x^2+1} = 4\cdot\tan^{-1}\Big|_0^\infty = 4\cdot\frac{\pi}{2} = 2\pi \quad \Rightarrow \quad \text{(D) is false]}$$

MATCH THE COLUMN TYPE QUESTION(S)

37. Match the following column :

Column I	Column II
(A) The area between the curves $y = 2x^4 - x^2$, the x-axis and the	(P) 2 sin 1

Exercise J-M

1. Let $y = y(x)$ be the solution of the differential equation

$$x\tan\left(\frac{y}{x}\right)dy = \left(y\tan\left(\frac{y}{x}\right) - x\right)dx,\ -1 \le x \le 1, y\left(\frac{1}{2}\right) = \frac{\pi}{6}$$

Then the area of the region bounded by the curves $x = 0, x = \frac{1}{\sqrt{2}}$ and $y = y(x)$ in the upper half plane is :

[JEE(Main)July2021]

माना अवकल समीकरण

$$x\tan\left(\frac{y}{x}\right)dy = \left(y\tan\left(\frac{y}{x}\right) - x\right)dx,\ -1 \le x \le 1, y\left(\frac{1}{2}\right) = \frac{\pi}{6}$$

का हल $y = y(x)$ है। तो वक्रों $x = 0, x = \frac{1}{\sqrt{2}}$ तथा $y = y(x)$ द्वारा ऊपरी आधे निर्देशांक तल में घिरे क्षेत्र का क्षेत्रफल है :

[JEE(Main)July2021]

(A) $\frac{1}{8}(\pi - 1)$ (B) $\frac{1}{12}(\pi - 3)$ (C) $\frac{1}{4}(\pi - 2)$ (D) $\frac{1}{6}(\pi - 1)$

Ans. A

Sol. $\tan\left(\frac{y}{x}\right)\frac{dy}{dx} = \frac{y}{x}\tan\left(\frac{y}{x}\right) - 1$

Put $y = vx$

$$\frac{dy}{dx} = v + \frac{xdv}{dx}$$

$$(\tan v)\left(v + \frac{xdv}{dx}\right) = v\tan v - 1$$

$$v\tan v + \left(\frac{xdv}{dx}\right)\tan v = v\tan v - 1$$

$$\int \tan v\, dv = -\int \frac{dx}{v}$$

$$\ln(\sec v) = -\ln x + \ln c$$

$$\sec v = \frac{c}{x}$$

$$\sec\left(\frac{y}{x}\right) = \frac{c}{x}$$

$x = 1/2 \Rightarrow y = \pi/6$

$\Rightarrow c = 1$

$$\cos\left(\frac{y}{x}\right) = x$$

$$y = x\cos^{-1}(x)$$

$$\text{Area} = \int_0^{1/\sqrt{2}} x\cos^{-1}(x)\,dx$$

$$x = \cos\theta d\theta$$

$$dx = -\sin\theta\, d\theta$$

$$= \frac{1}{4}\int_{\pi/2}^{\pi/2} t\sin t\,dt$$

$$= \frac{1}{8}\left((-t\cos t)_{\pi/2} + \int_{\pi/2}^{\pi}\cos t dt\right)$$

$$= \frac{1}{8}\left((\pi) + (\sin t)_{\pi/2}^{\pi}\right)$$

$$= \frac{(\pi-1)}{8}$$

2. The area (in sq. units) of the region, given by the set $\{(x, y) \in R \times R \mid x \geq 0, 2x^2 \leq y \leq 4 - 2x\}$ is :

[JEE(Main)July2021]

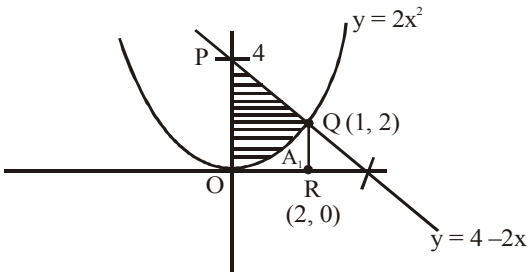
क्षेत्र $\{(x, y) \in R \times R \mid x \geq 0, 2x^2 \leq y \leq 4 - 2x\}$ का क्षेत्रफल (वर्ग इकाइयों में) बराबर है :

[JEE(Main)July2021]

(A) $\frac{13}{3}$ (B) $\frac{17}{3}$ (C) $\frac{7}{3}$ (D) $\frac{8}{3}$

Ans. C

Sol.



$$A_1 = \int_0^1 2x^2 dx = \left(\frac{2}{3}x^3\right)_0^1 = \frac{2}{3}$$

Req Area = Area of trapezium ORQP – A_1

$$= \frac{1}{2}(4+2)(1) - \frac{2}{3}$$

$$= 3 - \frac{2}{3} = \frac{7}{3}$$

3. If the area of the bounded region

$$R = \left\{ (x, y) : \max\{0, \log_e x\} \le y \le 2^x, \frac{1}{2} \le x \le 2 \right\}$$

is $\alpha(\log_e 2)^{-1} + \beta(\log_e 2) + \gamma$, then value of $(\alpha + \beta - 2\gamma)^2$ is equal to : **[JEE(Main)July2021]**

यदि परिबद्धित क्षेत्र

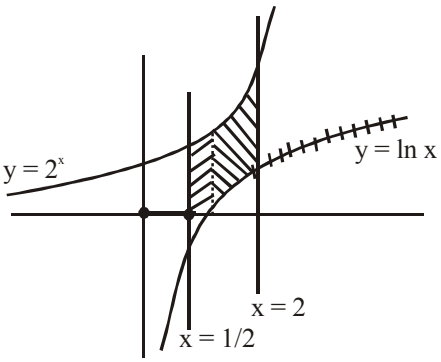
$$R = \left\{ (x, y) : \max\{0, \log_e x\} \le y \le 2^x, \frac{1}{2} \le x \le 2 \right\}$$

का क्षेत्रफल $\alpha(\log_e 2)^{-1} + \beta(\log_e 2) + \gamma$ है, तो $(\alpha + \beta - 2\gamma)^2$ का मान बराबर है : **[JEE(Main)July2021]**

(A) 4 (B) 8 (C) 1 (D) 2

Ans. D

Sol.



$$A = \int_{\frac{1}{2}}^{1} 2^x dx + \int_{1}^{2} \left(2^x - \ln x\right) dx$$

$$A = \left.\frac{2^x}{\ln 2}\right|_{\frac{1}{2}}^{2} + \left.\frac{2^x}{\ln 2} - (x \ln x - x)\right|_{1}^{2}$$

$$A = \left(4 - \sqrt{2}\right)\log_2 e - 2\log_2 e + 1$$

$$\alpha = 4 - \sqrt{2}, \beta = -2, \gamma = 1$$

$$(\alpha + \beta - 2\gamma)^2 = 2$$

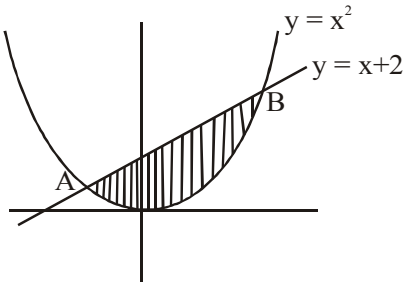
4. The area of the region bounded by $y - x = 2$ and $x^2 = y$ is equal to : **[JEE(Main)July2021]**

$y - x = 2$ तथा $x^2 = y$ द्वारा घिरे क्षेत्र का क्षेत्रफल बराबर है : **[JEE(Main)July2021]**

(A) $\frac{16}{3}$ (B) $\frac{4}{3}$ (C) $\frac{2}{3}$ (D) $\frac{9}{2}$

Ans. D

Sol.



For x co-ordinate of A and B

$x^2 = x + 2$

$\Rightarrow x^2 - x - 2 = 0$

$(x-2)(x+1) = 0$

$\therefore x = -1, 2$

Required Area :

$$= \int_{-1}^{2} \left((x+2) - x^2\right) dx$$

$$= \left[\frac{x^2}{2} + 2x - \frac{x^3}{3}\right]_{-1}^{2}$$

$$= \left[2 + 4 - \frac{8}{3}\right] - \left[\frac{1}{2} - 2 + \frac{1}{3}\right]$$

$$= \frac{10}{3} - \left[\frac{3 - 12 + 2}{6}\right]$$

$$= \frac{10}{3} - \left[\frac{3 - 12 + 2}{6}\right]$$

$$= \frac{9}{2}$$

5. Let T be the tangent to the ellipse E : $x^2 + 4y^2 = 5$ at the point P (1, 1). If the area of the region bounded by the tangent T, ellipse E, lines $x = 1$ and $x = \sqrt{5}$ is $\alpha\sqrt{5} + \beta + \gamma\cos^{-1}\left(\frac{1}{\sqrt{5}}\right)$, then $|\alpha + \beta + \gamma|$ is equal to _____.

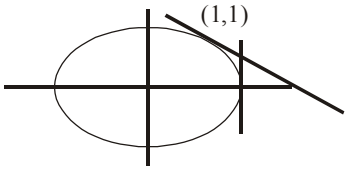
[JEE(Main)July2021]

माना दीर्घवृत E : $x^2 + 4y^2 = 5$ के बिंदु P (1, 1) पर स्पर्श रेखा T है। यदि स्पर्श रेखा T, दीर्घवृत E तथा रेखाओं x = 1 और $x = \sqrt{5}$ द्वारा घिरे क्षेत्र का क्षेत्रफल $\alpha\sqrt{5} + \beta + \gamma\cos^{-1}\left(\frac{1}{\sqrt{5}}\right)$ है, तो $|\alpha + \beta + \gamma|$ बराबर है _____

[JEE(Main)July2021]

Ans. 1.25

Sol.



E : $x^2 + 4y^2 - 5$

T : $x + 4y = 5$

$$\text{Area} = \int_1^{\sqrt{5}} (y_{\text{LINE}} - y_{\text{ELLIPSE}})\,dx$$

$$= \int_1^{\sqrt{5}} \left(\frac{5-x}{4} - \sqrt{\frac{5-x^2}{4}}\right) dx$$

$$= \frac{1}{4}(5x)_1^{\sqrt{5}} - \left(\frac{x^2}{8}\right)_1^{\sqrt{5}} - \frac{1}{4}\left(x\sqrt{5-x^2}\right)_1^{\sqrt{5}} - \frac{5}{4}\left(\sin^{-1}\left(\frac{x}{\sqrt{5}}\right)\right)_1^{\sqrt{5}}$$

$$= \frac{5\sqrt{5}}{4} - \frac{5}{4} - \frac{1}{2} - \frac{1}{4}(0-2) - \frac{5}{4}\left(\frac{\pi}{2} - \sin^{-1}\left(\frac{1}{\sqrt{5}}\right)\right)$$

$$= \frac{5\sqrt{5}}{4} - \frac{5}{4} - \frac{5}{4}\cos^{-1}\left(\frac{1}{\sqrt{5}}\right)$$

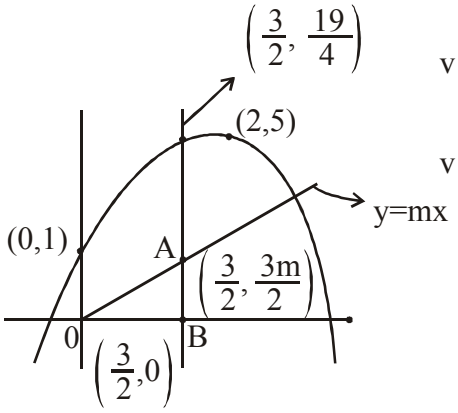
$$|\alpha + \beta + \gamma| = \frac{5}{4}$$

6. If the line y = mx bisects the area enclosed by the lines x = 0, y = 0, $x = \frac{3}{2}$ and the curve $y = 1 + 4x - x^2$, then 12 m is equal to_____. **[JEE(Main)Aug.2021]**

यदि रेखाओं x = 0, y = 0, $x = \frac{3}{2}$ तथा वक्र $y = 1 + 4x - x^2$ से घिरे क्षेत्र के क्षेत्रफल को रेखा y = mx समद्विभाजित करती है, तो 12 m बराबर है _____ **[JEE(Main)Aug.2021]**

Ans. 26

Sol.



$$\text{Total Area} = \int_0^{3/2} (1 + 4x - x^2)dx$$

$$= x + 2x^2 - \frac{x^3}{3}\Big|_0^{3/2}$$

$$= \frac{3}{2} + 2\left(\frac{9}{4}\right) - \frac{1}{3}\left(\frac{27}{8}\right)$$

$$= \frac{3}{2} + \frac{9}{2} - \frac{9}{8}$$

$$= \frac{12 + 36 - 9}{8} = \frac{39}{8}$$

$$\text{Area of } \Delta OAB = \frac{39}{16}$$

$$\frac{39}{16} = \frac{1}{2} \times \frac{3}{2} \times \frac{3m}{2}$$

$$\frac{13}{2} = 3m$$

$$12m = \frac{13}{2} \times 4 = 26$$

7. The area (in sq. units) of the region bounded by the curves $x^2 + 2y - 1 = 0$, $y^2 + 4x - 4 = 0$ and $y^2 - 4x - 4 = 0$, in the upper half plane is _____. **[JEE(Main)July2021]**

ऊपरी आधे निर्देशांक तल में वक्रों, $x^2 + 2y - 1 = 0$, $y^2 + 4x - 4 = 0$ तथा $y^2 - 4x - 4 = 0$ द्वारा परिबद्ध क्षेत्र का क्षेत्रफल (वर्ग इकाइयों में) बराबर है _____ **[JEE(Main)July2021]**

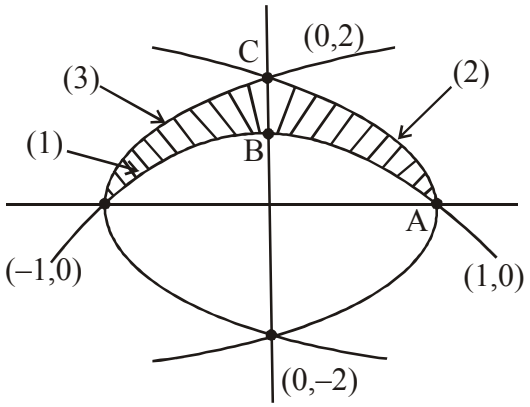
Ans. 2

Sol. $x^2 = -2y + 1$

$x^2 = -2(y - \frac{1}{2})$ .....(1)

$y^2 = -4(x - 1)$ .....(2)

$y^2 = 4(x + 1)$ .....(3)



Required area = 2 (area ABC)

$$= 2\int_0^1 \left(2\sqrt{1-x} - \frac{(1-x^2)}{2} \right) dx$$

= 2 square unit.

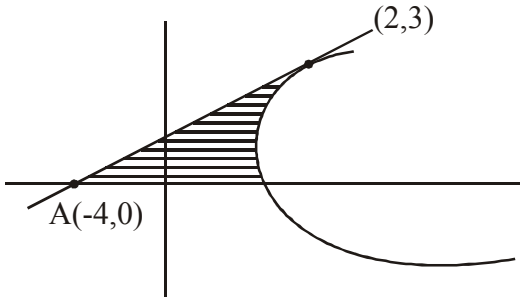
8. The area of the region bounded by the parabola $(y-2)^2 = (x-1)$, the tangent to it at the point whose ordinate is 3 and the x-axis is : **[JEE(Main)Aug.2021]**

परवलय $(y-2)^2 = (x-1)$, इसके उस बिन्दु जिसकी कोटि 3 है पर स्पर्श रेखा तथा x-अक्ष द्वारा परिबद्ध क्षेत्र का क्षेत्रफल है : **[JEE(Main)Aug.2021]**

(A) 6 (B) 4 (C) 9 (D) 10

Ans. C

Sol. $(y-2)^2 = (x-1)$



slope of tangent

$$2(y-2)\frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{2}$$

tangent $y - 3 = \frac{1}{2}(x-2)$

$2y - 6 = x - 2$

$x = 2y - 4$

Required area $= \int_0^3 \left[(y-2)^2 + 1 - (2y-4) \right] dy$

$$= \int_0^3 (y-2)^2 - 2y + 5 \, dy$$

$$= \frac{(y-2)^3}{3} - y^2 + 5y \Big]_0^3$$

$= \frac{1}{3} - 9 + 15 + \frac{8}{3}$

$= 6 + 3$

$= 9$ square unit

9. The area, enclosed by the curves $y = \sin x + \cos x$ and $y = |\cos x - \sin x|$ and the lines $x = 0,\ x = \frac{\pi}{2}$, is :

[JEE(Main)Aug.2021]

वक्रों $y = \sin x + \cos x$ एवं $y = |\cos x - \sin x|$ तथा रेखाओं $x = 0,\ x = \frac{\pi}{2}$ से घिरे क्षेत्र का क्षेत्रफल है :

[JEE(Main)Aug.2021]

(A) $2\sqrt{2}\left(\sqrt{2}+1\right)$ (B) $2\left(\sqrt{2}+1\right)$ (C) $2\sqrt{2}\left(\sqrt{2}-1\right)$ (D) $4\left(\sqrt{2}-1\right)$

Ans. C

Sol. $y = \sin x + \cos x$

$y = |\cos x - \sin x|$

For $x \in \left(0, \frac{\pi}{2}\right), (\sin x + \cos x) > |\cos x - \sin x|$

Required Area

$$= \int_0^{\frac{\pi}{2}} (\sin x + \cos x - |\cos x - \sin x|)dx$$

$$= \int_0^{\frac{\pi}{2}} (\sin x + \cos x - (\cos x - \sin x|)dx$$

$$+ \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (\sin x + \cos x - (\sin x - \cos x|)dx$$

$$= \int_0^{\frac{\pi}{4}} (2\sin x)dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (2\cos x)dx$$

$$= 2\left[\left[-\cos x\right]_0^{\frac{\pi}{4}}\right] + 2\left[\left[\sin x\right]_{\frac{\pi}{4}}^{\frac{\pi}{2}}\right]$$

$$= 2\left[\left[-\frac{1}{\sqrt{2}}+1\right]\right]+2\left[1-\frac{1}{\sqrt{2}}\right]$$

$$= -\sqrt{2}+2+2-\sqrt{2}$$

$$= 4-2\sqrt{2}$$

$$= 2(2-\sqrt{2})$$

$$= 2\sqrt{2}(\sqrt{2}-1)$$

10. The area of the region $S=\{(x,y):3x^2 \le 4y \le 6x+24\}$ is _____. **[JEE(Main)Aug.2021]**

क्षेत्र $S=\{(x,y):3x^2 \le 4y \le 6x+24\}$ का क्षेत्रफल है_____ **[JEE(Main)Aug.2021]**

Ans. 27

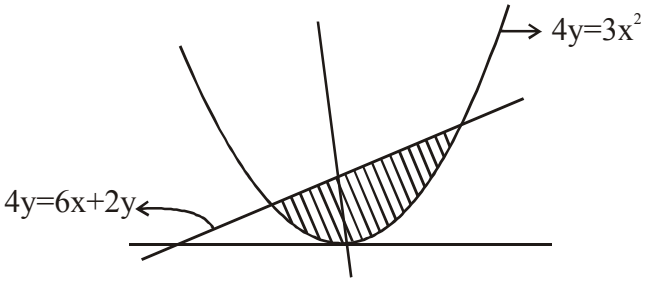
Sol. POM of intersection

$3x^2 = 6x + 24$

$\Rightarrow x = 4, x = -2$

$$\text{Area} = \int_{-2}^{4}\left(\frac{6x+24}{4}-\frac{3x^2}{4}\right)dx$$

$= 27$



11. Let a and b respectively be the points of local maximum and local minimum of the function $f(x) = 2x^3 - 3x^2 - 12x$. If A is the total area of the region bounded by $y = f(x)$, the x-axis and lines $x = a$ and $x = b$, then 4A is equal to _____. **[JEE(Main)Aug.2021]**

माना फलन $f(x) = 2x^3 - 3x^2 - 12x$ के स्थानीय उच्चतम तथा स्थानीय निम्नतम बिन्दु क्रमशः a तथा b है। यदि $y = f(x)$, x-अक्ष, तथा j§ kv ls x = a और x =b से घिरे क्षेत्र का क्षेत्रफल A है, तो 4A बराबर है_____ **[JEE(Main)Aug.2021]**

Ans. 114

Sol. $f(x) = 2x^3 - 3x^2 - 12x$

$f'(x) = 6x^2 - 6x - 12$

$= 6(x+1)(x-2)$

$x = -1, 2$

$$A = \int_{-1}^{0} \left(2x^3 - 3x^2 - 12x\right)dx - \int_{0}^{2} \left(2x^3 - 3x^2 - 12x\right)dx$$

$$A = \left|\frac{x^4}{2} - x^3 - 12\right)_{-1}^{0} - \left(\frac{x^4}{2} - x^3 - 12\right)_{0}^{2}$$

$$A = \frac{57}{2} \Rightarrow 4A = 114$$

12. The area of the region enclosed between the parabolas $y^2 = 2x - 1$ and $y^2 = 4x - 3$ is :

[JEE(Main)June2022]

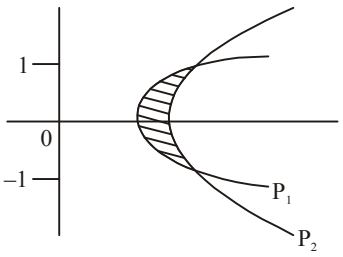
परवलयों $y^2 = 2x - 1$ तथा $y^2 = 4x - 3$ के बीच घिरे क्षेत्र का क्षेत्रफल है : **[JEE(Main)June2022]**

(A) $\frac{1}{3}$ (B) $\frac{1}{6}$ (C) $\frac{2}{3}$ (D) $\frac{3}{4}$

Ans. A

Sol. $P_1 : y^2 = 2x - 1$

$P_2 : y^2 = 4x - 3$



$$A = 2.\int_{0}^{1} \left(x_{P_2} - x_{P_1}\right)dy$$

$$A = 2\int_{0}^{1} \left(\frac{y^2+3}{4}\right) - \left(\frac{y^2+1}{2}\right)dy$$

$$A = \frac{1}{3}$$

13. The area bounded by the curve $y = |x^2 - 9|$ and the line $y = 3$ is : **[JEE(Main)June2022]**

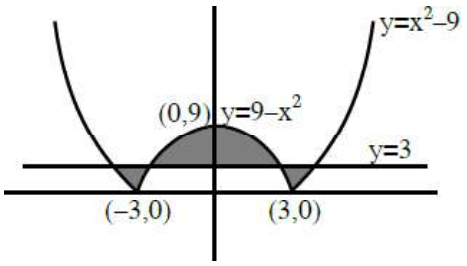
वक्र $y = |x^2 - 9|$ तथा रेखा $y = 3$ द्वारा घिरे क्षेत्र का क्षेत्रफल है : **[JEE(Main)June2022]**

(A) $4\left(2\sqrt{3} + \sqrt{6} - 4\right)$ (B) $4\left(4\sqrt{3} + \sqrt{6} - 4\right)$

(C) $8\left(4\sqrt{3} + 3\sqrt{6} - 9\right)$ (D) $8\left(4\sqrt{3} + 2\sqrt{6} - 9\right)$

Ans. D

Sol.



Area of shaded region

$$= 2\int_0^3 \left(\sqrt{9+y} - \sqrt{9-y}\right)dy + 2\int_3^9 \left(\sqrt{9-y}\right)dy$$

$$= 2\left[\int_0^3 (9+y)^{1/2} - \int_0^3 (9-y)^{1/2}\,dy + \int_3^9 \left(\sqrt{9-y}\right)^{1/2} dy\right]$$

$$= 2\left[\frac{2}{3}\left[(9+y)^{3/2}\right]_0^3 + \frac{2}{3}\left[(9-y)^{3/2}\right]_0^3 - \frac{2}{3}\left[(9-y)^{3/2}\right]_3^9\right]$$

$$= \frac{4}{3}\left[12\sqrt{12} - 27 + 6\sqrt{6} - 27 - \left(0 - 6\sqrt{6}\right)\right]$$

$$= \frac{4}{3}\left[24\sqrt{3} + 12\sqrt{6} - 54\right]$$

$$= 8\left(4\sqrt{3} + 2\sqrt{6} - 9\right)$$

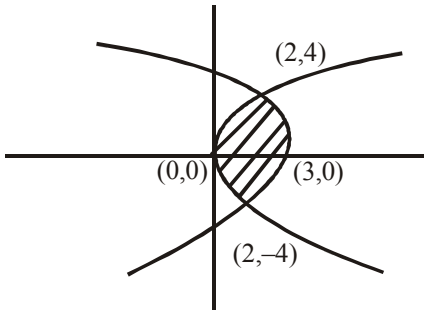
14. The area of the region bounded by $y^2 = 8x$ and $y^2 = 16(3 - x)$ is equal to : **[JEE(Main)June2022]**

वक्रों $y^2 = 8x$ तथा $y^2 = 16(3 - x)$ द्वारा घिरे क्षेत्र का क्षेत्रफल है : **[JEE(Main)June2022]**

(A) $\frac{32}{3}$ (B) $\frac{40}{3}$ (C) 16 (D) 19

Ans. C

Sol.



$$8x = 16(3 - x)$$

$x = 6 - 2x$

$3x = 6 \Rightarrow x = 2$

$y^2 = 16 \Rightarrow y = \pm 4$

$x = y^2/8,$

$x = \dfrac{48 - y^2}{16}$

$\text{Area} = 2\int_0^4 \left(\left(\dfrac{48 - y^2}{16} \right) - \left(\dfrac{y^2}{8} \right) \right) dy = 16$

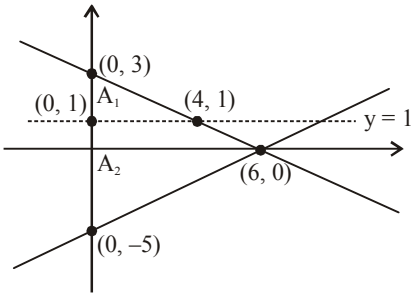
15. If a point A (x,y) lies in the region bounded by the y-axis, straight lines $2y + x = 6$ and $5x - 6y = 30$, then the probability that $y < 1$ is : **[JEE(Main)June2022]**

यदि y-अक्ष, सरल रेखाओं $2y + x = 6$ तथा $5x - 6y = 30$ से घिरे (bounded) क्षेत्र में एक बिन्दु A(x, y) है, तो $y < 1$ होने की i kf; d r kgS **[JEE(Main)June2022]**

(A) $\dfrac{1}{6}$ (B) $\dfrac{5}{6}$ (C) $\dfrac{2}{3}$ (D) $\dfrac{6}{7}$

Ans. B

Sol.



$P(y < 1) = \dfrac{A_2}{A_1 + A_2}$

$= 1 - \dfrac{A_1}{A_1 + A_2}$

$= 1 - \dfrac{(2)(4)}{6 \times 8}$

$1 - \dfrac{1}{6} \quad = \dfrac{5}{6}$

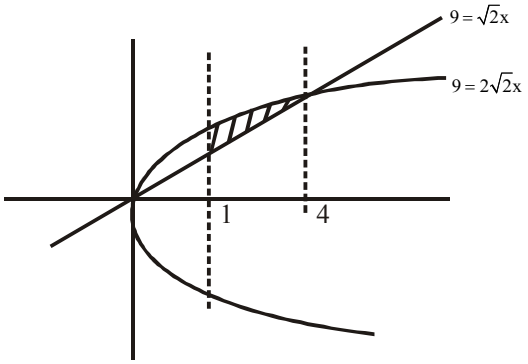
16. The area of the region $S = \left\{ (x, y) : y^2 \le 8x, y \ge \sqrt{2}x, x \ge 1 \right\}$ is : **[JEE(Main)June2022]**

क्षेत्र $S = \left\{ (x, y) : y^2 \le 8x, y \ge \sqrt{2}x, x \ge 1 \right\}$ का क्षेत्रफल है : **[JEE(Main)June2022]**

(A) $\frac{13\sqrt{2}}{6}$ (B) $\frac{11\sqrt{2}}{6}$ (C) $\frac{5\sqrt{2}}{6}$ (D) $\frac{19\sqrt{2}}{6}$

Ans. B

Sol. Required area



$$= \int_1^4 \left(2\sqrt{2x} - \sqrt{2}x\right)dx = \sqrt{2}\left[\frac{4.x^{3/2}}{3} - \frac{x^2}{2}\right]_1^4$$

$$= = \sqrt{2}\left[\frac{32}{3} - 8 - \frac{4}{3} + \frac{1}{2}\right]$$

$$= = \sqrt{2}\left[\frac{64 - 48 - 8 + 3}{6}\right] = \frac{11\sqrt{2}}{6}$$

17. The area of the bounded region enclosed by the curve $y = 3 - \left|x - \frac{1}{2}\right| - |x+1|$ and the x-axis is :

[JEE(Main)June2022]

वक्र $y = 3 - \left|x - \frac{1}{2}\right| - |x+1|$ तथा x-अक्ष से घिरे क्षेत्र का क्षेत्रफल है : **[JEE(Main)June2022]**

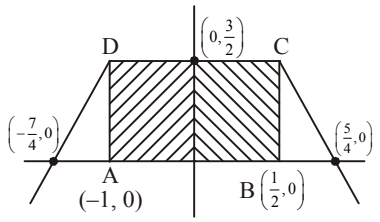
(A) $\frac{4}{9}$ (B) $\frac{45}{16}$ (C) $\frac{27}{8}$ (D) $\frac{63}{16}$

Ans. C

Sol. $y = 3 - \left|x - \frac{1}{2}\right| - |x+1|$

$$y = \begin{cases} 2x - \frac{7}{2} & x < -1 \\ \frac{3}{2} & -1 \le x \le \frac{1}{2} \\ \frac{5}{2} - 2x & x > \frac{1}{2} \end{cases}$$

graph of y.
required area is area of trapezium ABCD



$= \frac{1}{2}$ [sum of parallel sides] × distance between parallel sides

$= \frac{1}{2}\left(3+\frac{3}{2}\right).\frac{3}{2} = \frac{27}{8}$ sq. units

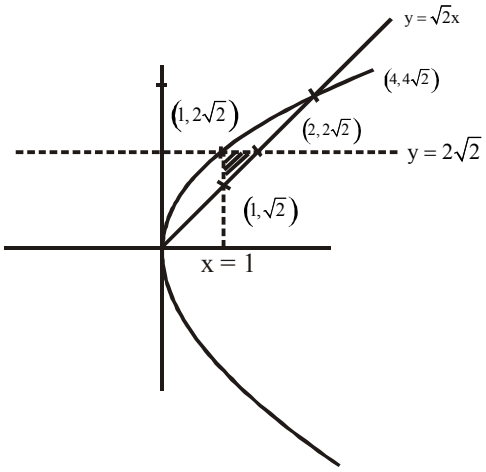
18. The area enclosed by $y^2 = 8x$ and $y = \sqrt{2}x$ that lies outside the triangle formed by $y = \sqrt{2}x$, $x = 1$, $y = 2\sqrt{2}$, is equal to : **[JEE(Main)June2022]**

वक्रों $y^2 = 8x$ तथा $y = \sqrt{2}x$ से घिरे तथा रेखाओं $y = \sqrt{2}x$, $x = 1$, $y = 2\sqrt{2}$ से बने त्रिभुज के बाहर के क्षेत्र का क्षेत्रफल है : **[JEE(Main)June2022]**

(A) $\frac{16\sqrt{2}}{6}$ (B) $\frac{11\sqrt{2}}{6}$ (C) $\frac{13\sqrt{2}}{6}$ (D) $\frac{5\sqrt{2}}{6}$

Ans. C

Sol.



Area of triangle $= \frac{1}{2} \times \sqrt{2} \times 1 = \frac{1}{\sqrt{2}}$

req. area $= \int_0^4 \left(2\sqrt{2}\sqrt{x} - \sqrt{2}x\right) dx - \frac{1}{\sqrt{2}}$

$$= \left(2\sqrt{2}\frac{x^{3/2}}{\frac{3}{2}} - \sqrt{2}\frac{x^2}{2} \right)_0^4 - \frac{1}{\sqrt{2}}$$

$$= \frac{4\sqrt{2}(4)^{3/2}}{3} - \frac{16\sqrt{2}}{2} - \frac{1}{\sqrt{2}}$$

$$= \frac{32\sqrt{2}}{3} - \frac{16\sqrt{2}+\sqrt{2}}{2}$$

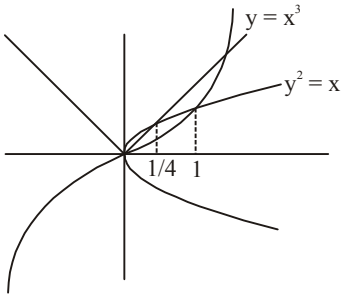
$$= \frac{64\sqrt{2}-51\sqrt{2}}{6} = \frac{13\sqrt{2}}{6} \text{ Ans.}$$

19. Let S be the region bounded by the curves $y = x^3$ and $y^2 = x$. The curve $y = 2|x|$ divides S into two regions of areas R_1 and R_2. If max $\{R_1, R_2\} = R_2$, then $\frac{R_2}{R_1}$ is equal to _____. **[JEE(Main)June2022]**

माना वक्रों $y = x^3$ तथा $y^2 = x$ द्वारा घिरा क्षेत्र S है। वक्र $y = 2|x|$, क्षेत्र S को R_1 तथा R_2 क्षेत्रफल के दो क्षेत्रों में बाँटता है। यदि max $\{R_1, R_2\} = R_2$ है, तो $\frac{R_2}{R_1}$ बराबर है _____ **[JEE(Main)June2022]**

Ans. 19

Sol.



$C_1 : y = x^3$
$C_2 : y^2 = x$ and
C_1 and C_2 intersect at (1, 1)

C_2 and C_3 intersect at $\left(\frac{1}{4}, \frac{1}{2}\right)$

Clearly $R_1 = \int_0^{1/4} \left(\sqrt{x} - 2x\right) dx = \frac{2}{3}\left(\frac{1}{8}\right) - \frac{1}{16} = \frac{1}{48}$ and $R_1 + R_2 = \int_0^1 \left(\sqrt{x} - x^3\right) dx = \frac{2}{3} - \frac{1}{4} = \frac{5}{12}$

so, $\frac{R_1 + R_2}{R_1} = \frac{5/12}{1/48} \Rightarrow 1 + \frac{R_2}{R_1} = 20$

$$\Rightarrow \frac{R_2}{R_1} = 19$$

20. The area (in sq. units) of the region enclosed between the parabola $y^2 = 2x$ and the line $x + y = 4$ is _____.

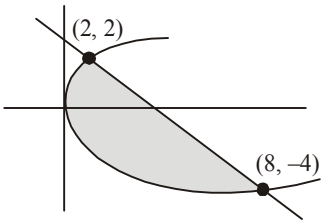
[JEE(Main)June2022]

परवलय $y^2 = 2x$ तथा रेखा $x + y = 4$ से घिरे क्षेत्र का क्षेत्रफल (वर्ग इकाईयों में) है _____

[JEE(Main)June2022]

Ans. 18

Sol.



$$\text{Required area} = \int_{-4}^{2} (x \text{ line} - x \text{ parabola})\, dy$$

$$= \int_{-4}^{2} \left(4 - y - \frac{y^2}{2}\right) dy$$

$$= 4y - \frac{y^2}{2} - \frac{y^3}{6}\Bigg]_{-4}^{2}$$

$$= 18 \text{ square units}$$

21. Let $f(x) = \max\{|x+1|, |x+2|, ..., |x+5|\}$. Then $\int_{-6}^{0} f(x)dx$ is equal to _____.

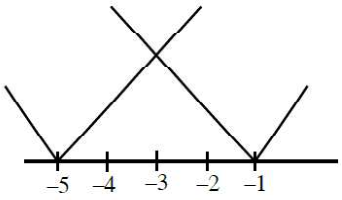
[JEE(Main)June2022]

माना $f(x) = \max\{|x+1|, |x+2|, ..., |x+5|\}$ है। तो $\int_{-6}^{0} f(x)dx$ बराबर है _____

[JEE(Main)June2022]

Ans. 21

Sol. $f(x) = \max\{|x+1|, |x+2|, |x+3|, |x+4|, |x+5|\}$



$$\int_{-6}^{0} f(x)dx = \int_{-6}^{-3} |x+1|dx + \int_{-3}^{0} |x+5|dx$$

$$= -\int_{-6}^{-3} (x+1)dx + \int_{-3}^{0} (x+5)dx$$

$$= -\left[\frac{x^2}{2}+x\right]_{-6}^{-3} + \left[\frac{x^2}{2}+5x\right]_{-3}^{0}$$

$$= -\left[\left(\frac{9}{2}-3\right)-(18-6)\right] + \left[0-\left(\frac{9}{2}-15\right)\right]$$

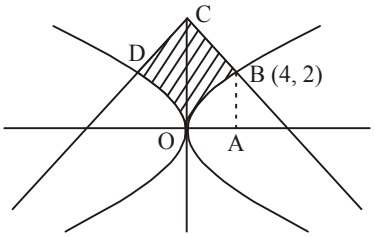
$$= -\left[\frac{3}{2}-12\right] + \frac{21}{2} = \frac{21}{2}+\frac{21}{2} = 21$$

22. Let $A_1 = \{(x,y) : |x| \le y^2, |x|+2y \le 8\}$ and $A_2 = \{(x,y) : |x|+|y| \le k\}$. If 27 (Area A_1) = 5 (Area A_2), then k is equal to _____. **[JEE(Main)June2022]**

माना $A_1 = \{(x,y) : |x| \le y^2, |x|+2y \le 8\}$ तथा $A_2 = \{(x,y) : |x|+|y| \le k\}$ हैं। यदि 27 (क्षेत्रफल A_1) = 5 (क्षेत्रफल A_2) है, तो k बराबर है _____ **[JEE(Main)June2022]**

Ans. 6

Sol. $A_1 = \{(x, y) : |x| \le y^2, |x| + 2y \le 8\}$



for point B $y^2 = x$ & $x + 2y = 8$

$\left(\frac{8-x}{2}\right)^2 = x$ $y = \frac{8-x}{2}$

$x^2 - 16x + 64 = 4x$

$x^2 - 20x + 64 = 0$

$x = 4, y = 2$

$$A_1 = 2\int_0^4 \left(\left(\frac{8-x}{2}\right) - \sqrt{x}\right)dx$$

$$A_1 = 2\left[\frac{8x}{2} - \frac{X^2}{4} - \frac{2x^{\frac{3}{2}}}{3}\right]_0^4$$

$$A_1 = \frac{40}{3}$$

$A_2 = \{(x, y) : |x| + |y| \le k\}$

$A_2 = 4(\text{Area of } \Delta OAB)$

$$= 4 \times \frac{1}{2} k \times k$$

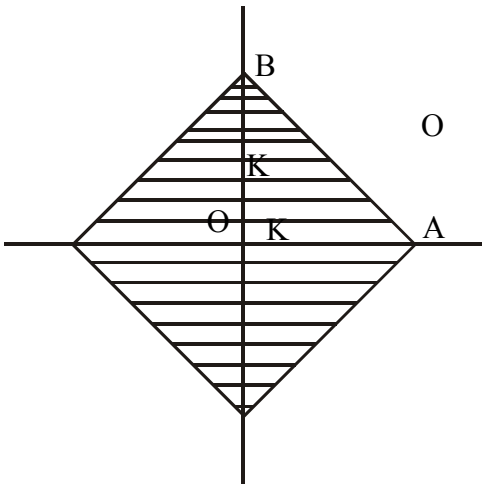
$= 2k^2$

$27(A_1) = 5(A_2)$

$$27 \times \frac{40}{3} = 5 \times 2k^2$$

$k^2 = 36$

$k = 6$



23. If the area of region $\left\{(x, y) : x^{\frac{2}{3}} + y^{\frac{2}{3}} \le 1, x + y \ge 0, y \ge 0\right\}$ is A, then $\frac{256A}{\pi}$ is equal to _____.

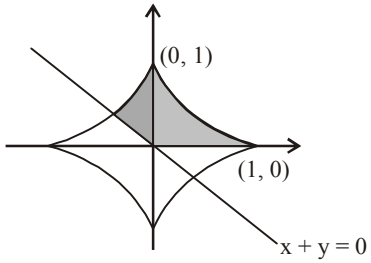
[JEE(Main)June2022]

यदि क्षेत्र $\left\{(x, y) : x^{\frac{2}{3}} + y^{\frac{2}{3}} \le 1, x + y \ge 0, y \ge 0\right\}$ का क्षेत्रफल A है, तो $\frac{256A}{\pi}$ बराबर है _____

[JEE(Main)June2022]

Ans. 36

Sol.



$$A = \frac{3}{2}\int_0^1 \left(1 - x^{\frac{2}{3}}\right)^{\frac{3}{2}} dx$$

put $x = \sin^3\theta$

$$A = \frac{3}{2}\int_0^{\frac{\pi}{2}} \left(1 - \sin^2\theta\right)^{\frac{3}{2}} .3\sin^2\theta\cos\theta\, d\theta$$

$$A = \frac{3}{2}\int_0^{\frac{\pi}{2}} 3\sin^2\theta\cos^4\theta\, d\theta$$

$$A = \frac{9}{2}\times\frac{(1.3)(1)}{(6)(4)(2)}.\frac{\pi}{2}$$

$$A = \frac{9\pi}{64}$$

$$\Rightarrow \frac{64A}{\pi} = 9$$

$$\Rightarrow \frac{256A}{\pi} = 36$$

24. For real numbers a, b (a > b > 0), let Area $\left\{(x,y): x^2+y^2 \le a^2 \ \& \ \frac{x^2}{a^2}+\frac{y^2}{b^2} \ge 1\right\} = 30\pi$ and

Area $\left\{(x,y): x^2+y^2 \ge b^2 \ \& \ \frac{x^2}{a^2}+\frac{y^2}{b^2} \le 1\right\} = 18\pi$. Then the value of $(a-b)^2$ is equal to _____.

[JEE(Main)June2022]

वास्तविक संख्याओं a, b (a > b > 0) के लिए यदि क्षेत्रफल $\left\{(x,y): x^2+y^2 \le a^2 \ \& \ \frac{x^2}{a^2}+\frac{y^2}{b^2} \ge 1\right\} = 30\pi$ तथा

{ksQy $\left\{(x,y): x^2+y^2 \ge b^2 \ \& \ \frac{x^2}{a^2}+\frac{y^2}{b^2} \le 1\right\} = 18\pi$ हैं, तो $(a-b)^2$ का मान बराबर है _____

[JEE(Main)June2022]

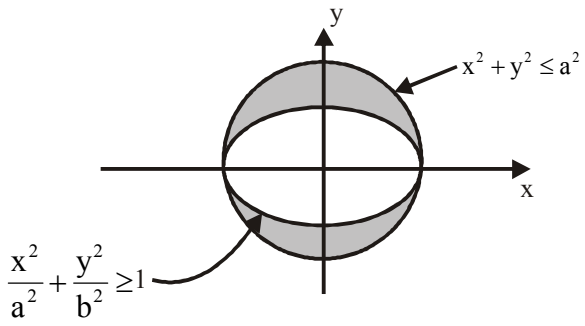
Ans. 12

Sol. Given a,b ∈ R or (a > b > 0)

$$\text{Area}\left\{(x,y): x^2+y^2 \le a^2 \text{and } \frac{x^2}{a^2}+\frac{y^2}{b^2} \le 1\right\} = 30\pi$$

$$\text{Area}\left\{(x,y): x^2+y^2 \le b^2 \text{and } \frac{x^2}{a^2}+\frac{y^2}{b^2} \le 1\right\} = 18\pi$$

Now $x^2+y^2 \le a^2 \ \ \& \ \frac{x^2}{a^2}+\frac{y^2}{b^2} \ge 1$



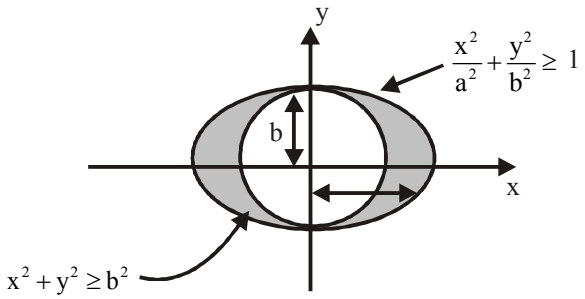
So, shaded region = area of circle - area of ellipse

$\Rightarrow \pi a^2 - \pi ab = 30\pi$

$\Rightarrow \pi a(a-b) = 30\pi$

$\Rightarrow a(a-b) = 30\pi$ .....(1)

Similarly $x^2 + y^2 \ge b^2$ & $\frac{x^2}{a^2} + \frac{y^2}{b^2} \le 1$



So, shaded region = area of ellipse - area of circle

$\Rightarrow \pi ab - \pi b^2 = 18\pi$

$\Rightarrow \pi b(a-b) = 18\pi$

$\Rightarrow b(a-b) = 18\pi$ .....(2)

From (1) – (2)

(a – b) (a – b) = 30 – 18

$(a-b)^2 = 12$

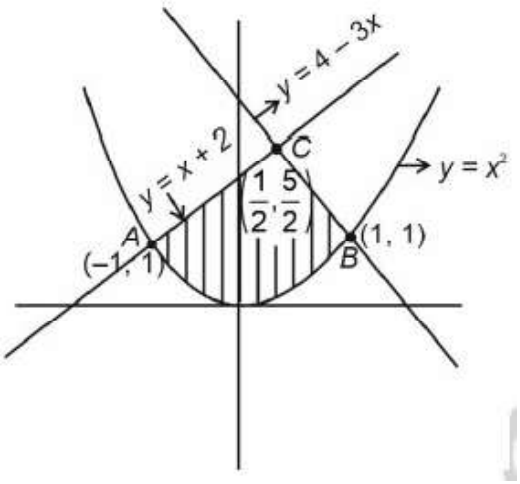
25. The area of the region given by $A = \{(x, y) : x^2 \le y \le \min\{x + 2, 4 - 3x\}$ is : **[JEE(Main)July2022]**

$A = \{(x, y) : x^2 \le y \le \min\{x + 2, 4 - 3x\}$ द्वारा दिए गए क्षेत्र का क्षेत्रफल है : **[JEE(Main)July2022]**

(A) $\frac{31}{8}$ (B) $\frac{17}{6}$ (C) $\frac{19}{6}$ (D) $\frac{27}{8}$

Ans. B

Sol. $A=\left\{(x,y): x^2 \le y \le \min\{x+2, 4-3x\}\right\}$



So area of requierd region

$$A=\int_{-1}^{\frac{1}{2}}\left(x+2-x^2\right)dx+\int_{\frac{1}{2}}^{1}\left(4-3x-x^2\right)dx$$

$$=\left[\frac{x^2}{2}+2x-\frac{x^3}{3}\right]_{-1}^{\frac{1}{2}}+\left[4x-\frac{3x^2}{2}-\frac{x^3}{3}\right]_{\frac{1}{2}}^{1}$$

$$=\left(\frac{1}{8}+1-\frac{1}{24}\right)-\frac{1}{2}-2+\frac{1}{3}+\left(1-\frac{3}{2}-\frac{1}{3}\right)-\left(2-\frac{3}{8}-\frac{1}{24}\right)$$

$$=\frac{17}{6}$$

26. The tangents at the points A(1, 3) and B(1, –1) on the parabola $y^2 - 2x - 2y = 1$ meet at the point P. Then the area (in unit^2) of the triangle PAB is : **[JEE(Main)July2022]**

माना परवलय $y^2 - 2x - 2y = 1$ के बिन्दुओं A(1, 3) तथा B(1, –1) पर स्पर्श रेखाएँ बिन्दु P पर मिलती हैं। तो त्रिभुज PAB का {kkQy (वर्ग इकाई में) है : **[JEE(Main)July2022]**

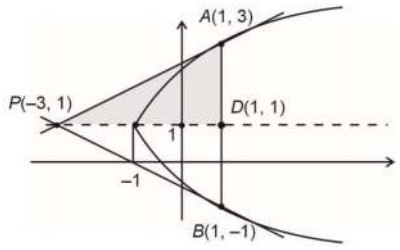
(A) 4 (B) 6 (C) 7 (D) 8

Ans. D

Sol. Given curve : y2 – 2x – 2y = 1.

Can be written as

$(y-1)^2 = 2(x+1)$



And, the given information

Can be plotted as shown in figure

Tangent at A : $2y - x - 5 = 0$ {using $T = 0$}

Intersection with $y = 1$ is $x = -3$

Hence, point P is (–3, 1)

Taking advantage of symmetry

Area of $\Delta PAB = 2 \times \frac{1}{2} \times (1 - (-3)) \times (3 - 1)$

= 8 sq. units

27. The odd natural number a, such that the area of the region bounded by $y = 1$, $y = 3$, $x = 0$, $x = y^a$ is $\frac{364}{3}$, is equal to : **[JEE(Main)July2022]**

विषम पूर्णांक संख्या a, जिसके लिए $y = 1$, $y = 3$, $x = 0$, $x = y^a$ से घिरे क्षेत्र का क्षेत्रफल $\frac{364}{3}$ है, बराबर है:

[JEE(Main)July2022]

(A) 3 (B) 5 (C) 7 (D) 9

Ans. B

Sol. a is a odd natural number and

$$\left|\int_1^3 y^a dy\right| = \frac{364}{3}$$

$$\Rightarrow \left|\frac{1}{a+1}\left(y^{a+1}\right)_1^3\right| = \frac{364}{3}$$

$$\Rightarrow \frac{3^{a+1} - 1}{a+1} = \pm\frac{364}{3}$$

Solving with (–) sign,

$\frac{3^{a+1} - 1}{a+1} = \frac{-364}{3}$, No a exist

$\therefore$ (a = 5)

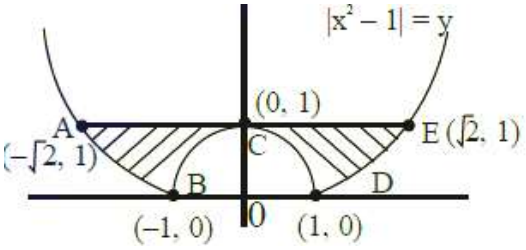
28. The area bounded by the curves $y = |x^2 - 1|$ and $y = 1$ is : **[JEE(Main)July2022]**

वक्रों $y = |x^2 - 1|$ तथा $y = 1$ द्वारा घिरे क्षेत्र का क्षेत्रफल है : **[JEE(Main)July2022]**

(A) $\frac{2}{3}\left(\sqrt{2}+1\right)$ (B) $\frac{4}{3}\left(\sqrt{2}-1\right)$ (C) $2\left(\sqrt{2}-1\right)$ (D) $\frac{8}{3}\left(\sqrt{2}-1\right)$

Ans. D

Sol. $y = |x^2 - 1|$



Arera = ABCDEA

$$= 2\left(\int_0^1 \left(1-\left(1-x^2\right)\right)dx + \int_1^{\sqrt{2}} \left(1-\left(x^2-1\right)\right)dx\right)$$

$$= \frac{8}{3}\left(\sqrt{2}-1\right)$$

29. The area of the smaller region enclosed by the curves $y^2 = 8x + 4$ and $x^2 + y^2 + 4\sqrt{3}x - 4 = 0$ is equal to : **[JEE(Main)July2022]**

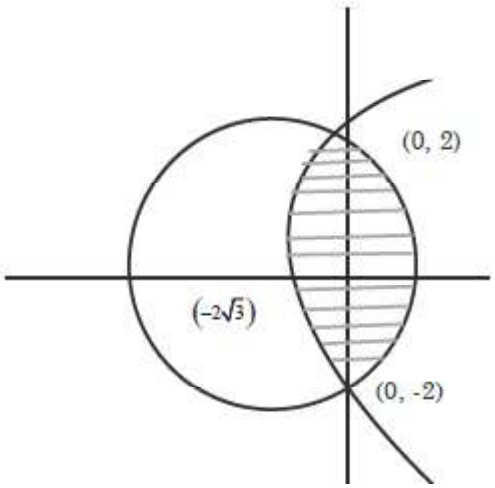
वक्रों $y^2 = 8x + 4$ तथा $x^2 + y^2 + 4\sqrt{3}x - 4 = 0$ से घिरे छोटे क्षेत्र का क्षेत्रफल है : **[JEE(Main)July2022]**

(A) $\frac{1}{3}\left(2-12\sqrt{3}+8\pi\right)$ (B) $\frac{1}{3}\left(2-12\sqrt{3}+6\pi\right)$

(C) $\frac{1}{3}\left(4-12\sqrt{3}+8\pi\right)$ (D) $\frac{1}{3}\left(4-12\sqrt{3}+6\pi\right)$

Ans. C

Sol.



$x^2 + y^2 + 4\sqrt{3}x - 4 = 0$

$y^2 = 8x + 4$

Point of intersections are (0, 2) & (0, –2)

Both are symmetric about x-axis

$$\text{Area} = 2\int_0^2 \left(\sqrt{16 - y^2} - 2\sqrt{3}\right) - \left(\frac{y^2 - 4}{8}\right) dy$$

$$\text{On solving Area} = \frac{1}{3}\left[8\pi + 4 - 12\sqrt{3}\right]$$

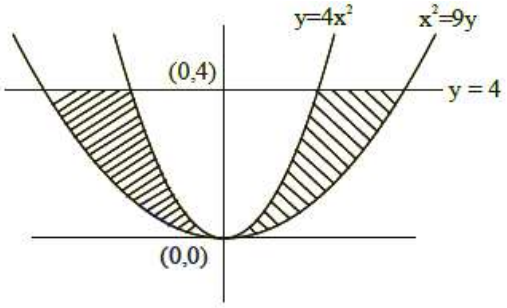
30. The area of the region enclosed by $y \le 4x^2$, $x^2 \le 9y$ and $y \le 4$, is equal to : **[JEE(Main)July2022]**

$y \le 4x^2$, $x^2 \le 9y$ तथा $y \le 4$ घिरे क्षेत्र का क्षेत्रफल है : **[JEE(Main)July2022]**

(A) $\frac{40}{3}$ (B) $\frac{56}{3}$ (C) $\frac{112}{3}$ (D) $\frac{80}{3}$

Ans. D

Sol.



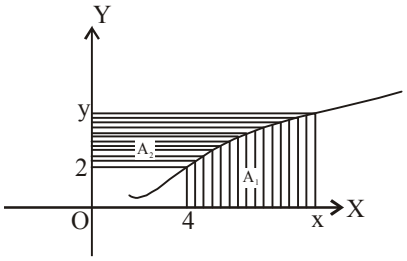
$$\Delta = 2.\int_0^4 \left(3\sqrt{y} - \frac{\sqrt{y}}{2}\right) dy$$

$$= 2.\int_0^4 \frac{5}{2}\sqrt{y}\,dy = \frac{80}{3}$$

31. Consider a curve y = y(x) in the first quadrant as shown in the figure. Let the area A_1 is twice the area A_2. Then the normal to the curve perpendicular to the line 2x – 12y = 15 does NOT pass through the point :

[JEE(Main)July2022]

चित्र में दिखाए अनुसार y = y(x) प्रथम चतुर्थांश में एक वक्र है। माना क्षेत्रफल A_1, क्षेत्रफल A_2 का दो गुना है। तो वक्र पर, रेखा 2x – 12y = 15 के लंबवत, अभिलंब किस बिन्दु से होकर नहीं जाता है : **[JEE(Main)July2022]**



(A) (6, 21)

(B) (8, 9)

(C) (10, –4)

(D) (12, –15)

Ans. C

Sol. Given that $A_1 = 2A_2$

from the graph $A_1 + A_2 = xy - 8$

$$\Rightarrow \frac{3}{2}A_1 = xy - 8$$

$$\Rightarrow A_1 = \frac{2}{3}xy - \frac{16}{3}$$

$$\Rightarrow \int_4^x f(x)\,dx = \frac{2}{3}xy - \frac{16}{3}$$

$$\Rightarrow f(x) = \frac{2}{3}\left(x\frac{dy}{dx} + y\right)$$

$$\Rightarrow 2\int \frac{dy}{y} = \int \frac{dx}{x}$$

$$\Rightarrow 2\ell ny = \ell nx + \ell nc$$

$$\Rightarrow y^2 = cx$$

As f(4) = 2 $\Rightarrow$ c = 1

so $y^2 = x$

slope of normal = –6

$$y = -(x) - \frac{1}{2}(-6) - \frac{1}{4}(-6)^3$$

$\Rightarrow y = -6x + 3 + 54$

$\Rightarrow y + 6x = 57$

Now check options and (3) will not satisfy.

32. The area enclosed by the curve $y = \log_e(x + e^2)$, $x = \log_e\left(\frac{2}{y}\right)$ and $x = \log_e 2$, above the line $y = 1$ is :

[JEE(Main)July2022]

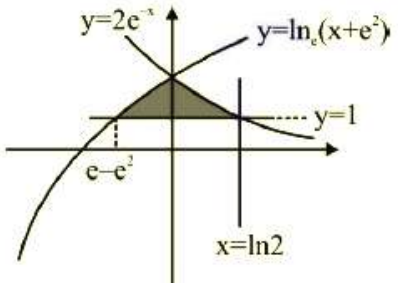
रेखा $y = 1$ के ऊपर वक्रों $y = \log_e(x + e^2)$, $x = \log_e\left(\frac{2}{y}\right)$ तथा $x = \log_e 2$ से घिरे क्षेत्र का क्षेत्रफल है :

[JEE(Main)July2022]

(A) $2 + e - \log_e 2$ (B) $1 + e - \log_e 2$ (C) $e - \log_e 2$ (D) $1 + \log_e 2$

Ans. B

Sol.



Required area is

$$= \int_{e-e^2}^{0} \ell n\left(x + e^2\right) - 1dx + \int_{0}^{\ell n2} 2e^{-x} - 1dx = 1 + e - \ell n2$$

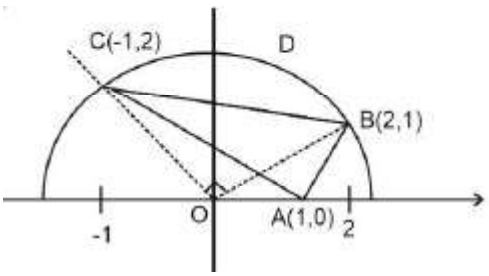
33. The area of the region $\left\{(x, y) : |x - 1| \le y \le \sqrt{5 - x^2}\right\}$ is equal to : **[JEE(Main)July2022]**

क्षेत्र $\left\{(x, y) : |x - 1| \le y \le \sqrt{5 - x^2}\right\}$ का क्षेत्रफल बराबर है : **[JEE(Main)July2022]**

(A) $\frac{5}{2}\sin^{-1}\left(\frac{3}{5}\right) - \frac{1}{2}$ (B) $\frac{5\pi}{4} - \frac{3}{2}$ (C) $\frac{3\pi}{4} + \frac{3}{2}$ (D) $\frac{5\pi}{4} - \frac{1}{2}$

Ans. D

Sol.



$|x-1| < y < \sqrt{5-x^2}$

When $|x-1| = \sqrt{5-x^2}$

$\Rightarrow (x-1)^2 = 5-x^2$

$\Rightarrow x^2 - x - 2 = 0$

$\Rightarrow x = 2, -1$

Required Area = Area of ΔABC + Area of region BCD

$$= \frac{1}{2}\begin{vmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \\ -1 & 2 & 1 \end{vmatrix} + \frac{\pi}{4}(\sqrt{5})^2 - \frac{1}{2}(\sqrt{5})^2$$

$$= \frac{5\pi}{4} - \frac{1}{2}$$

Exercise J - A

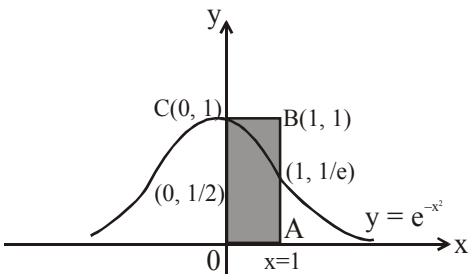
1. Let S be the area of the region enclosed by $y = e^{-x^2}, y = 0, x = 0$ and $x = 1$. Then : **[JEE-2012]**

यदि S वक्रों $y = e^{-x^2}, y = 0, x = 0$ तथा $x = 1$ के मध्य घिरा हुआ क्षेत्रफल हो तो : **[JEE-2012]**

(A) $S \geq \frac{1}{e}$ (B) $S \geq 1 - \frac{1}{e}$ (C) $S \leq \frac{1}{4}\left(1 + \frac{1}{\sqrt{e}}\right)$ (D) $S \leq \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{e}}\left(1 - \frac{1}{\sqrt{2}}\right)$

Ans. ABD

Sol. Area (OABC) = 1



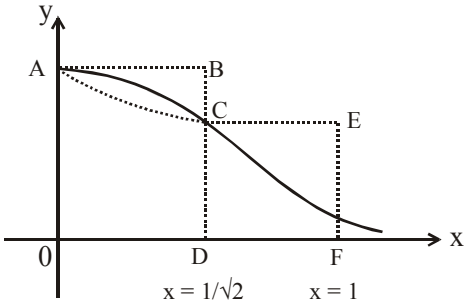
Shaded area is S.

Clearly S < 1

and $\int_0^1 e^{-x^2} dx > \int_0^1 e^{-x} dx$

$\Rightarrow \quad S > 1 - \frac{1}{e}$ ($\therefore$ (B) is correct

Again S ≥ Area (trapezium ACDO)



$\Rightarrow \quad S \geq \frac{1}{2}\left(1 + \frac{1}{\sqrt{e}}\right)\left(\frac{1}{\sqrt{2}}\right)$

$\Rightarrow \quad S \geq \frac{1}{2\sqrt{2}}\left(1 + \frac{1}{\sqrt{e}}\right)$

$\therefore \quad$ C is wrong

Also S ≤ Sum of areas of rectangles ABDO & CEFD

$\Rightarrow \quad S \leq \frac{1}{\sqrt{2}} \times 1 + \left(1 - \frac{1}{\sqrt{2}}\right)\left(\frac{1}{\sqrt{e}}\right)$

$\Rightarrow \quad S \leq \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{e}}\left(1 - \frac{1}{\sqrt{2}}\right)$

($\therefore$ (D) is correct)

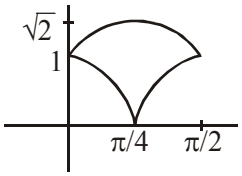
2. The area enclosed by the curves $y = \sin x + \cos x$ and $y = |\cos x - \sin x|$ over the interval $\left[0, \frac{\pi}{2}\right]$ is : **[JEE(Adv.)2013]**

अन्तराल $\left[0,\frac{\pi}{2}\right]$ पर वक्रों $y = \sin x + \cos x$ तथा $y = |\cos x - \sin x|$ द्वारा परिबद्ध क्षेत्रफल है : **[JEE(Adv.)2013]**

(A) $4\left(\sqrt{2}-1\right)$ (B) $2\sqrt{2}\left(\sqrt{2}-1\right)$ (C) $2\left(\sqrt{2}+1\right)$ (D) $2\sqrt{2}\left(\sqrt{2}+1\right)$

Ans. B

Sol. $y = \sin x + \cos x = \sqrt{2}\sin\left(x+\frac{\pi}{4}\right)$



$$y = |\cos x - \sin x| = \sqrt{2}\left(\cos\left(x+\frac{\pi}{4}\right)\right)$$

Area

$$= \int_0^{\pi/4}\left[(\sin x + \cos x) - (\cos x - \sin x)\right]dx$$

$$+ \int_{\pi/4}^{\pi/2}\left[(\sin x + \cos x) - (\sin x - \cos x)\right]dx$$

$$= \int_0^{\pi/4} 2\sin x\,dx + \int_{\pi/4}^{\pi/2} 2\cos x\,dx$$

$$= \left[-2\cos x\right]_0^{\pi/4} + \left[2\sin x\right]_{\pi/4}^{\pi/2}$$

$$= 2\sqrt{2}\left(\sqrt{2}-1\right)$$

3. Let $f:\left[\frac{1}{2},1\right]\to R$ (the set of all real numbers) be a positive, non-constant and differentiable function such that $f'(x) < 2f(x)$ and $f\left(\frac{1}{2}\right) = 1$. Then the value of $\int_{1/2}^{1} f(x)\,dx$ lies in the interval : **[JEE(Adv.)2013]**

माना कि $f:\left[\frac{1}{2},1\right]\to R$ (सभी वास्तविक संख्याओं का समुच्चय) एक धनात्मक अचरेतर (non constant) तथा अवकलनीय फलन gSft 1 d sfy ; sf'(x) < 2 f(x) तथा $f\left(\frac{1}{2}\right) = 1$ है, तब $\int_{1/2}^{1} f(x)\,dx$ का मान निम्न अन्तराल में है : **[JEE(Adv.)2013]**

(A) $(2e-1, 2e)$ (B) $(e-1, 2e-1)$ (C) $\left(\frac{e-1}{2}, e-1\right)$ (D) $\left(0, \frac{e-1}{2}\right)$

Ans. D

Sol. $f'(x) - 2f(x) < 0$

Multiply both side by e^{-2x}

$$e^{-2x} f'(x) - 2e^{-2x} f(x) < 0$$

$$\frac{d}{dx}\left(e^{-2x} f(x)\right) < 0$$

Now, $g(x) = e^{-2x} f(x)$

$\therefore$ g(x) is a decreasing function.

$$x > \frac{1}{2}$$

$$g(x) < g\left(\frac{1}{2}\right)$$

$$\Rightarrow \quad e^{-2x} f(x) < \frac{1}{e}$$

$$\Rightarrow \quad f(x) < e^{2x-1}$$

$$\Rightarrow \quad \int_{1/2}^{1} f(x)dx < \frac{1}{e}\int_{1/2}^{1} e^{2x}dx$$

$$= \left[\frac{1}{2e}e^{2x}\right]_{1/2}^{1} = \frac{1}{2e}\left(e^2 - e\right) = \frac{1}{2}\left(e-1\right)$$

$$\Rightarrow \quad \int_{1/2}^{1} f(x)dx < \frac{e-1}{2}$$

obviously $f(x)$ is positive

$$\therefore \quad \int_{1/2}^{1} f(x)dx > 0$$

4. Let $F(x) = \int_{x}^{x^2+\frac{\pi}{6}} 2\cos^2 t\, dt$ for all $x \in R$ and $f: \left[0, \frac{1}{2}\right] \to [0, \infty)$ be a continuous function. For $a \in \left[0, \frac{1}{2}\right]$, if $F'(a) + 2$ is the area of the region bounded by $x = 0$, $y = 0$, $y = f(x)$ and $x = a$, then $f(0)$ is. **[JEE(Adv.)2015]**

माना कि सभी $x \in R$ के लिए, $F(x) = \int_{x}^{x^2+\frac{\pi}{6}} 2\cos^2 t\,dt$ तथा $f: \left[0, \frac{1}{2}\right] \to [0, \infty)$ एक संतत फलन है। यदि उन सभी $a \in \left[0, \frac{1}{2}\right]$ के लिए $F'(a) + 2$ उस क्षेत्र का क्षेत्रफल है, जो कि $x = 0$, $y = 0$, $y = f(x)$ और $x = a$ से घिरा हुआ है, तब $f(0)$ का मान है।

[JEE(Adv.)2015]

Ans. 3

Sol. $F(x) = \int_{x}^{x^2+\frac{\pi}{6}} 2\cos^2 + dt$

$$F'(x) = 2\left(\cos\left(x^2 + \frac{\pi}{6}\right)\right)^2 2x - 2\cos^2 x$$

$$\because F'(a) + 2 = \int_0^a f(x)dx$$

$$2\left(\cos\left(a^2 + \frac{\pi}{6}\right)\right)^2 2a - 2\cos^2 a + 2 = \int_0^a f(x)dx$$

$$\Rightarrow 4\cos^2\left(a^2 + \frac{\pi}{6}\right) + 4a2\cos\left(a^2 + \frac{\pi}{6}\right).\left(-\sin\left(a^2 + \frac{\pi}{6}\right)\right) \times 2a + 4\cos a \sin a = f(a)$$

$$\therefore f(0) = 4\left(\frac{\sqrt{3}}{2}\right)^2 = 3$$

5. Match the following column : **[JEE(Adv.)-2015]**

Column I	**Column II**
(A) In a triangle ΔXYZ, let a, b and c be the lengths of the sides opposite to the angles X, Y and Z, respectively. If $2(a^2 - b^2) = c^2$ and $\lambda = \frac{\sin(X-Y)}{\sin Z}$, then possible values of n for which $\cos(n\pi\lambda) = 0$ is (are)	(P) 1
(B) In a triangle ΔXYZ, let a, b and c be the lengths of the sides opposite to the angles X, Y and Z, respectively. If $1 + \cos 2X - 2\cos 2Y = 2 \sin X \sin Y$, then possible value(s) of a/b is (are)	(Q) 2
(C) In R^2, let $\sqrt{3}\,\hat{i} + \hat{j}, \hat{i} + \sqrt{3}\,\hat{j}$ and $\beta\,\hat{i} + (1-\beta)\hat{j}$ be the position vectors of X, Y and Z with respect to the origin O, respectively. If the distance of Z	(R) 3

from the bisector of the acute angle of $\overrightarrow{OX}$ with $\overrightarrow{OY}$ is $\frac{3}{\sqrt{2}}$, then possible value(s) of $|\beta|$ is (are)

(D) Suppose that $F(\alpha)$ denotes the area of the region bounded by $x = 0$, $x = 2$, $y^2 = 4x$ and $y = |\alpha x - 1| + |\alpha x - 2| + \alpha x$, where $\alpha \in \{0, 1\}$. Then the value(s) of $F(\alpha) + \frac{8}{3}\sqrt{2}$, when $\alpha = 0$ and $\alpha = 1$, is (are) (S) 5

(T) 6

निम्न कॉलम कों सुमेहिलत कीजिए : **[JEE(Adv.)-2015]**

Column I	**Column II**
(A) माना कि एक त्रिभुज ΔXYZ में कोणों X, Y और Z के सामने की भुजाओं की लम्बाईयाँ क्रमशः a, b और c है। मानाकि $2(a^2 - b^2) = c^2$ और $\lambda = \frac{\sin(X-Y)}{\sin Z}$ है। यदि $\cos(n\pi\lambda) = 0$ तब n के संभव मान है।	(P) 1
(B) माना कि एक त्रिभुज ΔXYZ मे कोणों X, Y और Z के सामने की भुजाओं की लम्बाईयाँ क्रमशः a, b और c है। यदि $1 + \cos 2X - 2\cos 2Y = 2 \sin X \sin Y$ तब a/b के संभव मान है	(Q) 2
(C) माना कि R^2 में, मूल बिन्दु O के सापेक्ष $\sqrt{3}\,\hat{i} + \hat{j}, \hat{i} + \sqrt{3}\,\hat{j}$ और $\beta\,\hat{i} + (1-\beta)\hat{j}$ क्रमशः ग एल और Z के स्थिति सदिश है। यदि $\overrightarrow{OX}$ और $\overrightarrow{OY}$ के न्यून कोण के समद्विभाजक से Z की दूरी $\frac{3}{\sqrt{2}}$ हो, तो $\|\beta\|$ का संभव मान है।	(R) 3
(D) माना कि $F(\alpha)$ उस क्षेत्र के क्षेत्रफल को दर्शाता है जो $x = 0$, $x = 2$, $y^2 = 4x$ और $\psi = \|\alpha x - 1\| + \|\alpha x - 2\| + \alpha x$, से घिरा हुआ है, जहाँ $\alpha \in \{0, 1\}$ है। $\alpha = 0$ और $\alpha = 1$ के लिए $F(\alpha) + \frac{8}{3}\sqrt{2}$ का मान है।	(S) 5
	(T) 6

Ans. A -PRS , B -P , C - PQ, D - ST

Sol. (A) $2(a^2 - b^2) = c^2$

$2(4R^2(\sin^2 x \times - \sin^2 y)) = 4R^2 \sin^2 z$

$2(\sin(x-y)\sin(x+y)) = \sin^2 z$

$\frac{\sin(x-y)}{\sin z} = \frac{1}{2} = \lambda$

$$\cos\left(n\pi\lambda\right)=\cos\left(\frac{n\pi}{2}\right)=0$$

n = odd integer

(B) $1+\cos 2x=2\cos 2y=2\sin x\sin y$

$$\sin^2 x+\sin x\sin y-2\sin^2 y=0$$

$$\left(\sin x-\sin y\right)\left(\sin x+2\sin y\right)=0$$

$\sin x=\sin y$

$a=b$

$$\frac{a}{b}=1$$

(C) distance of z from bisector of $\overrightarrow{OX}$ & $\overrightarrow{OY}=\frac{3}{\sqrt{2}}$

$$\left(\beta-\frac{1}{2}\right)^2+\left(\beta-\frac{1}{2}\right)^2=\frac{9}{2}$$

$\beta=2,-1$

$|\beta|=2,1$

(D) when $\alpha=0$

$$A=6-\int_0^2 2\sqrt{x}dx$$

$$=6-\frac{8\sqrt{2}}{3}$$

when $\alpha=1$

$$A=\int_0^1\left(3-x-2\sqrt{x}\right)dx+\int_1^2\left(x+1-2\sqrt{x}\right)dx$$

$$=\left|3x-\frac{x^2}{2}-\frac{4}{3}x^{\frac{3}{2}}\right|_0^1+\left|\frac{x^2}{2}+x-\frac{4}{3}x^{\frac{3}{2}}\right|_1^2$$

$$= 5 - \frac{8\sqrt{2}}{3}$$

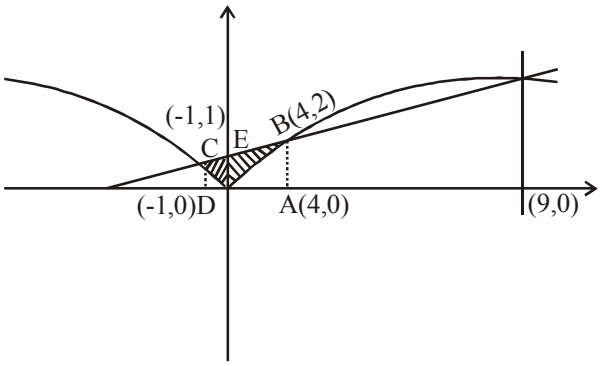
6. Area of the region $\left\{(x, y) \in R^2 : y \geq \sqrt{|x+3|}, 5y \leq x+9 \leq 15\right\}$ is equal to : **[JEE(Adv.)2016]**

क्षेत्र $\left\{(x, y) \in R^2 : y \geq \sqrt{|x+3|}, 5y \leq x+9 \leq 15\right\}$ का क्षेत्रफल है : **[JEE(Adv.)2016]**

(A) $\frac{1}{6}$ (B) $\frac{4}{3}$ (C) $\frac{3}{2}$ (D) $\frac{5}{3}$

Ans. C

Sol. Shifting origin to (–3, 0)



Area $\left\{(x, y) \in R^2 : y \geq \sqrt{|x|}, 5y \leq x+6 \leq 15\right\}$

Desired Area = Area of trapezium ABCDA – [Area DCOD + Area OBAO]

$$= \frac{1}{2} \times (1+2) \times 5 - \left(\int_{-1}^{0} \sqrt{-x}\ dx + \int_{0}^{4} \sqrt{x}\ dx \right)$$

$$= \frac{15}{2} - \left(\frac{2}{3} + \frac{16}{3} \right)$$

$$= \frac{3}{2}$$

7. If the line $x = \alpha$ divides the area of region $R = \{(x, y) \in R^2 : x^3 \leq y \leq x, 0 \leq x \leq 1\}$ into two equal parts, then : **[JEE(Adv.)-2017]**

यदि रेखा $x = \alpha$ क्षेत्र $R = \{(x, y) \in R^2 : x^3 \leq y \leq x, 0 \leq x \leq 1\}$ के क्षेत्रफल को दो बराबर भागों में विभाजित करती है, तब : **[JEE(Adv.)-2017]**

(A) $0 < \alpha \le \frac{1}{2}$ (B) $2\alpha^4 - 4\alpha^2 + 1 = 0$ (C) $\alpha^4 + 4\alpha^2 - 1 = 0$ (D) $\frac{1}{2} < \alpha < 1$

Ans. BD

Sol. $y = x^3$

$$\int_0^1 \left(x - x^3\right)dx = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$$

$$\int_0^\alpha \left(x - x^3\right)dx = \frac{1}{8}$$

$$4\alpha^2 - 2\alpha^4 = 1$$

$$2\alpha^4 - 4\alpha^2 + 1 = 0$$

$2t^2 - 4t + 1 = 0$ (taking $t = \alpha^2$)

$$t = \frac{4 \pm \sqrt{16-8}}{4}$$

$$t = \frac{4 \pm 2\sqrt{2}}{4}$$

$$t = \alpha^2 = 1 \pm \frac{1}{\sqrt{2}}$$

$$\therefore \alpha^2 = 1 - \frac{1}{\sqrt{2}} \qquad \Rightarrow \frac{1}{2} < \alpha < 1$$

8. Let $f: [0, \infty) \to R$ be a continuous function such that $f(x) = 1 - 2x + \int_0^x e^{x-t} f(t)\,dt$ for all $x \in [0, \infty)$. Then, which of the following statement(s) is (are) TRUE ? **[JEE(Adv.)2018]**

(A) The curve $y = f(x)$ passes through the point $(1, 2)$

(B) The curve $y = f(x)$ passes through the point $(2, -1)$

(C) The area of the region $\left\{(x, y) \in [0,1] \times R : f(x) \le y \le \sqrt{1-x^2}\right\}$ is $\frac{\pi - 2}{4}$

(D) The area of the region $\left\{(x, y) \in [0,1] \times R : f(x) \le y \le \sqrt{1-x^2}\right\}$ is $\frac{\pi - 1}{4}$

माना कि $f: [0, \infty) \to R$ एक ऐसा संतत फलन है कि $f(x) = 1 - 2x + \int_0^x e^{x-t} f(t)\,dt$ सभी $x \in [0, \infty)$ के लिये। तब निम्नलिखित esa lsa dkSu l k(से) कथन सत्य है (हैं)? **[JEE(Adv.)2018]**

(A) वक्र $y = f(x)$ बिन्दु $(1, 2)$ से गुजरता है

(B) वक्र $y = f(x)$ बिन्दु $(2, -1)$ से गुजरता है

(C) क्षेत्र $\left\{(x, y) \in [0,1] \times R : f(x) \le y \le \sqrt{1-x^2}\right\}$ का क्षेत्रफल $\frac{\pi-2}{4}$ है

(D) क्षेत्र $\left\{(x, y) \in [0,1] \times R : f(x) \le y \le \sqrt{1-x^2}\right\}$ का क्षेत्रफल $\frac{\pi-1}{4}$ है

Ans. BC

Sol. $f(x) = 1 - 2x + \int_0^x e^{x-t} f(t)dt \Rightarrow f(x).e^{-x} = (1-2x).e^{x} + \int_0^x e^{-t} f(t)dt$

$\Rightarrow f'(x)e^{-x} - e^{-x} \cdot f(x) = -2 \cdot e^{-x} - (1-2x) \cdot e^{-x} + e^{-x} \cdot f(x)$

$\Rightarrow f'(x) - 2f(x) = (2x-3)$

I.F. $= e^{-2x}$

$\therefore y.e^{-x} = \int \underset{|}{(2x-3)}.\underset{||}{e^{-2x}}dx$

$\Rightarrow y.e^{-2x} = (2x-3).\frac{e^{-2x}}{-2} - 2\int \frac{e^{-2x}}{-2}dx \qquad \Rightarrow y.e^{-2x} = -\frac{(2x-3)e^{-2x}}{2} - \frac{e^{-2x}}{2} + c$

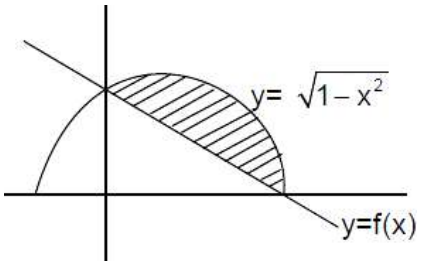
$\Rightarrow y.e^{-2x} = -\frac{(2x-3)-1}{2} + c.e^{2x} \qquad \Rightarrow y = (1-x) + c.e^{2x}$

$\Rightarrow y = (1-x) + c.e^{2x}$

put $x = 0$

$1 = 1 + c \qquad \Rightarrow c = 0$

$\therefore y = 1 - x$ which passes through point $(2, -1)$



Now

required area $= \frac{1}{4}.\pi.(1)^2 - \frac{1}{2}.1.1 = \frac{\pi}{4} - \frac{1}{2}$

9. A farmer F_1 has a land in the shape of a triangle with vertices at P(0,0), Q(1,1) and R(2,0). From this land, a neighbouring farmer F_2 takes away the region which lies between the side PQ and a curve of the form $y = x^n$ $(n > 1)$. If the area of the region taken away by the farmer F_2 is exactly 30% of the area of ΔPQR, then the value of n is _____ . **[JEE(Adv.)2018]**

एक किसान F_1 के पास एक त्रिभुजाकार भूमि है जिसके शीर्ष P(0,0), Q(1,1) तथा R(2,0) पर हैं। एक पड़ोसी किसान F_2 इस भूमि से उस क्षेत्र को ले लेता है जो कि भुजा PQ और $y = x^n$ $(n > 1)$ के रूप वाले वक्र के बीच स्थित है। यदि किसान F_2 द्वारा लिये गये क्षेत्र का क्षेत्रफल ΔPQR के क्षेत्रफल का ठीक 30% है, तब n का मान है। **[JEE(Adv.)2018]**

Ans. 4

Sol. $\frac{30}{100}A = \frac{3}{10}$

$$A = \int_0^1 \left(x - x^n\right) dx = \frac{3}{10}$$

$$\frac{x^2}{2} - \frac{x^{n+1}}{n+1}\Bigg|_0^1 = \frac{3}{10}$$

$$\frac{1}{2} - \frac{1}{n+1} = \frac{3}{10}$$

$n = 4$

10. The area of the region $\{(x,y): xy \le 8, 1 \le y \le x^2\}$ is : **[JEE(Adv.)2019]**

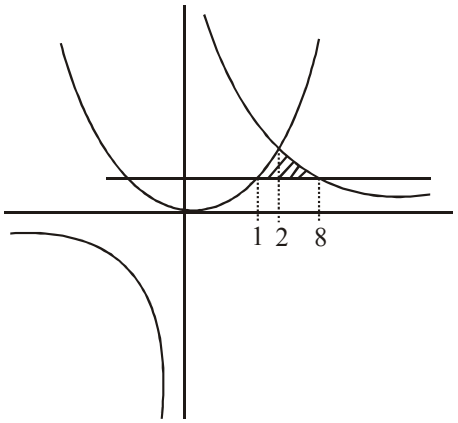
क्षेत्र $\{(x,y): xy \le 8, 1 \le y \le x^2\}$ का क्षेत्रफल है : **[JEE(Adv.)2019]**

(A) $16\log_e 2 - 6$ (B) $8\log_e 2 - \frac{14}{3}$ (C) $16\log_e 2 - \frac{14}{3}$ (D) $8\log_e 2 - \frac{7}{3}$

Ans. C

Sol. $x^3 = 8$

$x = 2$



$$\text{Area} = \int_1^2 \left(x^2 - 1\right) dx + \int_2^8 \left(\frac{8}{x} - 1\right) dx$$

$$= \left(\frac{x^3}{3} - x\right)_1^2 + \left(8\ln x - x\right)_2^8$$

$$= \left(\frac{8}{3} - 2 - \frac{1}{3} + 1\right) + \left[8\ln 8 - 8\ln 2 - 8 + 2\right]$$

$$= \left(\frac{7}{3} - 1\right) + 8\ln 4 - 6$$

$$= 16\ln 2 + \frac{7}{3} - 7$$

$$= 16\ln 2 - \frac{14}{3}$$

11. Let the functions $f: \mathbb{R} \to \mathbb{R}$ and $g: \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = e^{x-1} - e^{-|x-1|}$ and $g(x) = \frac{1}{2}(e^{x-1} + e^{1-x})$. Then the area of the region in the first quadrant bounded by the curves $y = f(x)$, $y = g(x)$ and $x = 0$ is :

[JEE(Adv.)2020]

माना कि फलनों $f: \mathbb{R} \to \mathbb{R}$ एवं $g: \mathbb{R} \to \mathbb{R}$ को $f(x) = e^{x-1} - e^{-|x-1|}$ एवं $g(x) = \frac{1}{2}(e^{x-1} + e^{1-x})$ के द्वारा परिभाषित किया जाता

gSr c i Ze pr dkZ kes aOkg $y = f(x)$, $y = g(x)$ एवं $x = 0$ के द्वारा प्रतिबद्ध क्षेत्र का क्षेत्रफल है? **[JEE(Adv.)2020]**

(A) $\left(2-\sqrt{3}\right)+\frac{1}{2}\left(e-e^{-1}\right)$

(B) $\left(2+\sqrt{3}\right)+\frac{1}{2}\left(e-e^{-1}\right)$

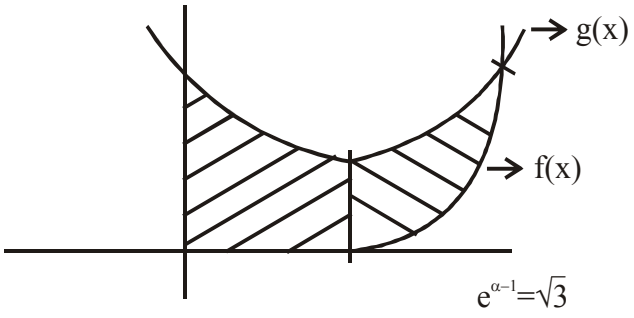
(C) $\left(2-\sqrt{3}\right)+\frac{1}{2}\left(e+e^{-1}\right)$

(D) $\left(2+\sqrt{3}\right)+\frac{1}{2}\left(e+e^{-1}\right)$

Ans. A

Sol. $f(x) \begin{cases} \to e^{x-1} - e^{1-x} & x \geq 1 \\ \to 0 & x < 1 \end{cases}$

$g(x) = \frac{1}{2}\left(e^{x-1} + e^{1-x}\right)$



$P\left(1+ln\sqrt{3}, \frac{2}{\sqrt{3}}\right) \equiv P\left(\alpha, \frac{2}{\sqrt{3}}\right)$

$\text{Required Area} = \int_0^1 \frac{e^{x-1}+e^{1-x}}{2}\,dx + \int_1^\alpha f(x) - f(x)\,dx$

$= \frac{1}{2}\left(e-\frac{1}{e}\right) + \int_1^\alpha \left(\frac{e^{x-1}+e^{1-x}}{2} - \left(e^{x-1}-e^{1-x}\right)\right)dx$

$= \frac{1}{2}\left(e-\frac{1}{e}\right) + \int_1^\alpha \left(\frac{3}{2}e^{1-x} - \frac{e^{x-1}}{2}\right)dx = \left(2-\sqrt{3}\right)+\frac{1}{2}\left(e-e^{-1}\right)$

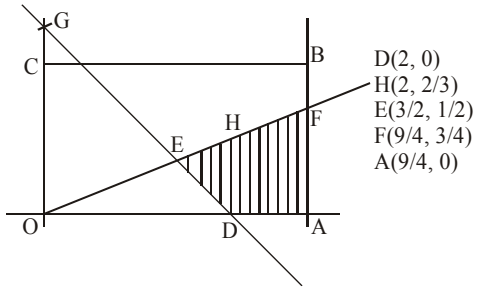
12. The area of the region $\left\{(x,y): 0 \leq x \leq \frac{9}{4},\quad 0 \leq y \leq 1,\quad x \geq 3y,\quad x+y \geq 2\right\}$ is : **[JEE(Adv.)2021]**

क्षेत्र $\left\{(x,y): 0 \leq x \leq \frac{9}{4},\quad 0 \leq y \leq 1,\quad x \geq 3y,\quad x+y \geq 2\right\}$ का क्षेत्रफल है : **[JEE(Adv.)2021]**

(A) $\frac{11}{32}$ (B) $\frac{35}{96}$ (C) $\frac{37}{96}$ (D) $\frac{13}{32}$

Ans. A

Sol.



$AB \equiv x = \frac{9}{4}$

$BC \equiv y = 1$

$DG \equiv x + y = 2$

$OF \equiv x = 3y$

$DG \equiv x = 2$

we will divide shaded region in two parts

required area = Area of ΔDEH + area of trapezium AFHD

$$= \frac{1}{2}\left(\frac{2}{3}\right)\left(\frac{1}{2}\right) + \frac{1}{2}\left(\frac{2}{3}+\frac{3}{4}\right)\left(\frac{1}{4}\right) = \frac{11}{32}$$

13. Consider the functions $f, g : R \to R$ defined by $f(x) = x^2 + \frac{5}{12}$ and $g(x) = \begin{cases} 2\left(1 - \frac{4|x|}{3}\right) & |x| \le \frac{3}{4}, \\ 0, & |x| > \frac{3}{4} \end{cases}$. If α is the area of the region $\left\{(x, y) \in R \times R : |x| \le \frac{3}{4}, 0 \le y \le \min\{f(x), g(x)\}\right\}$ then the value of 9α is ____. **[JEE(Adv.)2022]**

फलनों $f, g : R \to R$ पर विचार कीजिए जो $f(x) = x^2 + \frac{5}{12}$ एवं $g(x) = \begin{cases} 2\left(1 - \frac{4|x|}{3}\right) & |x| \le \frac{3}{4}, \\ 0, & |x| > \frac{3}{4} \end{cases}$ द्वारा परिभाषित है।

यदि क्षेत्र $\left\{(x, y) \in R \times R : |x| \le \frac{3}{4}, 0 \le y \le \min\{f(x), g(x)\}\right\}$ का क्षेत्रफल α है, तब 9α का मान _____ है।

[JEE(Adv.)2022]

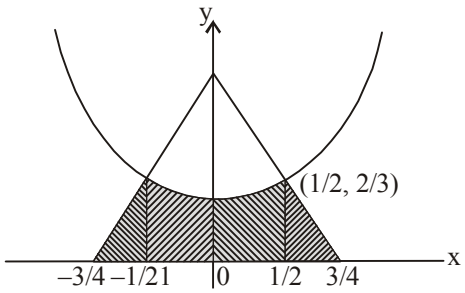
Ans. 6

Sol. $f(x)=x^2+\frac{5}{12}$

$g(x)=2\left(1-\frac{4}{3}|x|\right)$

$|x|\le\frac{3}{4}$ and $0\le y\le(f(x),g(x))$

for $x\ge 0$



$x^2+\frac{5}{12}=2-\frac{8}{3}x$

$\Rightarrow 12x^2+32x-19=0$

$12x^2+38x-6x-19=0\Rightarrow x=\frac{1}{2}$

Required area $=2\left(\int_0^{1/2}\left(x^2+\frac{5}{12}\right)dx+\frac{1}{2}\times\frac{1}{4}\times\left(2-\frac{4}{3}\right)\right)$

$\Rightarrow\alpha=2\left(\frac{1}{24}+\frac{5}{24}+\frac{2}{24}\right)=\frac{2}{3}$ sq.units $\Rightarrow 9\alpha=9\times\frac{2}{3}=6$

14. Let $f:[0,1]\to[0,1]$ be the function defined by $f(x)=\frac{x^3}{3}-x^2+\frac{5}{9}x+\frac{17}{36}$. Consider the square region $S=[0,1]\times[0,1]$. Let $G=\{(x,y)\in S: y>f(x)\}$ be called the green region and $R=\{(x,y)\in S: y<f(x)\}$ be called the red region. Let $L_h=\{(x,h)\in S: x\in[0,1]\}$ be the horizontal line drawn at a height $h\in[0,1]$. Then which of the following statements is(are) true? **[JEE(Adv.)2023]**

(A) There exists an $h\in\left[\frac{1}{4},\frac{2}{3}\right]$ such that the area of the green region above the line L_h equals the area of the green region below the line L_h

(B) There exists an $h\in\left[\frac{1}{4},\frac{2}{3}\right]$ such that the area of the red region above the line L_h equals the area of the red region below the line L_h

(C) There exists an $h\in\left[\frac{1}{4},\frac{2}{3}\right]$ such that the area of the green region above the line L_h equals the area of the red region below the line L_h

(D) There exists an $h\in\left[\frac{1}{4},\frac{2}{3}\right]$ such that the area of the red region above the line L_h equals the area of the green region below the line L_h

माना कि फलन $f:[0,1]\to[0,1]$, $f(x)=\frac{x^3}{3}-x^2+\frac{5}{9}x+\frac{17}{36}$ से परिभाषित है। वर्गाकार क्षेत्र(square region) $S=[0,1]\times[0,1]$, पर विचार कीजिए। माना कि $G=\{(x,y)\in S: y>f(x)\}$ हरित क्षेत्र(green region) एवं $R=\{(x,y)\in S: y<f(x)\}$ लाल क्षेत्र (red region) कहलाता है। मान लीजिये की $h\in[0,1]$ की ऊँचाई (height) पर खींची गई {kSfrt jsf[kk(horizontal line) $L_h=\{(x,h)\in S: x\in[0,1]\}$ है। तब निम्न में से कौनसा (से) कथन सत्य है(हैं)?

[JEE(Adv.)2023]

(A) एक ऐसा $h\in\left[\frac{1}{4},\frac{2}{3}\right]$ है कि रेखा L_h के ऊपर के लाल क्षेत्र का क्षेत्रफल रेखा L_h के नीचे के हरित क्षेत्र के क्षेत्रफल के बराबर है

(B) एक ऐसा $h\in\left[\frac{1}{4},\frac{2}{3}\right]$ है कि रेखा L_h के ऊपर के हरित क्षेत्र का क्षेत्रफल रेखा L_h के नीचे के लाल क्षेत्र के क्षेत्रफल के बराबर है

(C) एक ऐसा $h\in\left[\frac{1}{4},\frac{2}{3}\right]$ है कि रेखा L_h के ऊपर के लाल क्षेत्र का क्षेत्रफल रेखा L_h के नीचे के लाल क्षेत्र के क्षेत्रफल के बराबर है

(D) एक ऐसा $h\in\left[\frac{1}{4},\frac{2}{3}\right]$ है कि रेखा L_h के ऊपर के हरित क्षेत्र का क्षेत्रफल रेखा L_h के नीचे के हरित क्षेत्र के क्षेत्रफल के बराबर है

Ans. BCD

Sol. $f(x)=\frac{x^3}{3}-x^2+\frac{5x}{9}+\frac{17}{36}$

$f'(x)=x^2-2x+\frac{5}{9}$

$f'(x)=0$ at $x=\frac{1}{3}$ in $[0,1]$

$A_R=$ Area of Red region

$A_G=$ Area of Green region

$A_R=\int_0^1 f(x)\,dx=\frac{1}{2}$

Total area = 1

$\Rightarrow A_G=\frac{1}{2}$

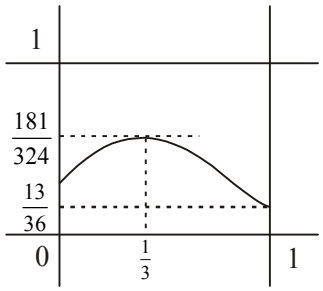
$\int_0^1 f(x)\,dx=\frac{1}{2}$

$A_G=A_R$

$f(0)=\frac{17}{36}$

$$f(1)=\frac{13}{36}\approx 0.36$$

$$f\left(\frac{1}{3}\right)=\frac{181}{324}\approx 0.558$$



(A) Given statement would be correct when $h=\frac{3}{4}$ but $h\in\left[\frac{1}{4},\frac{2}{3}\right]$

$\Rightarrow$ (A) is incorrect

(B) Given statement would be correct when $h=\frac{1}{4}$

$\Rightarrow$ (B) is correct

(C) When $h=\frac{181}{324}, A_R=\frac{1}{2}, A_G<\frac{1}{2}$

$h=\frac{13}{36}, A_R<\frac{1}{2}, A_G=\frac{1}{2}$

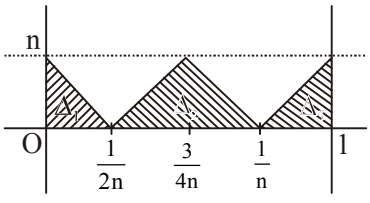
$\Rightarrow A_R=A_G$ for some $h\in\left(\frac{13}{36},\frac{181}{324}\right)$

$\Rightarrow$ (C) is correct

(D) Option (D) is remaining coloured part of option (C), hence option (D) is also correct.

15. Let $n\geq 2$ be a natural number and $f:[0,1]\to R$ be the function defined by

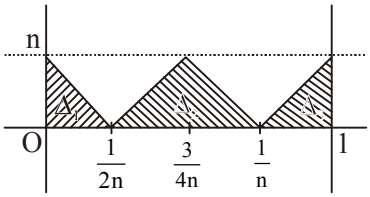
$$f(x)=\begin{cases} n(1-2nx) & \text{if } 0\leq x\leq\frac{1}{2n} \\ 2n(2nx-1) & \text{if } \frac{1}{2n}\leq x\leq\frac{3}{4n} \\ 4n(1-nx) & \text{if } \frac{3}{4n}\leq x\leq\frac{1}{n} \\ \frac{n}{n-1}(nx-1) & \text{if } \frac{1}{n}\leq x\leq 1 \end{cases}$$



If n is such that the area of the region bounded by the curves $x = 0$, $x = 1$, $y = 0$ and $y = f(x)$ is 4 , then the maximum value of the function f is _____. **[JEE(Adv.)2023]**

माना कि $n \geq 2$ एक प्राकृत संख्या (natural number) है एवं फलन $f : [0,1] \to R$ इस प्रकार परिभाषित है

$$f(x) = \begin{cases} n(1-2nx) & \text{if } 0 \leq x \leq \frac{1}{2n} \\ 2n(2nx-1) & \text{if } \frac{1}{2n} \leq x \leq \frac{3}{4n} \\ 4n(1-nx) & \text{if } \frac{3}{4n} \leq x \leq \frac{1}{n} \\ \frac{n}{n-1}(nx-1) & \text{if } \frac{1}{n} \leq x \leq 1 \end{cases}$$



यदि n इस प्रकार है कि वक्रों $x = 0$, $x = 1$, $y = 0$ एवं $y = f(x)$ से घिरे क्षेत्र का क्षेत्रफल 4 है तब फलन f का महत्तम मान (maximum value) है _____. **[JEE(Adv.)2023]**

Ans. 8

Sol. $$f(x) = \begin{cases} n(1-2nx) & 0 \leq n \leq \frac{1}{2n} \\ 2n(2nx-1) & \frac{1}{2n} \leq x \leq \frac{3}{4n} \\ 4n(1-nx) & \frac{3}{4n} \leq x \leq \frac{1}{n} \\ \frac{n}{n-1}(nx-1) & \frac{1}{n} \leq x \leq 1 \end{cases}$$

$\Delta_1 + \Delta_2 + \Delta_3 = 4$

$$\frac{1}{2} n \times \frac{1}{2n} + \frac{1}{2} \times n \times \frac{1}{2n} + \frac{1}{2} \times n \times \left(1 - \frac{1}{n}\right) = 4$$

$$4 = \frac{n}{2}$$

$n = 8$

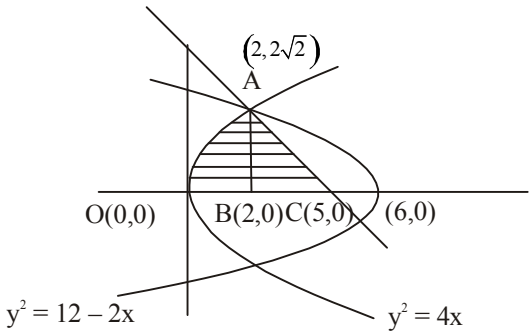
16. Let $S = \left\{(x, y) \in R \times R : x \geq 0,\ y \geq 0,\ y^2 \leq 4x,\ y^2 \leq 12 - 2x \text{ and } 3y + \sqrt{8}x \leq 5\sqrt{8}\right\}$ If the area of the region S is $\alpha\sqrt{2}$, then α is eqaul to : **[JEE(Adv.)2024]**

माना कि $S = \left\{(x, y) \in R \times R : x \geq 0,\ y \geq 0,\ y^2 \leq 4x,\ y^2 \leq 12 - 2x \text{ और } 3y + \sqrt{8}x \leq 5\sqrt{8}\right\}$ है। यदि क्षेत्र (region) S का {ksks Qy $\alpha\sqrt{2}$ है, तब α बराबर है : **[JEE(Adv.)2024]**

(A) $\frac{17}{2}$ (B) $\frac{17}{3}$ (C) $\frac{17}{4}$ (D) $\frac{17}{5}$

Ans. B

Sol. $3y+\sqrt{8}x=5\sqrt{8}$



$4x = 12 - 2x \Rightarrow 6x = 12 \Rightarrow x = 2$

$x = 2 \Rightarrow y^2 = 8 \Rightarrow y = \pm 2\sqrt{2}$

line $3y+\sqrt{8}x=5\sqrt{8}$ passes through $A\left(2,2\sqrt{2}\right)$ & C(5,0)

Required area = Area of OAB + Area of ΔABC

$$=\int_0^2 2\sqrt{x}dx+\frac{1}{2}(5-2)\left(2\sqrt{2}\right)$$

$$=\frac{4}{3}\left(x^{\frac{3}{2}}\right)_0^2+3\sqrt{2}$$

$$=\frac{8}{3}\sqrt{2}+3\sqrt{2}$$

$$=\frac{17}{3}\sqrt{2}$$

CHALLENGER EXERCISES

Exercise C - I

SINGLE CORRECT TYPE QUESTIONS

1. The area bounded by regions $y=\sqrt{x}$ and $x=-\sqrt{y}$ and the curve $x^2+y^2=2$ is : (above x-axis)

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}-\frac{1}{3}$ (C) $\frac{\pi}{2}$ (D) None

वक्रों $y=\sqrt{x}$, $x=-\sqrt{y}$ तथा $x^2+y^2=2$ से घिरा हुआ क्षेत्रफल होगा : (x-अक्ष के ऊपर)

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}-\frac{1}{3}$ (C) $\frac{\pi}{2}$ (D) कोई नहीं

Ans. C

Sol. $y=\sqrt{x}$ ...(1)

$x=-\sqrt{y}$ ...(2)

x^2+y^2+2 ...(3)

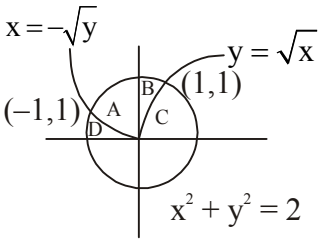
POI of (1) & (3)

$x^2+x+-2=0$

$x=-2, 1$

In 1st quadrant

(1,1)



POI of (2) & (3)

$y^2+y-2=0$

$y=-2,1$

In IInd quadrant

$y=1$

$x=-1$ (–1,1)

Let areas are A, B, C, D

from symmetry

A = C & B = D

we need

$= A + B$

$= C + D = \text{area of quadrant} = \dfrac{\pi r^2}{4} = \dfrac{\pi \cdot \left(\sqrt{2}\right)^2}{4} = \dfrac{\pi}{2}$

2. Area bounded by the curve $y = \max\{\sin x, \cos x\}$ and x-axis, between the lines $x = \dfrac{\pi}{4}$ and $x = 2\pi$ is equal to :

(A) $\dfrac{(4\sqrt{2}-1)}{\sqrt{2}}$ sq. units

(B) $(4\sqrt{2}-1)$ sq. units

(C) $\dfrac{(4\sqrt{2}-1)}{2}$ sq. units

(D) None of these

वक्रों $y = \max\{\sin x, \cos x\}$ तथा x-अक्ष, रेखाओं $x = \dfrac{\pi}{4}$ व $x = 2\pi$ के मध्य का क्षेत्रफल होगा :

(A) $\dfrac{(4\sqrt{2}-1)}{\sqrt{2}}$ वर्ग इकाई

(B) $(4\sqrt{2}-1)$ वर्ग इकाई

(C) $\dfrac{(4\sqrt{2}-1)}{2}$ वर्ग इकाई

(D) इनमें से कोई नहीं

Ans. A

Sol. $y = \max\{\sin x, \cos x\}$, x-axis $x = \dfrac{\pi}{4}$ & 2π

$$A = \int_{\frac{\pi}{4}}^{\pi} \sin x\,dx - \int_{\pi}^{\frac{5\pi}{4}} \sin x\,dx - \int_{\frac{5\pi}{4}}^{\frac{3\pi}{2}} \cos x\,dx + \int_{\frac{3\pi}{2}}^{2\pi} \cos x\,dx$$

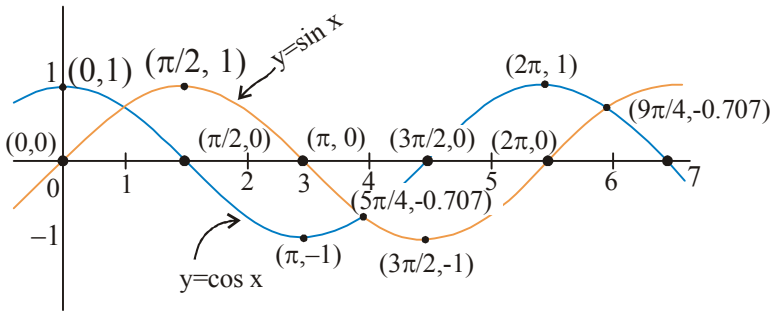
$$= \left(-\cos x\right]_{\frac{\pi}{4}}^{\pi} - \left[-\cos x\right]_{\pi}^{\frac{5\pi}{4}} - \left[\sin x\right]_{\frac{5\pi}{4}}^{\frac{3\pi}{2}} + \left[\sin x\right]_{\frac{3\pi}{2}}^{2\pi}$$

$$= \left[-\cos x\right]_{\frac{\pi}{4}}^{\pi} - \left[-\cos x\right]_{\pi}^{\frac{5\pi}{4}} - \left[\sin x\right]_{\frac{5\pi}{4}}^{\frac{3\pi}{2}} + \left[\sin x\right]_{\frac{3\pi}{2}}^{2\pi}$$

$$= 1 + \frac{1}{\sqrt{2}} + 1 - \frac{1}{\sqrt{2}} + 1 - \frac{1}{\sqrt{2}} + 1$$

$$= 4 - \frac{1}{\sqrt{2}}$$

$$= \frac{4\sqrt{2} - 1}{\sqrt{2}}$$



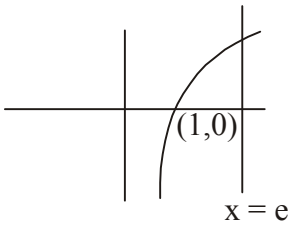
3. If A(n) represents the area bounded by the curve $y = n.\ell nx$, where $n \in N$ and $n > 1$, the x-axis and the lines $x = 1$ and $x = e$, then the value of $A(n) + nA(n-1)$ is equal to :

यदि A(n) वक्रों $y = n.\ell nx$, (जहाँ $n \in N$ तथा $n > 1$) x-अक्ष तथा रेखाओं $x = 1$ व $x = e$ के मध्य का क्षेत्रफल हो, तो $A(n) + nA(n-1)$ का मान होगा :

(A) $\frac{n^2}{e+1}$ (B) $\frac{n^2}{e-1}$ (C) n^2 (D) en^2

Ans. C

Sol. $y = n\ell nx$, $n \in N$, $n > 1$, x-axis



$$A(n) = n\int_1^e \ell n\, x \cdot dx$$

$$= n\left(x\,\ell n\, x - x\right)_1^e = n$$

$$\Rightarrow A(n) = n$$

$$A(n) + nA(n-1) = n + n(n-1)$$

$$= n^2$$

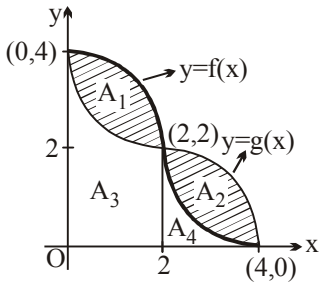
4. Suppose $y = f(x)$ and $y = g(x)$ are two continuous functions whose graphs intersect at the three points (0, 4), (2, 2) and (4, 0) with $f(x) > g(x)$ for $0 < x < 2$ and $f(x) < g(x)$ for $2 < x < 4$. If $\int_0^4 [f(x) - g(x)]dx = 10$ and $\int_2^4 [g(x) - f(x)]dx = 5$, the area between two curves for $0 < x < 2$, is :

माना $y = f(x)$ तथा $y = g(x)$ दो सतत् फलन है जिनके वक्र तीन बिन्दुओं (0, 4),(2, 2) तथा (4, 0) पर प्रतिच्छेद होते हैं साथ ही $f(x) > g(x)$ for $0 < x < 2$ तथा $f(x) < g(x)$ जब $2 < x < 4$. यदि $\int_0^4 [f(x) - g(x)]dx = 10$ तथा $\int_2^4 [g(x) - f(x)]dx = 5$, तो nksuksa oØksa ds e/; {ks=kQy] tks fd $0 < x < 2$ के लिए होगा :

(A) 5 (B) 10 (C) 15 (D) 20

Ans. C

Sol. Given $\int_0^4 f(x)\,dx - \int_0^4 g(x)\,dx = 10$



$(A_1 + A_3 + A_4) - (A_2 + A_3 + A_4) = 10$

$A_1 - A_2 = 10$ ....(1)

again $\int_2^4 g(x)\,dx - \int_2^4 f(x)\,dx = 5$

$(A_2 + A_4) - A_4 = 5$

$A_2 = 5$ .....(2)

$\therefore$ (1) + (2)

$A_1 = 15$

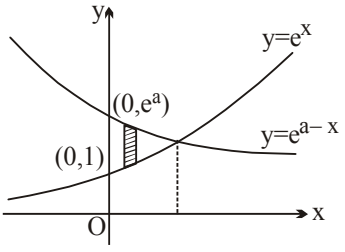
5. Let 'a' be a positive constant number. Consider two curves C_1: $y = e^x$, C_2 : $y = e^{a-x}$. Let S be the area of the part surrounding by C_1, C_2 and the y-axis, then $\lim_{a\to 0} \frac{S}{a^2}$ equals :

'a' एक धनात्मक अचर है। मान लीजिए दो वक्र है। C_1: $y = e^x$, C_2 : $y = e^{a-x}$. माना S उस भाग का क्षेत्रफल है जो C_1, C_2 तथा y-अक्ष }kjk ki fjc) g\$r ks $\lim_{a\to 0} \frac{S}{a^2}$ का मान होगा :

(A) 4 (B) 1/2 (C) 0 (D) 1/4

Ans. D

Sol. Solving $e^x = e^{a-x}$, we get



$e^{2x} = e^a \Rightarrow \quad x = \frac{a}{2}$

$S = \int_0^{a/2} (e^a \cdot e^{-x} - e^x)\,dx = \left[-(e^a \cdot e^{-x} + e^x)\right]_0^{a/2}$

$= (e^a + 1) - (e^{a/2} + e^{a/2}) = e^a - 2e^{a/2} + 1 = (e^{a/2} - 1)^2$

$\therefore \quad \frac{S}{a^2} = \left(\frac{e^{a/2}-1}{a}\right)^2 = \frac{1}{4}\left(\frac{e^{a/2}-1}{a/2}\right)^2$

$\therefore \quad \lim_{a\to 0} \frac{S}{a^2} = \frac{1}{4}$

6. The area of the region(s) enclosed by the curves $y = x^2$ and $y = \sqrt{|x|}$ is :

वक्रों $y = x^2$ तथा $y = \sqrt{|x|}$ द्वारा परिबद्ध क्षेत्र (क्षेत्रों) का क्षेत्रफल होगा :

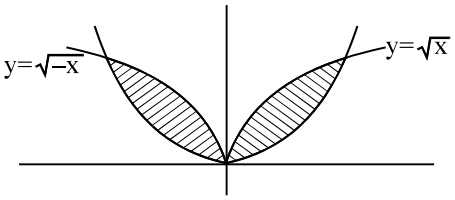
(A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $\frac{1}{6}$ (D) 1

Ans. B

Sol. $A = \left(\frac{16ab}{3}\right)\cdot 2$

$a = \frac{1}{4};\ b = \frac{1}{4}$

$A = \frac{2}{3}$



7. The slope of the tangent to a curve $y = f(x)$ at $(x, f(x))$ is $2x + 1$. If the curve passes through the point $(1, 2)$ then the area of the region bounded by the curve, the x-axis and the line $x = 1$ is :

वक्र $y = f(x)$ के बिन्दु $(x, f(x))$ पर स्पर्श रेखा की प्रवणता $2x + 1$ है। यदि वक्र बिन्दु $(1, 2)$ से गुजरता है तो वक्र का x-अक्ष तथा रेखा $x = 1$ द्वारा परिबद्ध क्षेत्र का क्षेत्रफल होगा :

(A) $\frac{5}{6}$ (B) $\frac{6}{5}$ (C) $\frac{1}{6}$ (D) 1

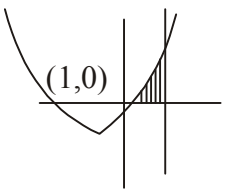
Ans. A

Sol. $\frac{dy}{dx} = 2x + 1$ (1,2)

$y = x^2 + x + c$

$y = x^2 + x$

$y + \frac{1}{4} = \left(x + \frac{1}{2}\right)^2$



$$A = \int_0^1 \left(x^2 + x\right) dx$$

$$A = \frac{x^3}{3} + \frac{x^2}{2}\Bigg|_0^1$$

$$= \frac{1}{3} + \frac{1}{2}$$

$$= \frac{5}{6}$$

8. The area bounded by the curves $y = -\sqrt{-x}$ and $x = -\sqrt{-y}$ where $x, y \le 0$:

(A) cannot be determined

(B) is 1/3

(C) is 2/3

(D) is same as that of the figure bounded by the curves $y = \sqrt{-x}$; $x \le 0$ and $x = \sqrt{-y}$; $y \le 0$

वक्रो $y = -\sqrt{-x}$ तथा $x = -\sqrt{-y}$ जहां $x, y \le 0$ द्वारा परिबद्ध क्षेत्रफल होगा :

(A) ज्ञात नहीं किया जा सकता

(B) 1/3

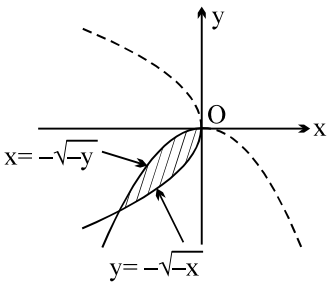
(C) 2/3

(D) वही होगा जो वक्रो $y = \sqrt{-x}$; $x \le 0$ तथा $x = \sqrt{-y}$; $y \le 0$ द्वारा परिबद्ध है

Ans. B

Sol. $y = -\sqrt{-x} \Rightarrow y^2 = -x$ where x & y both (–) ve

$x = -\sqrt{-y} \Rightarrow x^2 = -y$ where x & y both (–) ve



Hence $A = \frac{16ab}{3}$

where $a = b = \frac{1}{4}$

$\therefore \quad A = \frac{1}{3} \Rightarrow$ (B)

9. The area bounded by the curve $y = x e^{-x}$; $xy = 0$ and $x = c$ where c is the x-coordinate of the curve's inflection point, is :

वक्र $y = xe^{-x}$; $xy = 0$ तथा $x = c$ द्वारा घिरा क्षेत्रफल होगा, जहाँ वक्र के नति परिवर्तन बिन्दु का x-निर्देशांक c है :

(A) $1 - 3e^{-2}$ (B) $1 - 2e^{-2}$ (C) $1 - e^{-2}$ (D) 1

Ans. A

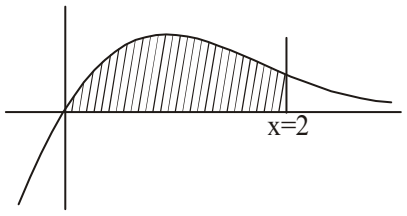
Sol. $y = xe^{-x}$

$y' = e^{-x} - x e^{-x} = (1 - x)e^{-x} \uparrow$ for $x < 1$

$y'' = - e^{-x} - [e^{-x} - xe^{-x}] = e^{-x}[-1 - 1 + x]$

$= (x - 2)e^{-x}$

for point of inflection $y'' = 0 \quad \Rightarrow \quad x = 2$



$$A = \int_0^2 xe^{-x}\,dx = -\,xe^{-x}\Big|_0^2 + \int_0^2 e^{-x}$$

$= (-2\,e^{-2}) - \left(e^{-x}\right)_0^2$

$= -2\,e^{-2} - (e^{-2} - 1) = 1 - e^{-2} - 2e^{-2} = 1 - 3e^{-2}$

10. Area enclosed by the curve $y = (x^2 + 2x)e^{-x}$ and the positive x-axis is :

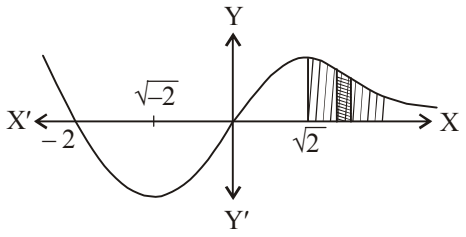
धनात्मक x-अक्ष तथा वक्र $y = (x^2 + 2x)e^{-x}$ द्वारा घिरा क्षेत्रफल होगा :

(A) 1 (B) 2 (C) 4 (D) 6

Ans. C

Sol. Required area $A = \int_0^{\infty} \underset{\text{(II)}}{e^{-x}} \underset{\text{(I)}}{(x^2 + 2x)}\,dx = 4$ Ans.

(I. B. P.)



11. Infinite rectangles each of width 1 unit and height $\left(\frac{1}{n} - \frac{1}{n+1}\right)$ $(n \in N)$ are constructed such that ends of exactly one diagonal of every rectangle lies along the curve $y = \frac{1}{x}$. The sum of areas of all such rectangles, is :

1 इकाई लम्बाई तथा $\left(\frac{1}{n} - \frac{1}{n+1}\right)$ इकाई चौड़ाई (जहाँ $n \in N$) के अनन्त आयत इस प्रकार है कि प्रत्येक आयत का केवल एक fod . kZ d s fl jk oØ $y = \frac{1}{x}$ पर स्थित है, तो इस प्रकार के सभी आयतों के क्षेत्रफल का योग होगा :

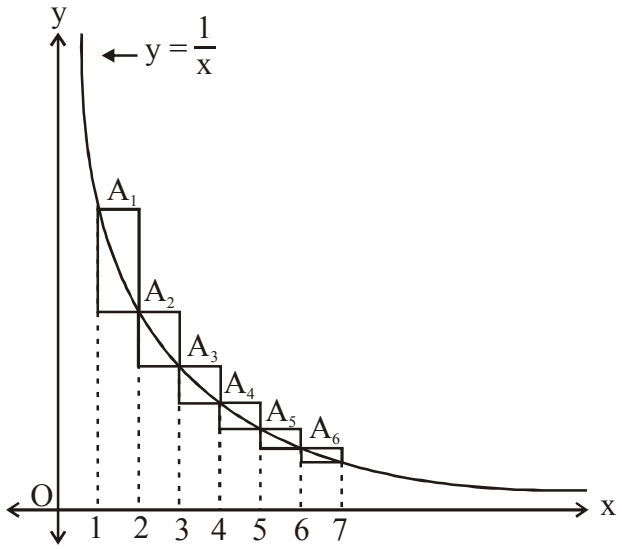
(A) $\frac{1}{2}$ (B) $\frac{2}{3}$ (C) $\frac{3}{4}$ (D) 1

Ans. D

Sol. Method-I:

$$\text{Required sum} = \lim_{n\to\infty}\left[\left(1-\frac{1}{2}\right)+\left(\frac{1}{2}-\frac{1}{3}\right)+\left(\frac{1}{3}-\frac{1}{4}\right)+.....+\left(\frac{1}{n}-\frac{1}{n+1}\right)\right] = \lim_{n\to\infty}\left(1-\frac{1}{n+1}\right) = 1$$

Method-II:



$$\text{Area of } r^{th} \text{ rectangle } A_r = (\text{height}) \times (\text{width}) = \left(\frac{1}{r}-\frac{1}{r+1}\right)\cdot 1$$

$$\text{Sum of areas} = \lim_{n\to\infty}\sum_{r=1}^{n} A_r = \lim_{n\to\infty}\left[\left(1-\frac{1}{2}\right)+\left(\frac{1}{2}-\frac{1}{3}\right)+\left(\frac{1}{3}-\frac{1}{4}\right)+.....+\left(\frac{1}{n}-\frac{1}{n+1}\right)\right]$$

$$= \lim_{n\to\infty}\left(1-\frac{1}{n+1}\right) = 1$$

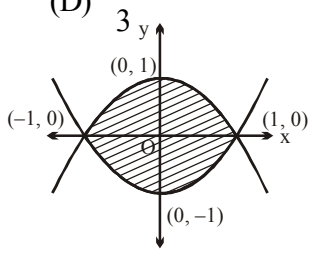
12. Area enclosed by the figure described by the equation $x^4 + 1 = 2x^2 + y^2$, is :

समीकरण $x^4 + 1 = 2x^2 + y^2$ से घिरे क्षेत्र का क्षेत्रफल होगा :

(A) 3 (B) $\frac{8}{3}$ (C) $\frac{7}{3}$ (D) $\frac{10}{3}$

Ans. B

Sol. We have $y^2 = (x^2 - 1)^2 \Rightarrow y = x^2 - 1$ or $y = 1 - x^2$



$$\text{So, required area} = 4\int_0^1 (1-x^2)dx = 4\left[x-\frac{x^3}{3}\right]_0^1 = \frac{8}{3}$$

13. Let $f:[-1,2]\to[0,\infty)$ be a continuous function such that $f(x)=f(1-x)$ for all $x\in[-1,2]$. Let $R_1=\int_{-1}^{2} x\,f(x)\,dx$, and R_2 be the area of the region bounded by $y=f(x)$, $x=-1$, $x=2$, and the x-axis. Then : **[JEE-2011]**

यदि $f:[-1,2]\to[0,\infty)$ एक सतत् फलन इस प्रकार है कि $f(x)=f(1-x)$ जहाँ $x\in[-1,2]$ यदि $R_1=\int_{-1}^{2} x\,f(x)\,dx$ तथा R_2 वक्रों $y=f(x)$, $x=-1$, $x=2$, तथा x-अक्ष से घिरा हुआ क्षेत्रफल है, तो : **[JEE-2011]**

(A) $R_1=2R_2$ (B) $R_1=3R_2$ (C) $2R_1=R_2$ (D) $3R_1=R_2$

Ans. C

Sol. $R_2=\int_{-1}^{2} f(x)dx$, $R_1=\int_{-1}^{2} x f(x)dx$

$=\int_{-1}^{2}(1-x)f(1-x)dx \quad \left(\because \int_a^b f(x)dx=\int_a^b f(a+b-x)dx\right)$

$=\int_{-1}^{2}(1-x)f(x)dx \quad (\text{given } f(x)=f(1-x))$

$=\int_{-1}^{2} f(x)dx-\int_{-1}^{2} x f(x)dx$

or $R_1=R_2-R_1 \quad \Rightarrow 2R_1=R_2$

Exercise C - II

SUBJECTIVE / INTEGER TYPE QUESTIONS

1. Find the area bounded by the curve $y = 2x - x^2$ & the straight line $y = -x$.

वक्र $y = 2x - x^2$ तथा सरल रेखा $y = -x$ द्वारा परिबद्ध क्षेत्रफल होगा।

Ans. $\frac{9}{2}$

Sol. $y = 2x - x^2$

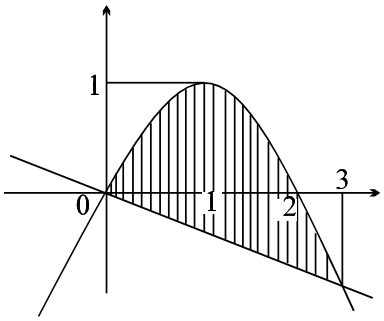
$$\frac{dy}{dx} = 2 - 2x = 0 \Rightarrow x = 1$$

solving $y = 2x - x^2$ and $y = -x$, we have

$$-x = 2x - x^2 \Rightarrow x^2 - 3x = 0 \Rightarrow x = 0 \text{ or } 3$$

$$A = \int_0^3 \left[\left(2x - x^2\right) - (-x)\right] dx = \int_0^3 \left(3x - x^2\right) dx$$

$$= \frac{3x^2}{2} - \frac{x^3}{3}\Big]_0^3 = \frac{27}{2} - 9 = \frac{9}{2} \text{ sq. unit}$$



2. Find the area enclosed by the parabola $y = 1 + x^2$ and a normal drawn to it with gradient -1.

परवलय $y = 1 + x^2$ तथा इसके अभिलम्ब जिसका ढाल -1 है द्वारा परिबद्ध क्षेत्रफल होगा।

Ans. $\frac{4}{3}$

Sol. $y = 1 + x^2 \quad \Rightarrow \quad \frac{dy}{dx} = 2x$ [Ans. $A = \frac{4}{3}$ sq. units]

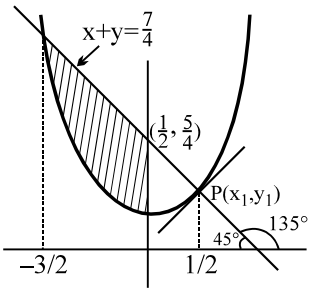
Hence $m = -\frac{1}{2x_1}$ (m = slope of normal at x_1, y_1)

$$\therefore \quad x_1 = \frac{-1}{2m} = \frac{1}{2} \quad ; \quad y_1 = 1 + \frac{1}{4m^2} = \frac{5}{4}$$

equation of normal $\quad y - \left(1 + \frac{1}{4m^2}\right) = m\left(x + \frac{1}{2m}\right)$

put $m = -1$, we get $y - \frac{5}{4} = -\left(x - \frac{1}{2}\right)$

$$x + y = \frac{5}{4} + \frac{1}{2} = \frac{7}{4} \qquad \ldots.(1)$$



$$A = \int_{x_1}^{x_2}\left(\frac{7}{4} - x\right) - (1 + x^2)\,dx = \int_{x_1}^{x_2}\left(\frac{3}{4} - x - x^2\right)dx$$

now $1 + x^2 = -x + \frac{7}{4}$

$4x^2 + 4x - 3 = 0 \Rightarrow x = -\frac{3}{2}$ and $x = \frac{1}{2}$

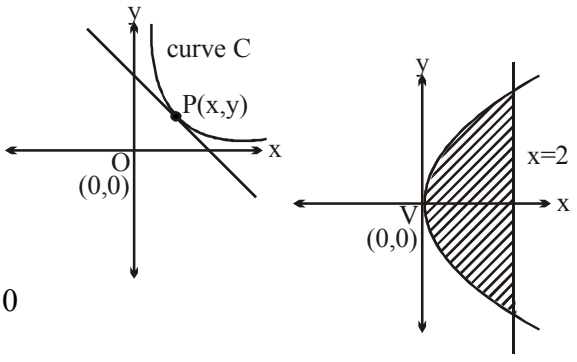
$\therefore \quad A = \int_{-3/2}^{1/2}\left(\frac{3}{4} - x - x^2\right)dx$ simplifying $A = \frac{4}{3}$ sq. units

3. Let C be a curve passing through M (2, 2) such that the slope of the tangent at any point to the curve is reciprocal of the ordinate of the point. If the area bounded by curve C and line $x = 2$ is expressed as a rational $\frac{p}{q}$ (where p and q are in their lowest form), then find $(p + q)$.

वक्र C बिन्दु M (2, 2) से गुजरता है जिसके किसी बिन्दु पर स्पर्श रेखा का ढाल उस बिन्दु की कोटि का व्युत्क्रम है। यदि वक्र C तथा रेखा $x = 2$ द्वारा परिबद्ध क्षेत्रफल $\frac{p}{q}$ है। (जहाँ p तथा q सह अभाज्य है।), तो $(p + q)$ का मान ज्ञात कीजिए।

Ans. 19

Sol. Let P (x, y) be any point on the curve C.



Now, $\frac{dy}{dx} = \frac{1}{y}$

$\Rightarrow ydy = dx \Rightarrow \frac{y^2}{2} = x + k$

Since the curve passes through M (2,2), so $k = 0$

$\Rightarrow y^2 = 2x$

Hence required area $= 2\int_0^2 \sqrt{2x}\,dx = 2\sqrt{2} \times \frac{2}{3}\left(x^{3/2}\right)_0^2 = \frac{4}{3}\sqrt{2} \times 2\sqrt{2} = \frac{16}{3}$ (square unit)

$\Rightarrow \quad p + q = 19$

4. Let l be the line created by rotating the tangent line to the parabola $y = x^2$ at the point A(1, 1), about A by an angle of $(-\pi/4)$. Let B be the other intersection of the line l with $y = x^2$. If the area enclosed by the line l and the parabola is $\frac{a_1}{a_2}$ where a_1 and a_2 are coprime, find $(a_1 + a_2)$.

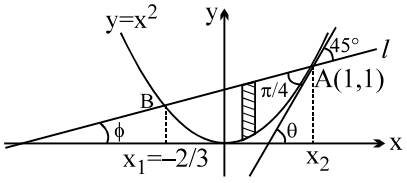
परवलय $y = x^2$ के बिन्दु A(1, 1) पर खीचीं स्पर्श रेखा को बिन्दु A के सापेक्ष $(-\pi/4)$ कोण घुमा देने पर नई रेखा l पुनः परवलय $y = x^2$ को बिन्दु B पर काटती है। यदि रेखा l तथा परवलय द्वारा परिबद्ध क्षेत्रफल $\frac{a_1}{a_2}$ हो (a_1 एवं a_2 सापेक्षतः अभाज्य है) तो $(a_1 + a_2)$ ज्ञात कीजिए।

Ans. 287

Sol. $y = x^2$

$$\left.\frac{dy}{dx}\right]_{A(1,1)} = 2x = 2; \quad \text{hence, } \tan\theta = 2 \qquad ....(1)$$

Let the line l makes an angle ϕ with the x-axis



then, $\phi = \theta - \frac{\pi}{4}; \quad m_l = \tan\phi = \frac{\tan\theta - 1}{1 + \tan\theta} = \frac{1}{3}$

$\therefore$ equation of the line l is,

$$y - 1 = \frac{1}{3}(x - 1) \quad \Rightarrow \quad 3y - 3 = x - 1 \Rightarrow \quad 3y = x + 2$$

solving line and parabola $y = x^2$,

$$3x^2 = x + 2 \Rightarrow 3x^2 - x - 2 = 0$$

$$x_1 x_2 = -\frac{2}{3}; \quad \text{but } x_2 = 1; \quad \therefore \quad x_1 = -\frac{2}{3}$$

$$\therefore \quad A = \int_{-2/3}^{1}\left(\frac{x+2}{3} - x^2\right)dx = \frac{1}{3}\int_{-2/3}^{1}\left(x + 2 - 3x^2\right)dx = \frac{1}{3}\left[\frac{x^2}{2} + 2x - x^3\right]_{-2/3}^{1}$$

$$= \frac{1}{3}\left[\left(\frac{1}{2} + 2 - 1\right) - \left(\frac{2}{9} - \frac{4}{3} + \frac{8}{27}\right)\right] = \frac{1}{3}\left[\frac{3}{2} - \left(\frac{6 - 36 + 8}{27}\right)\right] = \frac{1}{3}\left[\frac{3}{2} + \frac{22}{27}\right]$$

$$= \frac{1}{3}\left[\frac{81 + 44}{54}\right] = \frac{125}{162} = \frac{a_1}{a_2} \quad \Rightarrow \quad a_1 + a_2 = 287$$

5. Let $f(x) = x - x^2$ and $g(x) = ax, a < 0$. If the area bounded by $f(x)$ and $g(x)$ is equal to the area bounded by the curves $x = 3y - y^2$ and $x + y = 3$, then find the value of $|[a]|$

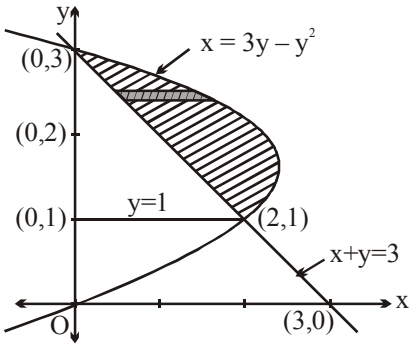
[Note : [k] denotes the greatest integer less than or equal to k.]

माना $f(x) = x - x^2$ एवं $g(x) = ax$ है। यदि $f(x)$ व $g(x)$ द्वारा परिबद्ध क्षेत्रफल वक्रों $x = 3y - y^2$ एवं $x + y = 3$ द्वारा परिबद्ध क्षेत्रफल के बराबर है, तो $|[a]|$ का मान ज्ञात कीजिए।

[नोट : $[k]$ महत्तम पूर्णांक को दर्शाता है जो 'k' के बराबर व 'k' से छोटा है]

Ans. 1

Sol. Area between $x = 3y - y^2$ and $x + y = 3$



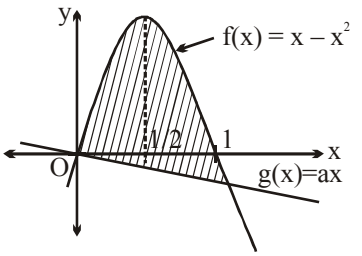
$$A = \int_1^3 \left(3y - y^2\right) - (3 - y)\,dy$$

$$= 4 \cdot \frac{y^2}{2} - \frac{y^3}{3} - 3y \Big]_1^3 = (2 \cdot 9 - 9 - 9) - \left(2 - \frac{1}{3} - 3\right)$$

$$A = \frac{4}{3}$$

Now area bounded by $f(x)$ above the x-axis is $= \int_0^1 \left(x - x^2\right)dx = \frac{1}{6}$, hence $a < 0$ (think)

Given, $\int_0^{1-a} \left((x - x^2) - (a\,x)\right)dx = \frac{4}{3}$



Note :
$x - x^2 = ax$
$x^2 + (a-1)x = 0$
$x_1 + x_2 = 1 - a$
but $x_1 = 0$
$\Rightarrow x_2 = 1 - a$

Solving $\frac{1}{6}(1-a)^3 = \frac{4}{3}$

$(1-a)^3 = 8$

$1 - a = 2 \Rightarrow a = -1$

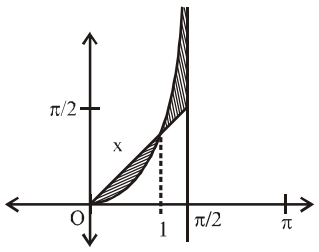
$\Rightarrow \quad |[a]| = 1$

6. If the area bounded by the curves $f(x) = \left(\cos^{-1}|\cos x|\right)^2$, $g(x) = \left(\cos^{-1}|\cos x|\right)$ and $|x| = \frac{\pi}{2}$ is $a\pi^3 + b\pi^2 + c$, then find the minimum value of $(|a| + |b| + |c|)$.

यदि वक्रों $f(x) = \left(\cos^{-1}|\cos x|\right)^2$, $g(x) = \left(\cos^{-1}|\cos x|\right)$ एवं $|x| = \frac{\pi}{2}$ द्वारा परिबद्ध क्षेत्रफल $a\pi^3 + b\pi^2 + c$ है, तो $(|a| + |b| + |c|)$ का न्यूनतम मान ज्ञात कीजिए।

Ans. 1

Sol. We have $f(x) = \begin{bmatrix} x^2 ; & x \in \left[0, \frac{\pi}{2}\right] \\ (\pi - x)^2 ; & x \in \left(\frac{\pi}{2}, \pi\right) \end{bmatrix}$; $g(x) = \begin{bmatrix} x ; & x \in \left[0, \frac{\pi}{2}\right] \\ (\pi - x); & x \in \left(\frac{\pi}{2}, \pi\right) \end{bmatrix}$



Both f and g are periodic with period = π and symmetric about x = π/2

$\therefore$ Required area $= 2\left(\int_0^1 (x - x^2)\,dx + \int_1^{\pi/2} (x^2 - x)\,dx\right)$

$$= 2\left(\left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{\pi^3}{24} - \frac{1}{3}\right) - \left(\frac{\pi^2}{8} - \frac{1}{2}\right)\right)$$

$$= 2\left(\frac{\pi^3}{24} - \frac{\pi^2}{8} + \frac{1}{3}\right) = \frac{\pi^3}{12} - \frac{\pi^2}{4} + \frac{2}{3} \equiv a\pi^3 + b\pi^2 + c \text{ (Given)}$$

So, $a = \frac{1}{12}$, $b = \frac{-1}{4}$, $c = \frac{2}{3}$

Hence, $(|a| + |b| + |c|) = \left|\frac{1}{12}\right| + \left|\frac{-1}{4}\right| + \left|\frac{2}{3}\right| = \frac{1}{12} + \frac{1}{4} + \frac{2}{3} = \frac{1+3+8}{12} = \frac{12}{12} = 1$

7. A polynomial function $f(x)$ satisfies the condition $f(x+1) = f(x) + 2x + 1$. Find $f(x)$ if $f(0) = 1$. Find also the equations of the pair of tangents from the origin on the curve $y = f(x)$ and compute the area enclosed by the curve and the pair of tangents.

एक बहुपदीय फलन $f(x)$, $f(x+1) = f(x) + 2x + 1$ का सन्तुष्ट करता है। यदि $f(0) = 1$ तब $f(x)$ ज्ञात कीजिए। साथ ही मूल बिन्दु से वक्र $y = f(x)$ पर स्पर्श रेखा युग्म के समीकरणों का ज्ञात कीजिए तथा वक्र $y = f(x)$ और स्पर्श रेखाओं से परिबद्ध क्षेत्रफल ज्ञात कीजिए।

Ans. $f(x) = x^2 + 1$; $y = \pm 2x$; $A = \frac{2}{3}$ sq. units

Sol. $f(x+1) = f(x) + 2x + 1$

$\Rightarrow f''(x+1) = f''(x) \quad \forall x \in R$

Let $f''(x) = a \quad \Rightarrow \quad f'(x) = ax + b$

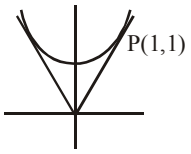
$\Rightarrow f(x) = \frac{ax^2}{2} + bx + c$

$\Rightarrow c = 1 \quad [\because f(0) = 1]$

Now $f(x+1) - f(x) = 2x + 1$

$\Rightarrow \left[\frac{a}{2}(x+1)^2 + b(x+1) + c\right] - \left[\frac{ax^2}{2} + bx + c\right] = 2x + 1$

$\Rightarrow ax + \frac{a}{2} + b = 2x + 1$



on comparing we get $a = 2$,

or $\frac{a}{2} + b = 1 \Rightarrow b = 0$

$\therefore f(x) = x^2 + 1$ ... (i)

Now let equation of tangent be $y = mx$ ... (ii)

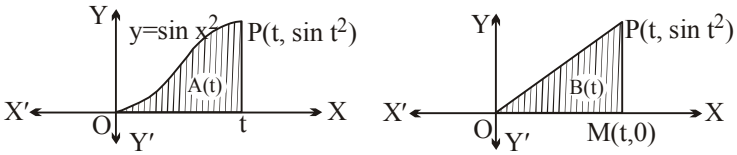
from (i) and (ii)

$x^2 - mx + 1 = 0 \Rightarrow m = \pm 2$

$\therefore$ tangent are $y = 2x$ or $y = -2x$

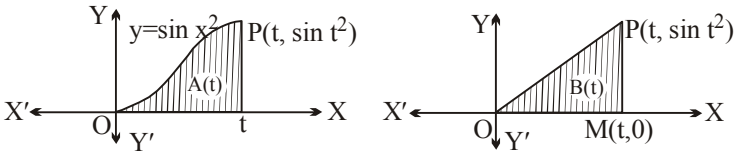
$$A = 2\int_0^1 (x^2 + 1 - 2x)dx = \frac{2}{3}$$

8. The figure shows two regions in the first quadrant.



A(t) is the area under the curve $y = \sin x^2$ from 0 to t and B(t) is the area of the triangle with vertices O, P and M(t, 0). Find $\lim_{t\to 0} \frac{A(t)}{B(t)}$.

चित्र प्रथम चतुर्थांश में दो क्षेत्रो को दर्शाते है।



वक्र $y = \sin x^2$ का $x = 0$ से t में क्षेत्रफल A(t) और B(t) उस त्रिभुज का क्षेत्रफल जिसके शीर्ष O, P और M(t, 0) है तब $\lim_{t\to 0} \frac{A(t)}{B(t)}$ का मान ज्ञात कीजिए।

Ans. 2/3

Sol. $A(t) = \int_0^t \sin x^2 dx$ & $B(t) = \frac{1}{2} \times t \sin t^2$

$$\lim_{t\to 0}\frac{A(t)}{B(t)}=\lim_{t\to 0}\frac{\int_0^t \sin x^2 dx}{\frac{1}{2}t\cdot\sin t^2}$$

$$=\lim_{t\to 0}\left[\frac{2\int_0^t \sin x^2 dx}{t^3}\right]$$

DLH

$$=\lim_{t\to 0}\frac{2\sin t^2}{3t^2}$$

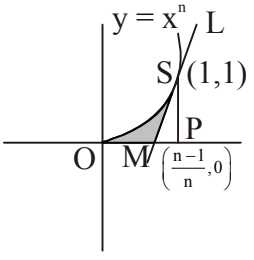
$$=\frac{2}{3}$$

9. Consider the curve $y = x^n$ where $n > 1$ in the 1st quadrant. If the area bounded by the curve, the x-axis and the tangent line to the graph of $y = x^n$ at the point (1, 1) is maximum then find the value of n.

माना प्रथम चतुर्थांशों में वक्र $y = x^n$ जहाँ $n > 1$ है। यदि वक्र $y = x^n$, x-अक्ष और बिन्दु (1, 1) पर स्पर्श रेखा से परिबद्ध क्षेत्र का क्षेत्रफल अधिकतम है तब n का मान ज्ञात कीजिए।

Ans. $\sqrt{2}+1$

Sol. equation of line L tangent at (1,1)



$y = x^n$

$$\frac{dy}{dx}=nx^{n-1}$$

$$m_T=n$$

$$L: y=nx-n+1 \Rightarrow \text{x-intercept}\left(\frac{n-1}{n},0\right)$$

shaded area

$$A=\int_0^1 x^n dx-\frac{1}{2}\times\left(1-\frac{n-1}{n}\right)\times 1$$

$$A = \frac{1}{n+1} - \frac{1}{2}\left(\frac{1}{n}\right)$$

$$A = \frac{1}{n+1} - \frac{1}{2n}$$

$$A = \frac{n-1}{2n(n+1)}$$

$$\frac{dA}{dn} = \frac{2n(n+1) - 2(n-1)n - 2(n-1)(n+1)}{4n^2(n+1)^2} = 0$$

$= n^2 + n - n^2 + n - n^2 + 1 = 0$

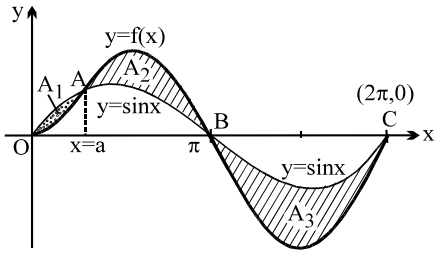
$-n^2 + 2n + 1 = 0$

$n^2 - 2n - 1 = 0$

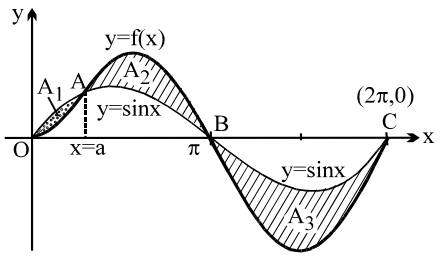
$$n = \frac{2 \pm \sqrt{4+4}}{2}$$

$n = 1 \pm \sqrt{2} \qquad \Rightarrow n \neq 1 - \sqrt{2}\,,\ n = 1 + \sqrt{2}$

10. In the adjacent figure, graphs of two functions y = f(x) and y = sinx are given. y = sinx intersects, y = f(x) at A(a, f(a)); B(π, 0) and C(2π, 0). A_i (i = 1, 2, 3,) is the area bounded by the curves y = f(x) and y = sinx between x=0 and x= a; i = 1, between x = a and x = π; i = 2, between x = π and x = 2π; i = 3. If $A_1 = 1 - \sin a + (a-1)\cos a$, determine the function f(x). Hence determine 'a' and A_1. Also calculate A_2 and A_3.



दिया गया चित्र फलन y = f(x) और y = sin x को दर्शाता है। यदि y = sin x, y = f(x) को A(a, f(a)); B(π, 0) और C(2π, 0) पर प्रतिच्छेदित करता है तथा A_i (i = 1, 2, 3,) वक्रों y = f(x) और y = sin x, x = 0 और x = a; i = 1, x = a और x = π; i = 2, और x = π और x = 2π; i = 3 से क्रमशः परिबद्ध क्षेत्रफल है। यदि $A_1 = 1 - \sin a + (a-1)\cos a$, तथा फलन f(x) ज्ञात कीजिए और a का मान बतलाईये, साथ ही A_2 और A_3 ज्ञात कीजिए।



Ans. $f(x) = x\sin x,\ a = 1;\ A_1 = 1 - \sin 1;\ A_2 = \pi - 1 - \sin 1;\ A_3 = (3\pi - 2)$ sq. units

Sol. $A_1 = \int_0^a \left(\sin x - f(x)\right)\cdot dx$ _____(i)

$1 - \sin A + (a-1)\cos a = -(\cos x)_0^a - \int_0^a f(x)\cdot dx$

$\int_0^a f(x)\cdot dx = -\cos a + 1 - 1 + \sin a - (a-1)\cos a$

$\int_0^a f(x)\cdot dx = \sin a - a\cos a$

By Newton Leibnitz Rule

$f(a) = \cos a - \left[a(-\sin a) + \cos a\right]$

$f(a) = \cos a + a\sin a - \cos a$

$\Rightarrow f(a) = a\sin a$

$\therefore f(x) = x\sin x$ _____(ii)

from $y = \sin x$ and $y = x\sin x$ point of intersection is "a"

$x\sin x = \sin x$

$(x-1)\sin x = 0$

$\Rightarrow \sin x = n\pi$ or $x = 1$

Hence $a = 1$

Now $A_1 = 1 - \sin 1$

$A_2 = \int_1^{\pi} (x\sin x - \sin x)\,dx$

$$A_2 = \int_1^{\pi} (x-1)\sin x \cdot dx$$

$$A_2 = \left(-(x-1)\cos x\right)_1^{\pi} + \int_1^{\pi} \cos x \cdot dx$$

$$A_2 = -\left((x-1)\cos x\right)_1^{\pi} + \sin\pi - \sin 1$$

$$A_2 = -\left[(\pi-1)\cos\pi\right] - \sin 1$$

$$A_2 = \pi - 1 - \sin 1$$

Again $$A_3 = \int_{\pi}^{2\pi} \left|x\sin x - \sin x\right| \cdot dx$$

$$A_3 = 3\pi - 2$$

11. Consider the two curves $y = 1/x^2$ and $y = 1/[4(x-1)]$.

(i) At what value of 'a' ($a > 2$) is the reciprocal of the area of the fig. bounded by the curves, the lines $x = 2$ and $x = a$ equal to 'a' itself?

(ii) At what value of 'b' ($1 < b < 2$) the area of the figure bounded by these curves, the lines $x = b$ and $x = 2$ equal to $1 - 1/b$.

माना दो वक्र $y = 1/x^2$ और $y = 1/[4(x-1)]$ दिये गये हैं।

(i) ५ ६; झ 2द्धके किस मान के लिये दिये गये वक्रों और रेखाओं $x = 2$ व $x = a$ से परिबद्ध क्षेत्र के क्षेत्रफल का प्रतिलोम भी 'a' स्वयं है ?

(ii) ५झ६ ;1 ढ इ ढ 2द्धके किस मान के लिये दिये गये वक्रों तथा रेखाओं $x = b$ और $x = 2$ से परिबद्ध क्षेत्र का क्षेत्रफल $1 - 1/b$ होगा।

Ans. $a = 1 + e^2$, $b = 1 + e^{-2}$

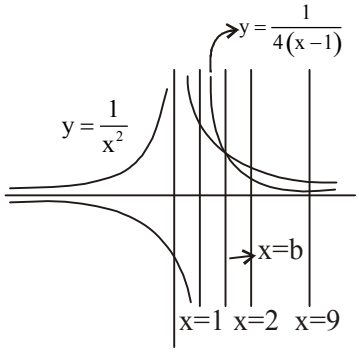
Sol. $y = \dfrac{1}{x^2}$ $\qquad$ $y = \dfrac{1}{4(x-1)}$

(i) Point of intersection

$$\frac{1}{x^2} = \frac{1}{4(x-1)}$$

$x^2 - 4x + 4 = 0$

$x = 2$



$$A = \int_2^a \left(\frac{1}{4(x-1)} - \frac{1}{x^2} \right) dx$$

$$= \frac{1}{4}\log(x-1) + \frac{1}{x}\Big|_2^a$$

$$= \frac{1}{4}\log(a-1) + \frac{1}{a} - \frac{1}{2}$$

Given $A = \frac{1}{a}$

$$\frac{1}{4}\log(a-1) + \frac{1}{a} - \frac{1}{2} = \frac{1}{a}$$

$a - 1 = e^2 \Rightarrow a = 1 + e^2$

(ii) $A = \int_b^2 \frac{1}{4(n-1)} - \left(\frac{1}{x^2} \right) dx$

$$= \frac{1}{4}\log(x-1) + \frac{1}{2}\Big|_b^2$$

$$= +\frac{1}{2} - \frac{1}{b} - \frac{1}{4}\log(b-1)$$

Given $A = 1 - \frac{1}{b}$

$$+\frac{1}{2} - \frac{1}{b} - \frac{1}{4}\log(b-1) = 1 - \frac{1}{b}$$

$$-\frac{1}{4}\log(b-1) = \frac{1}{2}$$

$\log(b-1) = -2 \Rightarrow b = 1 + e^{-2}$

12. Show that the area bounded by the curve $y = \frac{\ell n\, x - c}{x}$, the x-axis and the vertical line through the maximum point of the curve is independent of the constant c.

सिद्ध कीजिये कि वक्र $y = \frac{\ell n\, x - c}{x}$, x-अक्ष और वक्र के उच्चिष्ठ बिन्दु से खींची गयी उर्ध्वाधर रेखा से परिबद्ध क्षेत्र का क्षेत्रफल अचर 'c' से स्वतंत्र है।

Ans. 1/2

Sol. $y = \frac{\ell nx - C}{x}$

at $y = 0_{x\,=\,e^{c}}$

$$\frac{dy}{dx} = \frac{1-(\ell nx - c)}{x^2} = 0$$

$$\Rightarrow x = e^{1+c}$$

$$A = \int_{e^c}^{e^{1+c}} \left(\frac{\ell nx - c}{x}\right) dx$$

$$A = \frac{(\ell nx - c)^2}{2}\Bigg|_{e^c}^{e^{1+c}}$$

$$A = \frac{1}{2}$$

13. For what value of 'a' is the area of the figure bounded by the lines, $y = \frac{1}{x}$, $y = \frac{1}{2x-1}$, $x = 2$ and $x = a$, $a > 0$ equal to $\ell n \frac{4}{\sqrt{5}}$?

'a' के किस मान के लिये वक्रों, $y = \frac{1}{x}$, $y = \frac{1}{2x-1}$, $x = 2$ और $x = a$, $a > 0$ से परिबद्ध क्षेत्रफल $\ell n \frac{4}{\sqrt{5}}$ होगा?

Ans. $a = 8$ or $\frac{2}{5}\left(6 - \sqrt{21}\right)$

Sol. POI $y = \frac{1}{x}$; $y = \frac{1}{2x-1}$; $x = 2$ & $x = a$

$x = 2x - 1$

x = 1

Case-I : a > 2

$$A = \int_2^a \left(\frac{1}{x} - \frac{1}{2x-1}\right)dx$$

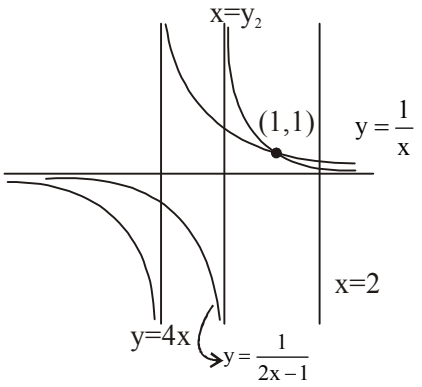
$$\ell n \frac{4}{\sqrt{5}} = \ell nx - \frac{1}{2}\ell n(2x-1)\Big|_2^a$$

$$= \ell na - \frac{1}{2}\ell n(2a-1) - \ell n2 + \frac{1}{2}\ell n3$$

$$\ell n \frac{4}{\sqrt{5}} = \ell n\frac{a}{2} - \frac{1}{2}\ell n\frac{2a-1}{3}$$

$$\frac{1}{2}\ell n\left(\frac{2a-1}{3}\right) = \ell n\frac{a}{2}\times\frac{\sqrt{5}}{4}$$

$$\frac{2a-1}{3} = \frac{a^2\times 5}{64}$$



$128a - 64 = 15a^2$

$15a^2 - 128a + 64 = 0$

$15a^2 - 120a - 8a + 64 = 0$

$$(15a - 8)(a - 8) = 0 \Rightarrow a = \frac{8}{15}, 8$$

a = 8

Case-II :

1 < a < 2

$$A = \int_a^2 \left(\frac{1}{x} - \frac{1}{2x-1}\right)dx$$

$$\ell n \frac{4}{\sqrt{5}} = \left(\ell nx - \frac{1}{2}\ell n(2x-1)\right)\Big|_a^2$$

$$\ell n \frac{4}{\sqrt{5}} = \frac{1}{2}\ell n\left(\frac{2a-1}{3}\right) - \ell n\frac{a}{2}$$

$$\ell n\frac{2a}{\sqrt{5}} = \frac{1}{2}\ell n\left(\frac{2a-1}{3}\right)$$

$$\frac{4a^2}{5} = \frac{2a-1}{3}$$

$12a^2 = 10a - 5$

$12a^2 - 10a + 5 = 0$

$D = 100 - 240 < 0$

Case-III :

$$\frac{1}{2} < a < 1$$

$$A = \int_a^1\left(\frac{1}{2x-1} - \frac{1}{x}\right)dx + \int_1^2\left(\frac{1}{x} - \frac{1}{2x-1}\right)dx$$

$$\ell n\frac{4}{\sqrt{5}} = \frac{1}{2}\ell n(2x-1) - \ell nx\Big|_a^1 + \left(\ell nx - \frac{1}{2}\ell n(2x-1)\right)\Big|_1^2$$

$$\ell n\frac{4}{\sqrt{5}} = \frac{1}{2}\ell n\frac{a^2}{2a-1} + \frac{1}{2}\ell n\left(\frac{4}{3}\right)$$

$$\frac{16}{5} = \left(\frac{a^2}{2a-1}\right)\frac{4}{3}$$

$12(2a - 1) = 5a^2$

$24a - 12 = 5a^2$

$5a^2 - 24a + 12 = 0$

$$a = \frac{24 \pm \sqrt{24^2 - 4\times 12\times 5}}{10}$$

$$a = \frac{24 \pm \sqrt{12\times 4\times 7}}{10}$$

$$a = \frac{12 \pm 2\sqrt{21}}{5}$$

$$a = \frac{2}{5}\left(6+\sqrt{21}\right), \frac{2}{5}\left(6-\sqrt{21}\right)$$

$$a \neq \frac{2}{5}\left(6+\sqrt{21}\right)$$

so, $a = \frac{2}{5}\left(6-\sqrt{21}\right)$

14. For the curve $f(x) = \frac{1}{1+x^2}$, let two points on it are $A\left(\alpha, f(\alpha)\right)$, $B\left(-\frac{1}{\alpha}, f\left(-\frac{1}{\alpha}\right)\right)$ $(\alpha > 0)$. Find the minimum area bounded by the line segments OA, OB and $f(x)$, where 'O' is the origin.

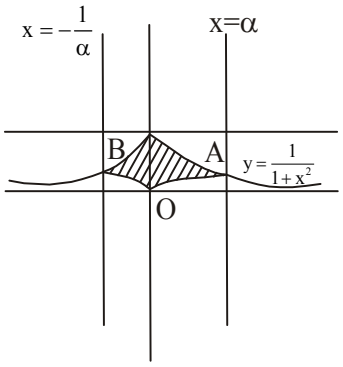
माना $A\left(\alpha, f(\alpha)\right)$, $B\left(-\frac{1}{\alpha}, f\left(-\frac{1}{\alpha}\right)\right)$ $(\alpha > 0)$ वक्र $f(x) = \frac{1}{1+x^2}$ पर दो बिन्दु है तथा रेखाओं OA, OB और वक्र $f(x)$, द्वारा परिबद्ध न्यूनतम क्षेत्रफल ज्ञात कीजिये। (जहाँ 'O' मूलबिन्दु है)

Ans. $\frac{(\pi-1)}{2}$

Sol. $f(\alpha) = \frac{1}{1+\alpha^2}$

$$f\left(-\frac{1}{\alpha}\right) = \frac{\alpha^2}{1+\alpha^2}$$

$$A = \int_{-\frac{1}{\alpha}}^{\alpha} \frac{1}{1+x^2} - \frac{1}{2}\times\frac{1}{\alpha}\times\frac{\alpha^2}{1+\alpha^2} - \frac{1}{2}\times\alpha\times\frac{1}{1+\alpha^2}$$



$$A = \tan^{-1}(x)\Big|_{-\frac{1}{\alpha}}^{\alpha} - \frac{1}{2}\frac{\alpha}{1+\alpha^2} - \frac{1}{2}\frac{\alpha}{1+\alpha^2}$$

$$A = \tan^{-1}\alpha + \tan^{-1}\left(\frac{1}{\alpha}\right) - \frac{\alpha}{1+\alpha^2}$$

$$A = \tan^{-1}\alpha + \cot^{-1}\alpha - \frac{\alpha}{1+\alpha^2}$$

$$A = \frac{\pi}{2} - \frac{\alpha}{1+\alpha^2}$$

$$\frac{dA}{d\alpha} = -\frac{\left(1+\alpha^2\right) - 2\alpha^2}{\left(1+\alpha^2\right)} = 0$$

$1 - \alpha^2 = 0$

$\alpha = 1 \qquad \alpha > 0$

$$A = \frac{\pi}{2} - \frac{1}{2}$$

$$A = \frac{\pi - 1}{2}$$

15. Let 'c' be the constant number such that $c > 1$. If the least area of the figure given by the line passing through the point (1, c) with gradient '*m*' and the parabola $y = x^2$ is 36 sq. units find the value of $(c^2 + m^2)$.

माना 'c' अचर संख्या इस प्रकार है कि $c > 1$. यदि बिन्दु (1, c) से जाने वाली तथा '*m*' प्रवणता की रेखा और परवलय $y = x^2$ से परिबद्ध न्यूनतम क्षेत्रफल 36 वर्ग इकाई है, तो $(c^2 + m^2)$ का मान ज्ञात कीजिये।

Ans. 104

Sol. Let line $y - c = m(x - 1)$

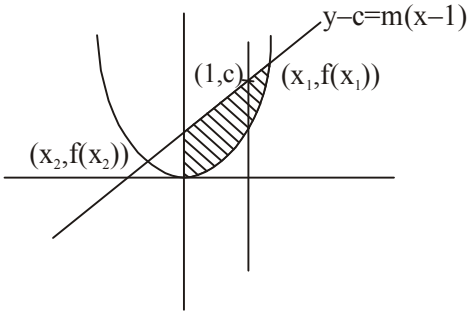
POI

$y = mx + c - m$

& $y = x^2$

$x^2 = mx + c - m$

$$x^2 - mx + m - c = 0 \begin{cases} x_1 \\ x_2 \end{cases}$$



$x_1 + x_2 = m$

$x_1 x_2 = m - c$

$$x_2 - x_1 = \sqrt{m^2 - 4m + 4c}$$

$$x_2^2 - x_1^2 = m\sqrt{m^2 - 4m + 4c}$$

$$x_2^3 - x_1^3 = (x_2 - x_1)\left(x_2^2 + x_1^2 + x_2x_1\right)$$

$$= (x_2 - x_1)\left((x_1 + x_2)^2 - x_1x_2\right)$$

$$= \sqrt{m^2 - 4m + 4c}\left(m^2 - m + c\right)$$

$$A = \int_{x_1}^{x_2}\left(mx + c - m - x^2\right)dx$$

$$A = \frac{m}{2}x^2 + (c - m)x - \frac{x^3}{3}\Bigg|_{x_1}^{x_2}$$

$$A = \frac{m}{2}\left(x_2^2 - x_1^2\right) + (c - m)(x_2 - x_1) - \frac{1}{3}\left(x_2^3 - x_1^3\right)$$

$$A = (x_2 - x_1)\left(\frac{m}{2}(x_2 + x_1) + (c - m) - \frac{1}{3}\left(x_1^2 + x_2^2 + x_1x_2\right)\right)$$

$$A = \sqrt{m^2 - 4m + 4c}\left(\frac{m^2}{2} + c - m - \frac{1}{3}\left(m^2 - m + c\right)\right)$$

$$A = \sqrt{m^2 - 4m + 4c}\left(\frac{m^2}{6} - \frac{2}{3}m + \frac{2}{3}c\right)$$

for minimum area

$$\frac{dA}{dm} = 0$$

$$\sqrt{m^2 - 4m + 4c}\left(\frac{m}{3} - \frac{2}{3}\right) + \left(\frac{m^2 - 4m + 4c}{6}\right)\frac{2m - 4}{\sqrt{m^2 - 4m + 4c}} = 0$$

$$\left(m^2 - 4m + 4c\right)\left(\frac{m - 2}{3}\right) + \frac{\left(m^2 - 4m + 4c\right)}{3}(m - 2) = 0$$

$m = 2$ or $m^2 - 4m + 4c = 0$ (not possible)

so, $m = 2$

$$A_{min.} = \sqrt{4c-4}\left(\frac{4}{6}-\frac{4}{3}+\frac{2}{3}c\right)$$

$$36 = 2\sqrt{c-1}\left(\frac{2}{3}c-\frac{2}{3}\right)$$

$$36 = \frac{4}{3}(c-1)^{\frac{3}{2}}$$

$$(c-1)^{\frac{3}{2}} = 27 \Rightarrow (c-1)^{\frac{1}{2}} = 3$$

$c - 1 = 9 \Rightarrow c = 10$

$c^2 + m^2 = 100 + 4 = 104$

16. Let A_n be the area bounded by the curve $y = (\tan x)^n$ & the lines $x = 0$, $y = 0$ & $x = \pi/4$. Prove that for $n > 2$, $A_n + A_{n-2} = 1/(n-1)$ & deduce that $1/(2n+2) < A_n < 1/(2n-2)$.

यदि A_n वक्र $y = (\tan x)^n$ तथा सरल रेखाओं $x = 0$, $y = 0$ और $x = \pi/4$ से परिबद्ध क्षेत्रफल है, तो सिद्ध कीजिए $n > 2$, $A_n + A_{n-2} = 1/(n-1)$ तथा $1/(2n+2) < A_n < 1/(2n-2)$.

Ans.

Sol. $A_n = \int_0^{\pi/4} (\tan x)^n \, dx$

$$A_n + A_{n-2} = \int_0^{\pi/4} \left[(\tan x)^n + (\tan x)^{n-2}\right] dx$$

$$= \int_0^{\pi/4} (\tan x)^{n-2} \sec^2 x \, dx = \left[\frac{t^{n-1}}{n-1}\right]_0^1 = \frac{1}{n-1}$$

Also $A_{n+2} < A_n < A_{n-2}$

$$\Rightarrow \frac{1}{n+1} < 2A_n < \frac{1}{n-1}$$

17. Consider the two curves $C_1 : y = 1 + \cos x$ & $C_2 : y = 1 + \cos(x-\alpha)$ for $\alpha \in (0, \pi/2)$; $x \in [0, \pi]$. Find the value of α, for which the area of the figure bounded by the curves C_1, C_2 & $x = 0$ is same as that of the figure bounded by C_2, $y = 1$ & $x = \pi$. For this value of α, find the ratio in which the line $y = 1$ divides the area of the figure by the curves C_1, C_2 & $x = \pi$.

दिया है $C_1 : y = 1 + \cos x$ और $C_2 : y = 1 + \cos(x-\alpha)$ for $\alpha \in (0, \pi/2)$ और $x \in [0, \pi]$ दो वक्र हैं। α का मान ज्ञात कीजिये यदि वक्रो C_1, C_2 और $x = 0$ से परिबद्ध क्षेत्रफल और C_2, $y = 1$ और $x = \pi$ से परिबद्ध क्षेत्रफल समान है। इस α के मान के लिये वह अनुपात ज्ञात कीजिये जबकि वक्रों C_1, C_2 और $x = \pi$ से परिबद्ध क्षेत्रफल को रेखा $y = 1$ विभाजित करती है।

Ans. $\alpha = \pi/3$, ratio $= 2 : \sqrt{3}$

Sol. $C_1 : y = 1 + \cos x$

$C_2 : y = 1 + \cos(x - \alpha) \qquad \alpha \in \left(0, \frac{\pi}{2}\right),\ x \in [0, \pi]$

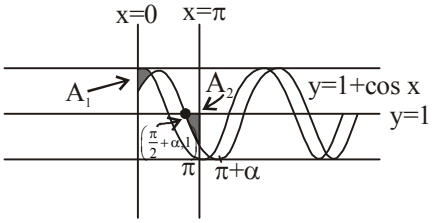
POI of C_1 & C_2

$\cos x = \cos(x - \alpha)$

$x = |x - \alpha|$

$2\xi = \alpha$

$x = \frac{\alpha}{2}$



Area

$A_1 : C_1, C_2$ & $x = 0$

$A_2 : C_2, y = 1$ & $x = \pi$

$A_1 = A_2$

$$\int_0^{\frac{\alpha}{2}} \left(\cos x - \cos(x-\alpha)\right) dx = \int_{\frac{\pi}{2}+\alpha}^{\pi} -\cos(x-\alpha)\, dx$$

$$\sin x - \sin(x-\alpha)\Big|_0^{\frac{\alpha}{2}} = \sin(x-\alpha)\Big|_{\pi}^{\frac{\pi}{2}+\alpha}$$

$$\sin\frac{\alpha}{2} + \sin\frac{\alpha}{2} - \sin\alpha = 1 - \sin\alpha$$

$$\sin\frac{\alpha}{2} = \frac{1}{2}$$

$$\frac{\alpha}{2} = \frac{\pi}{6}$$

$$\alpha = \frac{\pi}{3}$$

18. For what values of $a \in [0,1]$ does the area of the figure bounded by the graph of the function $y = f(x)$ and the straight lines $x = 0$, $x = 1$ & $y = f(a)$ is at a minimum & for what values it is at a maximum if $f(x) = \sqrt{1-x^2}$. Find also the maximum & the minimum areas.

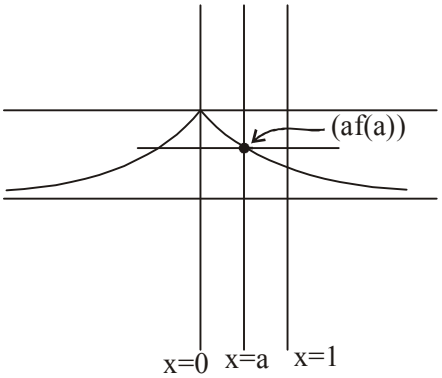
$\in [0,1]$ के किस मान के लिये वक्र $y = f(x)$ सरल रेखाऐं $x = 0$, $x = 1$ और $y = f(a)$ से परिबद्ध क्षेत्रफल न्यूनतम होगा तथा किस मान के लिये यह अधिकतम होगा यदि $f(x) = \sqrt{1-x^2}$ साथ ही अधिकतम और न्यूनतम क्षेत्रफल भी ज्ञात कीजिये।

Ans. $a = 1/2$ gives minima, $A\left(\frac{1}{2}\right) = \frac{3\sqrt{3}-\pi}{12}$; $a = 0$ gives local maxima $A(0) = 1 - \frac{\pi}{4}$; $a = 1$ gives maximum value, $A(1) = \pi/4$

Sol. Given $a \in [0,1]$

$$A = \int_0^a \left(\sqrt{1-x^2} - f(a)\right)dx + \int_a^1 \left(f(a) - \sqrt{1-x^2}\right)dx$$

$$A = \int_0^a \sqrt{1-x^2}\,dx - \int_0^a f(a)\,dx + \int_a^1 f(a)\,dx - \int_0^1 \sqrt{1-x^2}\,dx$$



$$\left[\frac{x}{2}\sqrt{1-x^2} - \frac{1}{2}\sin^{-1}(x)\right]_0^a - \left[\frac{x}{2}\sqrt{1-x^2} - \frac{1}{2}\sin^{-1}(x)\right]_a^1$$

$-a\,f(0) + (1-a)\,f(a)$

$$= \frac{a}{2}\sqrt{1-a^2} - \frac{1}{2}\sin^{-1}a + \frac{1}{2}\times\frac{\pi}{2} + \frac{a}{2}\sqrt{1\cdot a^2} - \frac{1}{2}\sin^{-1}a$$

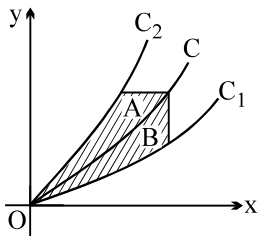
$$-a + (1-a)\sqrt{1-a^2}$$

$$A = \sqrt{1-a^2} - \sin^{-1}a + \frac{\pi}{4}$$

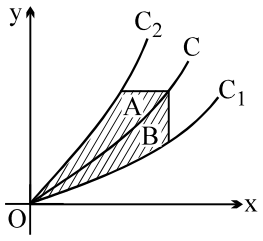
$$\frac{dA}{da} = \frac{-2a}{\sqrt{1-a^2}} - \frac{1}{\sqrt{1-a^2}} = 0$$

$$a = -\frac{1}{2}$$

19. Let C_1 and C_2 be two curves passing through the origin as shown in the figure. A curve C is said to "bisect the area" the region between C_1 and C_2, if for each point P of C, the two shaded regions A and B shown in the figure have equal areas. Determine the upper curve C_2, given that the bisecting curve C has the equation $y = x^2$ & that the lower curve C_1 has the equation $y = x^2/2$.



माना C_1 और C_2 मूल बिन्दु से जाने वाले वक्र चित्रनुसार है। यदि वक्र C, C_1 और C_2 के क्षेत्र को C पर प्रत्येक बिन्दु P के लिये समद्विभाजित करता है, तथा चित्र में दो परिबद्ध क्षेत्रफल A और B समान क्षेत्रफल वाले दिये गये हैं। तब ऊपरी वक्र C_2 ज्ञात कीजिये। जबकि वक्र C और C_1 के समीकरण क्रमशः $y = x^2$ और $y = x^2/2$ दिये गये हैं।



Ans. $(16/9)\,x^2$

Sol. $C : y = x^2$

$C_1 : \; y = \dfrac{x^2}{2}$

Let $C_2 : y = f(x)$

Let point of curve C is (α, α^2)

$A_1 = B$

$$\int_0^{\alpha^2} (C - C_2)\,dy = \int_0^{\alpha} (C - C_1)\,dx$$

$$\int_0^{\alpha^2} \left(\sqrt{y} - f^{-1}(y)\right)dy = \int_0^{\alpha} \left(x^2 - \frac{x^2}{2}\right)dx$$

$$\left(\alpha - f^{-1}\left(\alpha^2\right)\right)2\alpha = \frac{\alpha^2}{2}$$

$$\alpha - f^{-1}\left(\alpha^2\right) = \frac{\alpha}{4}$$

$$f^{-1}\left(\alpha^2\right) = \frac{3}{4}\alpha$$

$$f\left(\frac{3}{4}\alpha\right)=\alpha^2$$

$$f(x)=\frac{16}{9}x^2$$

$$\because \frac{3}{4}\alpha=x$$

$$\alpha=\frac{4}{3}x$$

$$\therefore\ y=\alpha^2=\frac{16}{9}x^2$$

20. Given f (x) = $\int_0^x e^t(\ln\sec t-\sec^2 t)\,dt$; g (x) = – $2e^x$ tan x. Find the area bounded by the curves $y=f(x)$ and $y=g(x)$ between the ordinates $x=0$ and $x=\frac{\pi}{3}$.

माना $f(x)=\int_0^x e^t(\ln\sec t-\sec^2 t)\,dt$; $g(x)=-2e^x\tan x$ तब $y=f(x)$, $y=g(x)$ और $x=0$ तथा $x=\frac{\pi}{3}$ से परिबद्ध क्षेत्रफल ज्ञात कीजिए।

Ans. $e^{\pi/3}\log 2$ sq. units

Sol. $f(x)=\int_0^x e^t\left(\ell n\sec t-\sec^2 t\right)$

$$=\int_0^x e^t\left(\ell n\sec t+\tan t\right)-e^t\left(\tan t+\sec^2 t\right)dt$$

$$=e^t\left(\ell n\sec t-\tan t\right)\Big|_0^x=e^x\left(\ell n\sec x-\tan x\right)$$

Area between f(x) & g(x)

$$A=\int_0^{\frac{\pi}{3}}\left(f(x)-g(x)\right)dx$$

$$= \int_0^{\frac{\pi}{3}} e^x \left(\ell n \sec x + \tan x\right) dx$$

$$= e^x \ell n \sec x \Big|_0^{\frac{\pi}{3}}$$

$$= e^{\frac{\pi}{3}} \ell n 2$$

21. Let $f: [0, \infty) \to R$ be a continuous and strictly increasing function such that $f^3(x) = \int_0^x t f^2(t)\, dt$, $\forall\ x > 0$. Find the area enclosed by $y = f(x)$, the x-axis and the ordinate at $x = 3$.

माना $f : [0, \infty) \to R$ सतत् तथा निरतंर वर्धमान फलन इस प्रकार है कि $f^3(x) = \int_0^x t f^2(t)\, dt$, $\forall\ x > 0$, तब $y = f(x)$, x-अक्ष और $x = 3$ से परिबद्ध क्षेत्र का क्षेत्रफल ज्ञात कीजिए।

Ans. 3/2

Sol. $f^3(x) = \int_0^x t f^2(t)\, dt$

$f(0) = 0$

$f \Rightarrow$ strictly increasing

$3f^2(x)\, f'(x) = x f^2(x)$

$$f'(x) = \frac{x}{3}$$

$$f(x) = \frac{x^2}{6} + C$$

$f(0) = C = 0$

$$f(x) = \frac{x^2}{6}$$

$$A = \int_0^3 \frac{x^2}{6} dx$$

$$= \frac{x^3}{18}\Big|_0^3 = \frac{3}{2}$$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question nos. 22 to 24

If f is smooth on [a, b], the length of the curve y = f(x) from a to b is $L = \int_a^b \sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx = \int_a^b \sqrt{1+(f'(x))^2}\,dx$ .....(i)

At point on a curve where dy/dx fails to exist, dx/dy may exist and we may be able to find the curve 's length by expressing x as a function of y and applying the following analogue of Eq.(i) Formula for the length of a smooth

curve $x = g(y),\ c \le y \le d$ $L = \int_c^b \sqrt{1+\left(\frac{dx}{dy}\right)^2}\,dy = \int_c^d \sqrt{1+(g'(y))^2}\,dy$ .....(ii)

यदि f अन्तराल [a, b] में अवकलनीय है तथा a से b में वक्र y = f(x) की लम्बाई निम्न प्रकार से परिभाषित की जाती है

$$L = \int_a^b \sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx = \int_a^b \sqrt{1+(f'(x))^2}\,dx \qquad \text{.....(i)}$$

यदि वक्र के किसी बिन्दु पर dy/dx विद्यमान नहीं है तथा dx/dy विद्यमान हो सकता है तो हम वक्र की लम्बाई x को y के फलन के रूप esa Q;Dr djds fuEu izdkj ls Kkr dj ldrs gSaA Eq.(i) की मदद से वक्र की लम्बाई $x = g(y),\ c \le y \le d$

$$L = \int_c^b \sqrt{1+\left(\frac{dx}{dy}\right)^2}\,dy = \int_c^d \sqrt{1+(g'(y))^2}\,dy \qquad \text{.....(ii)}$$

22. The length of the curve $y = \frac{4\sqrt{2}}{3}x^{3/2} - 1,\ 0 \le x \le 1$ is :

(A) $\frac{13}{6}$ (B) 2 (C) $\frac{1}{12}$ (D) none of these

वक्र $y = \frac{4\sqrt{2}}{3}x^{3/2} - 1,\ 0 \le x \le 1$ की लम्बाई होगी :

(A) $\frac{13}{6}$ (B) 2 (C) $\frac{1}{12}$ (D) कोई नहीं

Ans. A

Sol. $y = \frac{4\sqrt{2}}{3}x^{\frac{3}{2}} - 1 \qquad 0 \le x \le 1$

$$\frac{dy}{dx} = 2\sqrt{2}x^{\frac{1}{2}}$$

$$L = \int_0^1 \sqrt{1+8x}\,dx$$

$$= \frac{2}{3}\frac{(1+8x)^{\frac{3}{2}}}{8}\Bigg|_0^1 = \frac{2}{3}\times\frac{27}{8} - \frac{1}{12}$$

$$= \frac{27}{12} - \frac{1}{12}$$

$$= \frac{13}{6}$$

23. The length of the curve $y = (x/2)^{2/3}$ from $x = 0$ to $x = 2$ is :

(A) 3.37 (B) 2.27 (C) 2.0 (D) none of these

वक्र $y = (x/2)^{2/3}$ की $x = 0$ तथा $x = 2$ के मध्य की लम्बाई होगी :

(A) 3.37 (B) 2.27 (C) 2.0 (D) कोई नहीं

Ans. B

Sol. $y = \left(\frac{x}{2}\right)^{\frac{2}{3}} \Rightarrow x = 0 \text{ to } y = 2$

$$\frac{dy}{dx} = \frac{2}{3\left(\frac{x}{2}\right)^{\frac{1}{3}}} \qquad x = 0 \text{ not defined}$$

$x = 2y^{\frac{3}{2}} \Rightarrow y = 0 \text{ to } y = 1$

$$\frac{dx}{dy} = 3y^{\frac{1}{2}}$$

$$L = \int_0^1 \sqrt{1+9y}\,dy$$

$$= \frac{2}{3}\frac{(1+9y)^{\frac{3}{2}}}{9}\Bigg|_0^1$$

$$= \frac{2}{3}\times\frac{10^{\frac{3}{2}}}{9} - \frac{2}{3\times 9}$$

$$= \frac{2\times 31{\cdot}62}{27} - \frac{2}{27}$$

$= 2.266$

24. The length of the curve $y = (1/3)\,(x^2+2)^{3/2}$ from $x = 0$ to $x = 3$ is :

(A) 10 (B) 12 (C) $\frac{1}{12}$ (D) none of these

वक्र $y = (1/3)\,(x^2+2)^{3/2}$ की $x = 0$ से $x = 3$ के मध्य की लम्बाई होगी :

(A) 10 (B) 12 (C) $\frac{1}{12}$ (D) कोई नही

Ans. B

Sol. $y = \frac{1}{3}\left(x^2+2\right)^{\frac{3}{2}}$ $x = 0$ to $x = 3$

$$\frac{dy}{dx} = \frac{1}{2}\left(x^2+2\right)^{\frac{1}{2}}\cdot 2x$$

$$L = \int_0^3 \sqrt{1+\frac{1}{4}\left(x^2+2\right)\cdot 4x^2}\,dx$$

$$L = \int_0^3 \sqrt{1+x^4+2x^2}\,dx$$

$$= \int_0^3 \left(x^2+1\right)dx$$

$$= \frac{x^3}{3} + x \Big|_0^3$$

$= 9 + 3 = 12$

Paragraph for Question nos. 25 to 28

Let $f(x)$ be a differntiable function such that $f(x+y) = e^x f(y) + e^y f(x)$ all x,y and $f'(0) = 1$

माना $f(x)$ अवकलनीय फलन इस प्रकार है कि $f(x+y) = e^x f(y) + e^x f(x)$ सभी x,y के लिये तथा $f'(0) = 1$

25. f(x) has :

(A) maximum (B) minimum

(C) both maximum and minimum (D) neither maximum nor minimum

f(x) रखता है :

(A) उच्चिष्ठ (B) निम्निष्ठ

(C) उच्चिष्ठ तथा निम्निष्ठ (D) ना तो उच्चिष्ठ न ही निम्निष्ठ

Ans. B

Sol. $f(x + y) = e^x f(y) + e^y f(x)$

$f'(0) = 1 \qquad f(0) = 0$

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \to 0} \frac{e^x f(h) + e^h f(x) - f(x)}{h}$$

$$= \lim_{h \to 0} \frac{e^x f'(h) + e^h f(x)}{h}$$

$$= e^x f'(0) + e^0 f(x)$$

$$f'(x) = e^x + f(x)$$

$$\frac{dy}{dx} - y = e^x$$

I.F. $= e^{-x}$

$$y \cdot e^{-x} = \int dx$$

$y = (x + c)e^x$

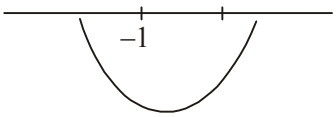
$x = 0, y = 0$

$0 = c \Rightarrow c = 0$

$y = x e^x$

$$\frac{dy}{dx} = e^x (x+1) = 0$$

$x = -1$



minima

26. The range of f(x) is :

f(x) का परिसर होगा :

(A) R (B) $[0, \infty]$ (C) $\left(-\frac{1}{e}, 1\right)$ (D) $\left[-\frac{1}{e}, \infty\right)$

Ans. D

Sol. $f_{min}(-1) = -\frac{1}{e}$

Range $\left[-\frac{1}{e}, \infty\right)$

27. $\lim\limits_{x \to -\infty} f(x)$ is :

(A) 0 (B) 1 (C) –1 (D) non existent

$\lim\limits_{x \to -\infty} f(x)$ होगा :

(A) 0 (B) 1 (C) –1 (D) विद्यमान नहीं है

Ans. A

Sol. $\lim\limits_{x \to -\infty} f(x) = \lim\limits_{x \to -\infty} xe^x$

$= \lim\limits_{x \to -\infty} \frac{x}{e^{-x}}$

$= \lim_{x \to -\infty} \frac{1}{-e^{-x}}$

$= 0$

28. The area bounded by the curve $y = f(x)$ and the x–axis is :

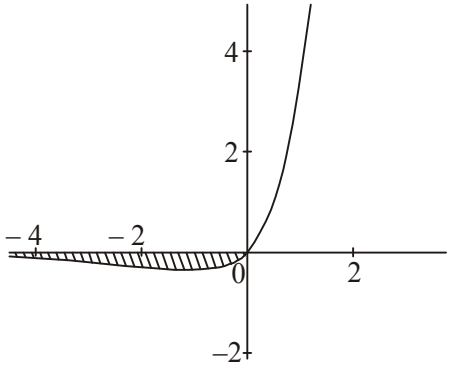
वक्र $y = f(x)$ तथा x–अक्ष द्वारा परिबद्ध क्षेत्रफल होगा :

(A) 1 (B) $\frac{1}{2}$ (C) 2 (D) e

Ans. A

Sol. Required area $= \int_{-\infty}^{0} \left| x \cdot e^x \right| dx = -\left(e^x (x-1) \right)_{-\infty}^{0} = -\left(e^0 (0-1) - 0 \right)$

= 1 sq. units



MATCH THE COLUMN TYPE QUESTION(S)

29. Match the following column :

Column I	**Column II**
(A) The area between the parabolas $y^2 = 4ax$ and $x^2 = 8ay$ is	(P) 27
(B) Area bounded by the curve $y = (x-1)(x-2)(x-3)$ and x-axis lying between the ordinates $x = 0$ and $x = 3$ is equal to	(Q) 2π
(C) The area cut off the parabola $4y = 3x^2$ by the straight line $2y = 3x + 12$ in sq. units is	(R) $\frac{32}{3}a^2$
(D) The area of the region satisfying $\lvert x \rvert + \lvert y \rvert \le \sqrt{\pi}$ is	(S) $\frac{11}{4}$

निम्न कॉलम को सुमेलित कीजिए :

Column I	**Column II**
(A) परवलयों $y^2 = 4ax$ तथा $x^2 = 8ay$ के मध्य का क्षेत्रफल होगा	(P) 27
(B) वक्र $y = (x - 1)(x - 2)(x - 3)$, x-अक्ष तथा रेखाओं $x = 0$ और $x = 3$ के मध्य क्षेत्रफल होगा	(Q) 2π
(C) परवलय $4y = 3x^2$ तथा सरल रेखा $2y = 3x + 12$ के मध्य का क्षेत्रफल होगा	(R) $\frac{32}{3}a^2$
(D) वक्र $\lvert x\rvert + \lvert y\rvert \le \sqrt{\pi}$ से घिरा हुआ क्षेत्रफल होगा	(S) $\frac{11}{4}$

Ans. A - R, B - S, C - P, D - Q

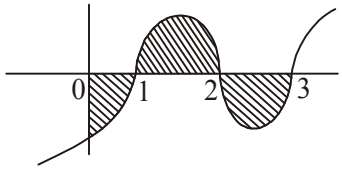
Sol. (A) $y^2 = 4ax$

$x^2 = 8ay$

$$A = \frac{16}{3} \times a \times b$$

$$A = \frac{32}{3}a^2$$

(B)



$$A = -\int_0^1 (x-1)(x-2)(x-3) + \int_1^2 (x-1)(x-2)(x-3) - \int_2^3 (x-1)(x-2)(x-3)\,dx$$

$$\int (x-1)(x-2)(x-3)\,dx = \int \left(x^3 - 6x^2 + 11x - 6\right)dx$$

$$= \frac{x^4}{4} - 2x^3 + \frac{11}{4}x^2 - 6x$$

$$A = -\left[\frac{x^4}{4} - 2x^3 + \frac{11}{4}x^2 - 6x\right]_0^1 + \left[\frac{x^4}{4} - 2x^3 + \frac{11}{4}x^2 - 6x\right]_1^2 - \left[\frac{x^4}{4} - 2x^3 + \frac{11}{4}x^2 - 6x\right]_2^3$$

$$A = \frac{9}{4} + \frac{1}{4} + \frac{1}{4} = \frac{11}{4}$$

(C) Point of Intersection

$4y = 3x^2$ & $2y = 3x + 12$

$3x^2 = 6x + 24$

$x^2 - 2x - 8 = 0$

$x = 4$ & $x = -2$

$y = 12$ $y = 3$

$$A = \int_{-2}^{4}\left(\frac{3x+12}{2} - \frac{3x^2}{4}\right)dx$$

$$= \left(\frac{3x^2}{4} + 6x - \frac{x^3}{4}\right)_{-2}^{4}$$

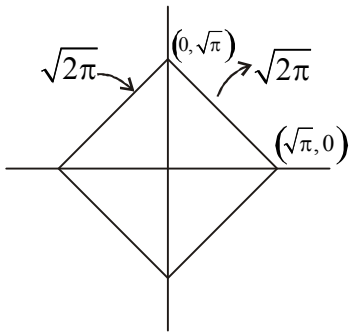
$= 20 - (-7)$

$= 27$

(D) $|x| + |y| \le \sqrt{\pi}$

$A = \sqrt{2\pi} \times \sqrt{2\pi}$

$= 2\pi$



30. Match the following column :

Column I	**Column II**
(A) The cosine of the angle between the curves $y = 3^{x-1} \ln x$ and $y = x^x - 1$ at their point of intersection on the line $y = 0$, is	(P) 0
(B) The area bounded by the curves $x = -4y^2$ and $(x - 1) = -5y^2$ is	(Q) 1
(C) The value of the integral $\int_0^{\pi/2} (\sin x)^{\cos x}(\cos x \cot x - \ln(\sin x)^{\sin x})\,dx$, is	(R) $\frac{4}{3}$

(D) A continuous function $f: [1, 6] \to [0, \infty)$ is such that $f'(x) = \frac{2}{x + f(x)}$ (S) $2\ ln\ 6$

and $f(1) = 0$, then the maximum value of f cannot exceed

निम्न कॉलम को सुमेलित कीजिए :

	Column I	**Column II**
(A)	वक्रों $y = 3^{x-1}\ ln\ x$ और $y = x^x - 1$ के मध्य रेखा $y = 0$ से प्रतिच्छेद बिन्दु पर कोज्या कोण का मान है	(P) 0
(B)	वक्रों $x = -4y^2$ और $(x - 1) = -5y^2$ से परिबद्ध क्षेत्रफल है	(Q) 1
(C)	समाकलन $\int_0^{\pi/2} (\sin x)^{\cos x} (\cos x \cot x - \ln(\sin x)^{\sin x})\, dx$ का मान है	(R) $\frac{4}{3}$
(D)	एक सतत् फलन $f: [1, 6] \to [0, \infty)$ इस प्रकार है कि $f'(x) = \frac{2}{x + f(x)}$ और $f(1) = 0$, तब f का मान अधिकतम मान से अधिक नहीं हो सकता	(S) $2\ ln\ 6$

Ans. (A) Q, (B) R, (C) Q, (D) S

Sol. (A) $y_1 = 3^{x-1} \ell nx$ $\qquad y_2 = x^x - 1$

Point of intersection $y = 0$

$x = 1$

$$\frac{dy_1}{dx} = \frac{3^{x-1}}{x} + 3x^{x-1} \ell n3 \ell nx$$

$(1, 0)$

$m_1 = 1$

$$\frac{dy_2}{dx} = x^x (1 + \ell nx)$$

$(1,0)$

$m_2 = 1$

Angle $\theta = 0$

$\cos \theta = 1$

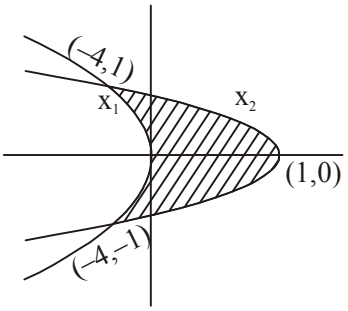
(B) $x = -4y^2$ $\qquad x - 1 = -5y^2$

Point of intersection $\qquad -4y^2 - 1 = -5y^2$

$y^2 = 1$

$y = \pm 1$

$$A = \int_{-1}^{1} (x_2 - x_1)\,dy$$



$$= \int_{-1}^{1} \left[1 - 5y^2 + 4y^2\right] dy$$

$$= \int_{-1}^{1} \left(1 - y^2\right) dy$$

$$= 2\int_{0}^{1} \left(1 - y^2\right) dy$$

$$= 2\left(y - \frac{y^3}{3}\right)_0^1 = \frac{4}{3}$$

(C) $$\int_{0}^{\frac{\pi}{2}} \sin x^{\cos x} \left(\cos x \cdot \cot x - \ell n\left(\sin x\right)^{\sin x}\right) dx$$

$$= \int_{0}^{\frac{\pi}{2}} \frac{d}{dx}\left(\sin x^{\cos x}\right) dx$$

$$= \left(\sin x\right)^{\cos x}\Big|_0^{\frac{\pi}{2}}$$

$$= 1$$

(D) $f'(x) > 0$ so f_{max} at $x = 6$

$$f'(x) = \frac{2}{x + f(x)}$$

$$\frac{dy}{dx} = \frac{2}{x + y}$$

$$x + y = t$$

$$1 + \frac{dy}{dx} = \frac{dt}{dx}$$

$$\frac{dt}{dx}-1=\frac{2}{t}$$

$$\frac{tdt}{2+t}=dx$$

$$\int\left(1-\frac{2}{2+t}\right)dt=\int dx$$

$$t-2\log(2+t)=x+C$$

$$x+y-2\log\left(2+x+y\right)=x+C$$

$$y-2\log\left(2+x+y\right)=C$$

$x=1 \qquad y=0$

$$-2\log 3=C$$

$$y-2\log\left(2+x+y\right)=-2\log 3$$

$$y=2\log\left(\frac{2+x+y}{3}\right)$$

at $x=6$

$$y=2\log\left(\frac{8+y}{3}\right)$$

$$\frac{8+y}{3}=e^{\frac{y}{2}}$$

$$8+y=3e^{\frac{y}{2}}$$

$$y<e^{\frac{y}{2}}$$

$$8+e^{\frac{y}{2}}\geq 3e^{\frac{y}{2}}$$

$$8\geq 2e^{\frac{y}{2}}$$

$$e^{\frac{y}{2}}\leq 4$$

$$\frac{y}{2}\leq \ell n4$$

$y \le 2\ell n4$

31. Match the following column :

Column I	**Column II**
(A) Area of region formed by points (x, y) satisfying $[x] = [y]$, $0 \le x, y \le 10$, [.] denote greatest integer function is	(P) 2
(B) Area of region formed by points (x, y) satisfying $\max(|x|, |y|) \le 1$ is	(Q) 4
(C) If area of region formed by points (x, y) satisfying $|y + x| \le 1$, $|y - x| \le 1$ and $2x^2 + 2y^2 \ge 1$ is $k - \frac{\pi}{2}$, then k =	(R) 6
(D) If area of region formed by points (x, y) satisfying $4 \le x^2 + y^2 \le 2(|x| + |y|)$ is 2k, then k =	(S) 10
	(T) 12

निम्न कॉलम को सुमेलित कीजिए :

Column I	**Column II**
(A) खा, ञ्ज ख्का ए$0 \le x, y \le 10$ को सन्तुष्ट करने वाले बिन्दुओं से निर्मित क्षेत्र का क्षेत्रफल होगा	(P) 2
(B) उं ग ; ब्बाद्यद्बद्बद्ब≤ 1 को सन्तुष्ट करने वाले बिन्दुओं से निर्मित क्षेत्र का क्षेत्रफल होगा	(Q) 4
(C) ब्का गद्य≤ 1, $|y - x| \le 1$ & $2x^2 + 2y^2 \ge 1$ को सन्तुष्ट करने वाले बिन्दुओं से निर्मित क्षेत्र का क्षेत्रफल $k - \frac{\pi}{2}$, है तो k =	(R) 6
(D) $4 \le x^2 + y^2 \le 2(|x| + |y|)$ को सन्तुष्ट करने वाले बिन्दुओं से निर्मित क्षेत्र का क्षेत्रफल 2k है तो k =	(S) 10
	(T) 12

Ans. A – S ; B – Q ; C – P ; D – Q

Sol. (A) Given : $[x] = [y]$

and $0 \le x, y \le 10$

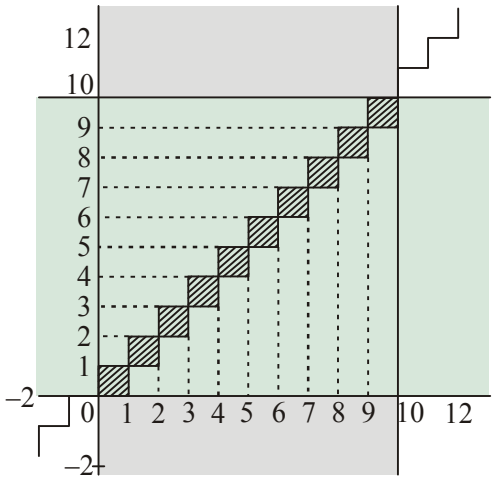
Case 1 $0 \le x < 1, 0 \le y < 1$

Case 2 $1 \le x < 2, 1 \le y < 2$

⋮

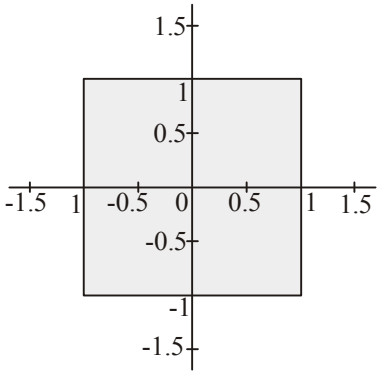
Case 10 $9 \le x < 10, 9 \le y < 10$

Similarly 10 cases wil for m; each case contain a square of area 1 sq. unit



Therefore required area is 10 sq. units.

(B) Given condition



$\max(|x|, |y|) \le 1$

There will be a square formed by lines

$x = \pm 1$ and $y = \pm 1$

So side of square = 2 units

required area = 4 sq. units

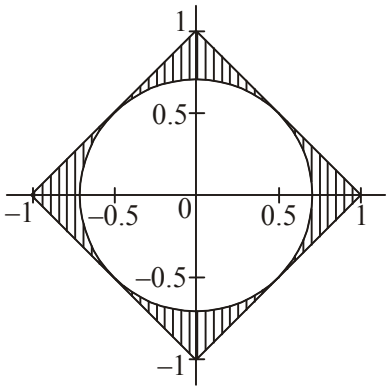
(C) Given constraints

$|y + x| \le 1$ _____(1)

$|y - x| \le 1$ _____(2)

fromo (1) and (2) a square will be formed having vertices $(0, \pm 1)$ and $(\pm 1, 0)$

So area of square $A_1 = 2$ sq. units



Also $x^2 + y^2 \ge \frac{1}{2}$

area of circle $A_2 = \pi \times \frac{1}{2}$

required area $= A_1 - A_2$

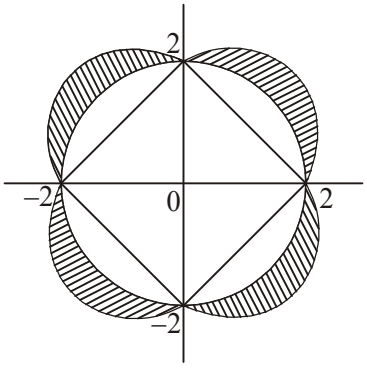
$= 2 - \frac{\pi}{2}$

$\therefore k = 2$

(D) Given constraints

$4 \le x^2 + y^2 \le 2(|x| + |y|)$

A_1 = area of square formed by $|x| + |y| \ge 2$



$A_1 = \left(2\sqrt{2}\right)^2 = 8$ square units

A_2 = area of circle formed by $x^2 + y^2 \ge 4$

$A_2 = \pi(4)$ square units

A_3 = area of shape for med by $x^2 + y^2 \le 2(|x| + |y|)$

$A_3 = 4 \times$ semi circles $= 4 \times \frac{\pi}{2} \times \left(\sqrt{2}\right)^2$

$= 4\pi$ square units

Required area $= A_3 - (A_2 - A_1)$

$= 4\pi - (4\pi - 8)$

$= 8$ square units

ATQ $2k = 8$

$k = 4$